Gravitation Theory 2

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Abstract

These are personal notes on the elective course *Gravitation Theory 2*, taught during the first semester of 2026 for master's and PhD students. They cover gauge systems and Yang–Mills theory, the first-order (Cartan) formulation of gravity in the language of differential forms, gravity as a gauge theory of the Poincaré group, and three-dimensional gravity as a Chern–Simons theory. Supporting mathematical material is collected in the appendices. These notes are not an official text of the course, and any errors are my own.

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Introduction

The gauge principle, the idea that interactions arise from demanding that a symmetry hold *locally*, independently at each point of spacetime, is the organizing principle of modern fundamental physics. It underlies electrodynamics and the Yang–Mills theories of the strong and electroweak interactions [1, 2]. Soon after the work of Yang and Mills it was also realized that the same principle sheds light on gravitation: gauging the Lorentz and Poincaré groups leads naturally to the vielbein, the spin connection and the Einstein–Cartan theory [3, 4, 5, 6]. Understanding precisely in which sense general relativity is, and is not, a gauge theory is one of the guiding questions of this course.

The natural language for this question is that of differential forms and the first-order (Cartan) formalism, in which the vielbein e^a and the Lorentz connection ω^{ab} are treated as independent fields [7, 8]. Far from being a mere rewriting of general relativity, this formalism is indispensable in current research. It is required to couple fermions to gravity; it is the setting in which torsion and the spin of matter enter the gravitational dynamics [5]; and it is the starting point of supergravity, of Lovelock and Chern–Simons gravities in higher dimensions [9, 10], and of the connection variables used in non-perturbative approaches to quantum gravity. Geometric identities that are cumbersome in components, such as the Cartan structure equations, the Bianchi identities or the Nieh–Yan identity [11], become one-line statements in this language.

Gravity in $2+1$ dimensions is a remarkable laboratory in which these ideas can be pushed all the way. Pure Einstein gravity in three dimensions has no local propagating degrees of freedom, yet it admits black holes [12] and a rich asymptotic structure: Brown and Henneaux showed that the asymptotic symmetries of AdS_3 gravity form two copies of the Virasoro algebra with a classical central charge [13], a result now regarded as a precursor of the AdS/CFT correspondence. Moreover, at the classical level three-dimensional gravity can be written as a Chern–Simons gauge theory [14, 15]. This makes it one of the few settings in which gravity is a gauge theory in the strict sense, and in which questions about quantum gravity and holography can be addressed in a controlled way [16, 17].

The course also shows how closely mathematical physics and the physics of gauge fields and gravity are intertwined. Holonomy, de Rham cohomology, invariant tensors and bilinear forms on Lie algebras, characteristic classes, the Chern–Weil theorem and the Chern–Simons forms [18] are not merely mathematical background: they control physical effects such as the Aharonov–Bohm phase [19], the existence of topologically non-trivial gauge configurations, and the structure of gravitational actions. A concrete example treated in these notes is the meron [20], a genuinely non-Abelian solution of the Yang–Mills equations which, once coupled to gravity, sources an exact black hole [21]. Constructing exact solutions of this kind, often with the help of computer algebra, is an active line of research.

The notes are organized as follows. Section 1 reviews the dynamical properties of gauge systems [22], Lie groups and algebras, and non-Abelian Yang–Mills theory, and ends with the meron black hole. Section 2 introduces manifolds, frames, the vielbein and the calculus of differential forms. Section 3 develops the geometry of general connections: torsion, curvature, Cartan's structure equations, the Bianchi identities and the contorsion. Section 4 studies the Palatini and Einstein–Cartan actions and shows how general relativity is recovered dynamically. Section 5 reformulates first-order gravity as a gauge theory of the Poincaré group and discusses the relation between local translations and

diffeomorphisms. Finally, Section 6 is devoted to gravity in $2+1$ dimensions and its formulation as a Chern–Simons theory. Background material on group theory [23] and on pullbacks and Stokes' theorem is collected in the appendices.

1 Gauge systems

1.1 Gauge theory for gravity

A gauge theory is a theory wich have a (local) redundancy in the description of it dynamical variables. The fields are defined only up-to local transformations that doesn't change the physical content. For example, in electrodynamics

$$\vec{B} = \vec{\nabla} \times \vec{A}, \tag{1}$$

with the transformations

$$\vec{A}' = \vec{A} + \vec{\nabla}\theta. \tag{2}$$

Yang-Mills theory: Gauge invariance principles, in wich the interaction result from the existence of a locally realized symmetry, wich requires a correlation between particles in different localizations in the spacetime. The correlations appears as an interaction wich is interpreted as a force field.

A **local symmetry** transformation in wich the parameters depend on the points of the spacetime, i.e

$$\phi(x) \to U(x)\phi(x), \tag{3}$$

where $U(x)$ depends on what point of spacetime we are.
Example: Dirac Lagrangian

$$\mathcal{L}_D = \bar{\Psi}\left(i\gamma^\mu \partial_\mu - m\right)\Psi, \tag{4}$$
$$\bar{\Psi} = \Psi^+ \gamma^0, \tag{5}$$
$$\Psi' = e^{iq\theta}\Psi, \tag{6}$$
$$\bar{\Psi}' = e^{-iq\theta}\bar{\Psi}. \tag{7}$$

If θ is constant the we have a global symmetry. ${\mathcal{L}'}_D = \mathcal{L}_D$. But we have to consider the speed of light, there is no instantaneous actions of this things. We promove the *global symmetry* to a *local symmetry*.

$$\Psi' = e^{iq\theta(x)}\Psi, \tag{8}$$
$$\bar{\Psi}' = e^{-iq\theta(x)}\bar{\Psi}, \tag{9}$$

and then

$$\mathcal{L}'_D = \mathcal{L}_D - q\bar{\Psi}\gamma^\mu \partial_\mu \theta \Psi. \tag{10}$$

In order to solve the non-invariance of the lagrangian, we introduce a new derivative (minimal coupling of gauge conection to Dirac theory)

$$\partial_\mu \to \mathcal{D}_\mu + iqA_\mu(x), \tag{11}$$

we have introduced a new field: **Gauge conection of** $U(1)$. How we know that is the gauge conection of $U(1)$ group? Replacing the derivative

$$\mathcal{L}_D = \bar{\Psi}(i\gamma^\mu \partial_\mu - m)\bar{\Psi} - q\bar{\Psi}\gamma^\mu A_\mu \Psi, \tag{12}$$

where the second term became from the interaction. In order to be the invariant, A_μ must transform in a particular way. In the derivative $\mathcal{D}_\mu$, the coviariant character must fulfill the covariant transformation

Covariant derivative

$$\begin{aligned}(\mathcal{D}_\mu \Psi)' &= e^{iq\theta}\mathcal{D}_\mu \\ \mathcal{D}'_\mu \Psi' &= (\partial_\mu + iqA'_\mu)e^{iq\theta}\Psi = e^{iq\theta}(\partial_\mu + iq(A'_\mu + \partial_\mu \theta))\Psi \\ \text{Then}: \; A'_\mu &= A'_\mu + \partial_\mu \theta\end{aligned}$$

this is the Maxwell field transformation. Finally, the full dynamic of this gauge theory is

$$\mathcal{L}_D = \bar{\Psi}(i\gamma^\mu \mathcal{D}_\mu - m)\Psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \tag{13}$$

1.2 Evaluations

Evaluations of this course will be homework, always finished before the next one.

- First part: 08/05
- Second part: 05/06
- Third part: 10/07
- Fourth part: 03/08

1.3 Introduction

It is possible to conclude that a gauge theory can be thought as a one in which the dynamical variables are specified w.r.t a reference frame (internal). The election of it is arbitrary for each time and position in the space. The physically relevant variables son the ones that are independent of the reference frame.

Dynamical properties of a gauge theory:
The action principle is given by

$$I[A^\mu] = -\frac{1}{4}\int d^4x \, F_{\mu\nu}F^{\mu\nu}, \tag{14}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and the metric is $\eta_{\mu\nu} = (-,+,+,+)$. The symmetries are global Poincaré and, in four dimensions, the conformal group. Which symmetries are present in the dynamical equations?

$$\delta I \to \partial_\mu F^{\mu\nu} = 0. \tag{15}$$

This equations are invariant under duality transformations $\mathbf{B} \leftrightarrow \mathbf{E}$ which can be seen as

$$F_{\mu\nu} \to F'_{\mu\nu} = \cos\theta F_{\mu\nu} + \sin\theta * F_{\mu\nu}. \tag{16}$$

From the equation of motion we can propose a plane wave solution

$$\partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = \Box A^\nu - \partial^\nu(\partial_\mu A^\mu) = 0, \tag{17}$$

we propose

$$A^\mu(x) = \int d^4k\, \hat{A}^\mu(k) e^{-ik_\lambda x^\lambda}. \tag{18}$$

Replacing this into the dynamical equation

$$(k^2\delta^\nu_\mu - k^\nu k_\mu)A^\mu = 0 \to M^\nu_\mu A^\mu = 0, \tag{19}$$

we need to analyze this operator. We can see this is a degenerated operator because it have an null eigenvector (from a zero eigenvalue)

$$M^\nu_\mu k^\mu = k^2 k^\nu - k^2 k^\nu = 0. \tag{20}$$

Given that k^μ is a null eigenvalue ($\lambda = 0$) of the operator $\mathbf{M}$. This means that the equation cannot be fully determined of a unique way the paralell component of the dynamical field

$$\hat{A}^\mu_{//} = k^\mu c(k), \tag{21}$$

then we can see that

$$M^\mu_\nu k^\nu c(k) = 0, \tag{22}$$

for all $c(k)$. Here appears the first arbitrariness of gauge theories. Because $k^\mu \in \ker(M)$, si $\hat{A}^\mu(k)$ is a solution

$$\hat{A}^\mu(k) \to \hat{A}^\mu(k) - ik^\mu\hat{\Lambda}(k), \tag{23}$$

is also a solution. In the coordinate space

$$ik^\mu\hat{\Lambda}(k) \to \partial^\mu\Lambda(x). \tag{24}$$

We have that

$$\hat{A}^\mu(k) = \hat{A}_\perp(k) + \hat{A}_{//}(k), \tag{25}$$

if we compute this, the $F_{\mu\nu}$ of the parallel component will be zero. There is no dynamics associated and will have no propagating degrees of freedom, then is **pure gauge**. We need to eliminate them to have the physics of our theory, but in the full theory we will have less degrees of freedom (the theory start with much more d.o.f that we need to describe the system). Now we consider $k^2 = 0$, i.e, $k^0 = |\vec{k}|$, then

$$k^2 A^\mu_\perp - k^\nu k_\mu A^\mu_\perp = 0, \tag{26}$$

$$k_\mu A^\mu_\perp = 0. \tag{27}$$

This is a second restriction and we can eliminate two degrees of freedom in the Maxwell theory resulting in the two propagating d.o.f.
In resume we have $\hat{A}^\mu_\perp$ is transversal to k^μ. Exists three vectors l.i transverse to k^μ. They are

$$\begin{aligned} e^\alpha_0 &= 0 \\ e^\alpha_i k^i &= 0 \\ e^\alpha_i e^i_\beta &= \delta^\alpha_\beta \end{aligned}$$

then $\hat{A}^\mu_\perp(k) = e^\mu_\alpha b_\alpha(k)$. Finally the solution is given by

$$A^\mu(x) = A^\mu_\perp + A^\mu_{//}(x), \tag{28}$$

where

$$A^\mu_\perp(x) = \frac{1}{(2\pi)^4}\int d^4k\, e^\mu_\alpha(k) b_\alpha(k) e^{-ik_\lambda x^\lambda}\,, \tag{29}$$

$$A^\mu_{//}(x) = \frac{1}{(2\pi)^4}\int d^4k\, k^\mu c(k) e^{-ik_\lambda x^\lambda}, \tag{30}$$

with c an arbitrary functions. As a conclusion we can say that a gauge theory has as a dynamical properties

- Dynamical equations doesn't fix uniquely all the variables.
- The general solution contains arbitrary functions.
- The arbitrary functions reveals a gauge symmetry.
- The number of variables exceeds the number of physical degrees of freedom.

1.4 The Aharonov–Bohm effect

Although the pure gauge quantities doesn't generate physical degrees of freedom (locally), we need to have certain subtlety in this affirmation. This is true in trivial topologies. Globally pure gauge potentials can have non trivial global information.

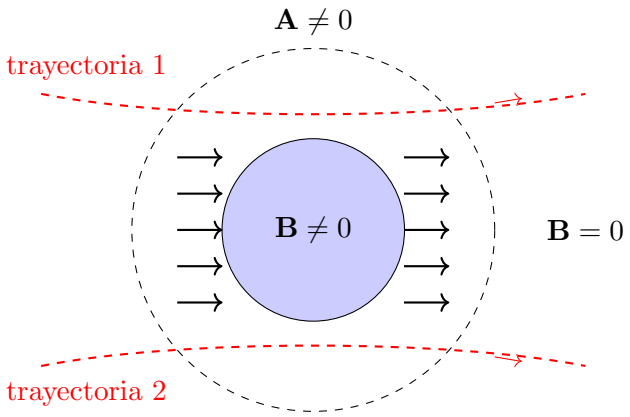
Example : Aharanov-Bohm effect: In this effect we consider a solenoid in which a electric current exist. We can compute the potential via magnetic field

$$\vec{\nabla} \times \vec{A} = \vec{B} \; / \int d\vec{S}, \tag{31}$$

$$\oint \vec{A} \cdot d\vec{l} = \int \vec{B} \cdot d\vec{S} = \Phi, \tag{32}$$

$$\implies \vec{A} = \frac{\Phi}{2\pi r_0}\hat{\phi} = \vec{\nabla}(\frac{\Phi\phi}{2\pi})|_r = r_0. \tag{33}$$

$\vec{A}$ is non null and a pure gauge contribution .What happens if a quantum particle interact in this region?



Solenoide (Efecto Aharonov-Bohm)

With a non minimal coupling as we used in Dirac, we have

$$H = \frac{1}{2m}\left(P_\phi - qA_q\right)^2 = \frac{1}{2mr_0^2}\left(-i\hbar\partial_\phi - \frac{q\Phi}{2\pi}\right)^2, \tag{34}$$

$$\implies H\psi = E_n\psi \;,\; \psi(\phi) = \frac{1}{\sqrt{2\pi r_0}}e^{in\phi}, n \in \mathbb{Z}. \tag{35}$$

This n comes from $\psi(\phi + 2\pi) = \psi(\phi)$. We consider

$$E_n = \frac{1}{2mr_0}\left(\hbar n - \frac{q\Phi}{2\pi}\right)^2 = \frac{\hbar^2}{2mr_0}\left(n - \frac{\Phi}{\Phi_0}\right) \;, \text{with } \Phi_0 = \frac{2\pi\hbar}{q}. \tag{36}$$

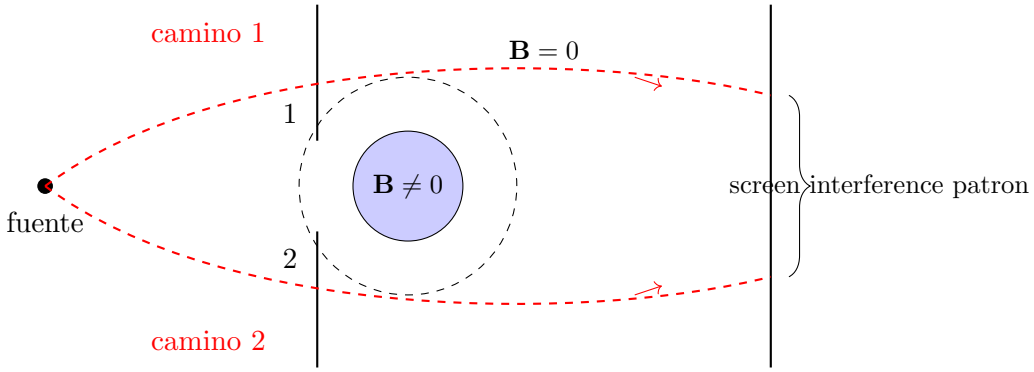
If $\Phi = m\Phi_0, m \in Z$ then the energy spectrum doesn't change. In the case in which Φ_0 is not integer multiple the energy spectrum is modified. We reinterpret this result noticing that outside the solenoid $\vec{A} = \vec{\nabla}\alpha, \alpha = \frac{\Phi\phi}{2\phi}$. We can remove the gauge by a gauge transformation. So we introduce a gauge transformation

$$\psi \to \bar{\psi} = e^{-iq\alpha/\hbar}\psi, \tag{37}$$

$$= e^{-iq\Phi\phi/\hbar}\psi. \tag{38}$$

Check Tong notes [2]. The proper gauge transformation allowed is the one in which Φ is integer multiplier of Φ_0. In the experiment of Aharanov-Bohm, the double slit is seen to be modified

Doble rendija con solenoide (Efecto Aharonov–Bohm)



$$\Delta\phi = \frac{q}{\hbar}\oint \mathbf{A}\cdot d\mathbf{l} = \frac{q}{\hbar}\Phi_B$$

Considering the Schrodinger equation

$$\frac{1}{2m}\left(-i\hbar\vec{\nabla} - q\vec{A}\right)^2\psi = E\psi, \tag{39}$$

we can remove the gauge field

$$\psi(x) = \exp\left(\frac{-iq}{\hbar}\int^x \vec{A}(x')d\vec{x}'\right)\phi(x), \tag{40}$$

this exponential argument is the **holonomy** of $U(1)$ gauge group. The value of the holonomy depends on the trajectory (global structure). The phase difference is given by

$$\begin{aligned}\Delta\theta &= \frac{q}{\hbar}\int_{P_1}\vec{A}\cdot d\vec{x} - \frac{q}{\hbar}\int_{p_2}\vec{A}\cdot d\vec{x}\,,\\ &= \frac{q}{\hbar}\oint\vec{A}\cdot d\vec{x} = \frac{q}{\hbar}\oint\vec{B}d\cdot\vec{S} = \frac{q}{\hbar}\Phi.\end{aligned} \tag{41}$$

1.5 Closed and exact forms: de Rham cohomology

Check de Rahm cohomology. Where is the subtlety of this? We can define the gauge field A as a 1-form.

$$A = A_\mu dx^\mu. \tag{42}$$

This is a base in the cotangent space. The curvature associated is a 2-form $F = dA$, where $d = dx^\mu\partial_\mu$.

$$F = \partial_\mu A_\nu dx^\mu\wedge dx^\nu = \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu)dx^\mu\wedge dx^\nu = \frac{1}{2}F_{\mu\nu}dx^\mu\wedge dx^\nu. \tag{43}$$

In a exterior region to the solenoid, $F = 0$ and locally $A = d\alpha$. However, the exterior is nos simply connected. A can be locally written as a gradient, but not necessarily it can be truth globally. Formally, the expression "*non trivial topology*" designs the non triviality of some topological invariant of the space. In the case of gauge theories, non trivial topology implies

$$\pi_1(X)\neq 0\,,\ \text{Fundamental group (first homotopy group)} \implies \text{non contractible closed curves}, \tag{44}$$

or equivalent

$$H^1(X) \neq 0\,,\ \text{De Rahm cohomology of first class} \implies \text{flat connections are locally exact but non globally trivial.} \tag{45}$$

In resume, if we can write $\vec{\nabla} \times \vec{A} = \vec{B}$, then we can write $F = dA = 0$. If $\vec{B} = 0$ locally $\vec{A} = \vec{\nabla}\alpha$. This affirmation is globally true if the topology is non trivial because

$$\oint d\vec{A} \cdot d\vec{l} = \Phi. \tag{46}$$

The subtlety is that not all the closed forms are exact. We can define a $1 - form$ A in a manifold M. A is exact exists a global function α such that

$$A = d\alpha, \tag{47}$$

and is closed if

$$dA = 0. \tag{48}$$

Given that for all differential forms $d^2A = 0$, then all exact form is closed. The reverse implicate is not true in general.

Poincaré lemma In a contractible region (the holonomy group is trivial) all closed form is exact

$$dA = 0 \to A = d\alpha. \tag{49}$$

In order to analyze when this fails, we introduce the de Rahm cohomology. The cohomology group is defined as

$$\mathcal{H}^1(M) = \frac{\{\text{closed}\, 1 - forms\, \text{in M}\}}{\{\text{exact}\, 1 - forms\, \text{in M}\}} \equiv \frac{V}{W}. \tag{50}$$

The quotient space is a new vector space if V is a vector space and $W \subseteq V$ is a subspace. V/W is the set of equivalence classes in V given $v.v' \in V$ they are considered equivalences if differs in a element of W

$$v \leftrightarrow v' \Leftrightarrow v - v' \in W, \tag{51}$$

$$[v] = \{u \in V; u - v\} = \{v + w : w \in W\}. \tag{52}$$

An example of equivalence classes. Consider the set $\mathbb{Z} \mod (3)$, i.e, all the integers whose difference is a multiple of 3.

$$a - b = 3k, \tag{53}$$

we can classify by the rest all the options

$$[0] = \{3, 6, 9, ...\},$$
$$[1] = \{1, 4, 7, 10, ...\}$$
$$[2] = \{2, 5, 8, 11...\}.$$

We can say that $\mathbb{Z} \mod (3)$ has three equivalence classes. If $\mathcal{H}^1(M) = 0$ and we denote $V = Z^1(M)$ (set of closed 1-forms) and $W = B^1(M)$ (set of the exact 1-forms). Then if the equivalence class is the identity, this means

$$[v] = [0] \Leftrightarrow v - 0 \in W \Leftrightarrow v \in W, \tag{54}$$

then $V \subseteq W$ and $W \subseteq V \Rightarrow Z^1 = B^1$,i.e, all the closed form are exact.

Example 1: If $M = \mathbb{R}$. In $\mathbb{R}$ all 1-form is

$$\omega = a(x)dx, \tag{55}$$

so we can notice $d\omega = a'(x)dx \wedge dx = 0 \Rightarrow \omega$ is closed. Given that $\omega = a(x)dx$ always we can define

$$f(x) = \int_{x_0}^{x} a(t)dt \rightarrow df = f'(x)dx = a(x)dx = \omega, \tag{56}$$

then there is not exists an topological obstruction in $\mathbb{R}$.

Example 2: $M = \mathcal{S}^1$. We can parametrize the elements by $\theta \in [0, 2\pi]$. We consider 1-form $d\theta$ and we notice that $d\, d\theta = 0$ so is a closed form. We can try to define globally a smooth function $f : \mathcal{S}^1 \rightarrow \mathbb{R}$, so that $df = d\theta$

$$\int_{\mathcal{S}^1} df = \int_{\mathcal{S}^1} d\theta, \tag{57}$$

we see that for any global function in a closed loop $\oint f = 0$, but $\int d\theta = 2\pi$, so it's not an exact form.

If we want to see how is this related to the Aharanov-Bohm effect we need to consider $M = \mathbb{R}^2 - \{0\}$. We define

$$\omega = \frac{-ydx + xdy}{(x^2 + y^2)}. \tag{58}$$

If we reparametrize as $x = \cos t$, $y = \sin t$, $t \in [0, 2\pi]$, then $\omega = dt$. This is closed but not exact $\mathcal{H}^1(M) \neq 0$. **Homework:** Proof that $\mathbb{R} - \{0\}$ is non trivial using the first group of homotopy,i.e, exists a closed curve that is not contractible to a point. (From the definition of homotopy).

1.6 Groups and Lie algebras

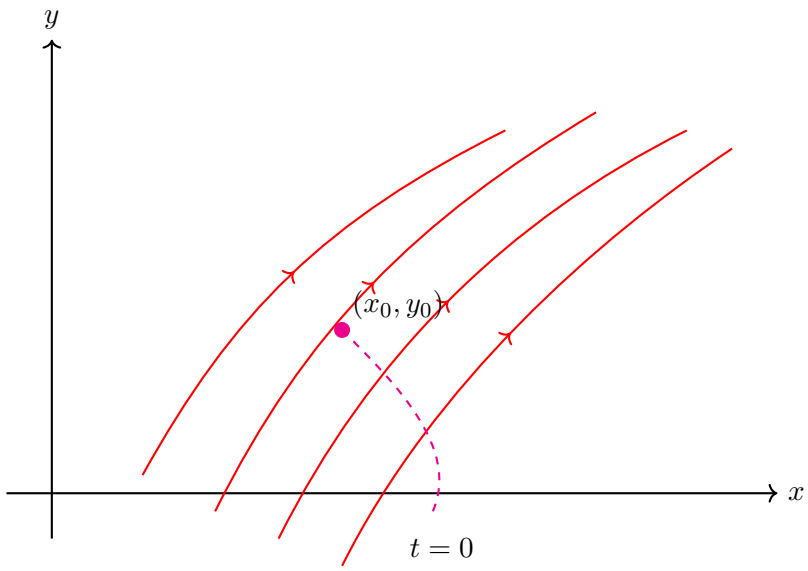
Definition: uniparametric groups of points transformations and their infinitesimal generators. Often we try to simplify a differential equation realizing an appropriated change of variables (A transformation of the independent variable x and dependent variable y)

$$x = \bar{x}(x, y)\ ;\ y = \bar{y}(x, y). \tag{59}$$

There are point transformations $(x, y) \rightarrow (\bar{x}, \bar{y})$. In the context of symmetries, consider they depend at least on one parameter, i.e, $\bar{x} = \bar{x}(x, y; t)$, $\bar{y} = \bar{y}(x, y; t)$. If this transformation satisfy

- Invertible.
- Closed.
- Exist an identity element.

Then the transformation have a group structure.



Consider uniparametric groups.

$$\bar{x} = \bar{x}(x, y; t), \tag{60}$$
$$\bar{y} = \bar{y}(x, y; t). \tag{61}$$

For example the flat rotations

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \tag{62}$$

where $R(t)$ is the rotation matrix. The satisfy

$$R(t + t') = R(t)R(t'). \tag{63}$$

We can try to understand this transformations as integral curves (see previous image) where each curve are related by this type of group transformations each other. The set of curves (orbits) is completely characterized by the vector field of the tangent vectors of each curve $\mathbb{X}$ and vice versa.

How to compute $\mathbb{X}$? We Taylor expand in the parameter. For example

$$\bar{x}(x, y : t) = x + t\xi(x, t) + \cdots = x + t\mathbb{X}x + \cdots, \tag{64}$$
$$\bar{y}(x, y : t) = y + t\eta(y, t) + \cdots = y + t\mathbb{X}y + \cdots, \tag{65}$$

where we have defined

$$\xi(x, y) = \partial_t \bar{x}|_{t=0} \ ; \ \eta(x, y) = \partial_t \bar{y}|_{t=0}. \tag{66}$$

Then the operator is simply

$$\mathbb{X} = \xi(x, y)\partial_x + \eta(x, y)\partial_y \ \rightarrow \text{ Transformation infinitesimal generator.} \tag{67}$$

An uniparametric group is a family $\{g(t)\}; t \in \mathbb{R}$ in some group G such that $g(0) = e\,;\, g(t,s) = g(t)g(s)\,\forall\, t,s \in \mathbb{R}$, i.e, $g : \mathbb{R} \to G$.v
Let be M a differential manifold and $\mathbb{X}$ a vector space in M. $\mathbb{X}$ defines the orbit equation

$$\frac{d}{dt}\gamma(t) = \mathbb{X}_{\gamma(t)}\,;\, \gamma(0) = p. \tag{68}$$

For each point $p \in M$, the solution $\gamma_p(t)$ is an integral curve of $\mathbb{X}$ that cross in p. If the solution exists, we define

$$\phi_t(p) = \gamma_p(t)\,;\, \phi_t : M \to M. \tag{69}$$

The family $\{\phi_t\}$ is the flux $\mathbb{X}$. The flux is the family of applications for each t, which tell where arrives each point that follows the integral curve.
Show that: The flux corresponds to an uniparametric group, i.e,

$$\phi_0 = e\,,\, \phi_{t+s} = \phi_t \odot \phi_s, \tag{70}$$

with the initial conditions

$$\phi(x.,0) = x\,,\, \phi(\phi(x,t),s) = \phi(x,s+t). \tag{71}$$

An example with the rotations in $\mathbb{R}^2$. We want to compute $\mathbb{X}$, so we use

$$\bar{x} = x\cos t - y\sin t, \tag{72}$$

$$\bar{y} = x\sin t + t\sin t, \tag{73}$$

then the vector field is

$$\mathbb{X} = -y\partial_x + x\partial_y, \tag{74}$$

at $t = 0$. It also can reverse the process and compute from

$$\frac{d}{dt}\gamma(t) = \mathbb{X}_{\gamma(t)}, \tag{75}$$

from this we need to see that

$$\dot{x}(t) = -y(y)\ ,\ \dot{y}(t) = x(t), \tag{76}$$

and consider the initial conditions

$$\gamma(0) = p \Rightarrow \phi(x_0,y_0) = (x,y). \tag{77}$$

This allows us to obtain

$$\phi_t(x,y) = (x\cos t - y\sin t, x\sin t + y\cos t)\,. \tag{78}$$

Lie groups are sets G which possess the properties

- Group structure.
- Are differential manifolds.
- Compatibility between both structures.

If we had a set and an internal operation $(G, \cdot)$ they need to be smooth and differentiable

$$m = G \times G \to G\,,\; m(g,h) = gh, \tag{79}$$

$$1 = G \to C\,,\; i(g) = g^{-1}. \tag{80}$$

Lie algebras are vector spaces g equipped with a bilinear operation $[\cdot,\cdot] : g \times g \to g$ called the *Lie bracket*. It is antisymmetric and is closed under Jacobi identity

$$[x,y] = -[y,x], \tag{81}$$

$$[x,[y,z]] + [y,[z,x]] + [z,[x,y]] = 0. \tag{82}$$

The second comes from the associativity of the Lie group. Given a Lie group G, the Lie algebra$g$ is defined as the tangent space on the identity element $g := T_eG$. The group describes the global structure while the tangent space close to e describes the infinitesimal structure. Why close to e? Because $L_g(h) = gh$ and $L_g(e) = g$. Any point $g \in G$ is obtained from e by the action of G. Each application $L_g : G \to G\; :\; h \to gh$ is a diffeomorphism. This means that not only point moves, also tangent vectors are transported.

$$(L_g)_{*e} : T_eG \to T_gG. \tag{83}$$

It is necessary to introduce a vector field that is defined globally. What we must do is to transport the vector $\curvearrowleft \in T_eG$ in each point $g \in G$ using the group structure. A possibility is to introduce left invariant vector fields

$$\mathbb{X}^L_g = (L_g)_{*e}\mathbb{X}. \tag{84}$$

It copies the same infinitesimal direction throughout the manifold. This field must satisfy the homogeneous condition

$$(L_g)_{*h}(\mathbb{X}^{\mathbb{L}}_{\widetilde{\ }}) = \mathbb{X}_{gh}. \tag{85}$$

1.7 Representations of Lie groups

The transformation law of the fields is determined by the representation of a group G in a vector space V. A homomorphism

$$T : G \longrightarrow GL(V),$$

where $GL(V)$ is the group of linear transformations in V. We see that $\forall\, g \in G$ is assigned a matrix as a linear transformation

$$T(g_1g_2) = T(g_1)\cdot T(g_2)$$
$$T(g^{-1}) = [T(g)]^{-1}.$$

Fix a basis e_a in V. If $T(g)$ is the representation of $g \in G$ then the action takes e_a for some vector of V such that

$$T(g)e_a = T_{ba}(g)e_b.$$

Here T_{ab} has dimension $n \times n$ and $\dim V = n$. Any vector $\psi \in V$ can be represented in the form $\psi = \psi_a e_a$. Then $T(g) = \psi_a(T(g)e_a) = \psi_a T_{ba}e_b$, so we can see that

$$(T(g)\psi)_a = T_{ab}(g)\psi_b.$$

Trivial representation: Is the rep. in which

$$T(g) = \mathbb{I},$$

with $\mathbb{I}$ the identity. Then $T(g)\psi_a = \delta_{ba}\psi_b = \psi_a$ (Singlets).
Fundamental representation: For a Lie group of $n \times n$ matrices (for example $SU(n)$,$SO(n)$) and V a vector space of columns ($\dim V = n$)

$$\psi = \begin{pmatrix} \psi_1 \\ . \\ . \\ . \\ \psi_n \end{pmatrix},$$

the fundamental representation acts on this space as

$$(T(g)\psi)_a = g_{ab}\psi_b.$$

Adjoint representation: Be $Ad(g)$ a linear operator corresponding to a element $g \in G$ over a matrix $\mathbb{A} \in \mathfrak{g}$, then

$$Ad(g)\mathbb{A} = g\mathbb{A}g^{-1}\,;\; Ad(g) : \mathfrak{g} \to \mathfrak{g}.$$

For example, consider the gauge field A_μ, then

$$A_\mu \to A_\mu = gA_\mu g^{-1} + gdg^{-1}, \tag{86}$$

$$F_{\mu\nu} \to gF_{\mu\nu}g^{-1}. \tag{87}$$

1.8 Non-Abelian gauge theories

Non Abelian gauge theories: SU(2)
Consider a theory for a complex scalar field

$$\mathcal{L} = \partial_\mu\varphi^\dagger\partial_\mu\varphi - m^2\varphi^\dagger\varphi - \lambda(\varphi^\dagger\varphi)^2,\ \varphi = \begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix}. \tag{88}$$

This lagrangian is invariant under $\varphi' \to g\varphi$ (fundamental rep. transformation) and $g \in SU(2)$. Consider now the transformation $\varphi \to \varphi' = g(x)\varphi(x)$. We see that

$$\partial_\mu\varphi' = \partial_\mu g \cdot \varphi + g \cdot \partial_\mu\varphi. \tag{89}$$

We would like that $\mathcal{L}' = \mathcal{L}$, so the derivative must transform under the fundamental rep.

$$(D_\mu\varphi)' = g(x)D_\mu\varphi, \tag{90}$$

now we introduce a local gauge field $A_\mu(x)$ and write

$$D_\mu\varphi = \partial_\mu\varphi + A_\mu\varphi, \tag{91}$$

and we see that the transformation for this $A_\mu(x)$

$$(D_\mu\varphi)' = D'_\mu\varphi' = (\partial_\mu + A_\mu)\varphi' = \partial_\mu g\varphi + g\partial_\mu\varphi + A'_\mu g\varphi, \tag{92}$$

and now we consider (91) equal to (92)

$$g\partial_\mu\varphi + gA_\mu\varphi = \partial_\mu g\varphi + g\partial_\mu\varphi + A'g\varphi, \tag{93}$$

so

$$A'_\mu g = gA_\mu - \partial_\mu g. \tag{94}$$

From this relations we see that the conclusion is

$$A'_\mu = gA_\mu g^{-1} - \partial_\mu g g^{-1}, \tag{95}$$

but

$$\partial_\mu(gg^{-1}) = \partial_\mu g \cdot g^{-1} + g\partial_\mu g^{-1} = 0, \tag{96}$$

so the transformation for the gauge field is

$$A'_\mu = gA_\mu g^{-1} + g\partial_\mu g^{-1}, \tag{97}$$

where we see that the first term transform in the adjoint and the second term makes inhomogeneous the transformation. Now we determine the shape of the field A_μ. For this we consider an infinitesimal gauge transformation that allow us to uste the exponential map

$$g = \mathbb{I} + \epsilon(x)\,;\ (g^{-1} = \mathbb{I} - \epsilon(x)), \tag{98}$$

where $\epsilon \in \mathfrak{su}(2)$. From the pure gauge term we can see that

$$g\partial_\mu g^{-1} = -\partial_\mu\epsilon(x) + O(\epsilon^2). \tag{99}$$

We can notice that A_μ take values in the Lie algebra

$$A_\mu = A^a_\mu(x)\sigma_a, \tag{100}$$

where σ_a are the Pauli matrices. also $gA_\mu g^{-1} \in \mathfrak{su}(2)$. In resume the transformation rules for the scalar and gauge fields are given by

$$A_\mu \to A'_\mu = gA_\mu g^{-1} + g\partial_\mu g^{-1}\,,\ \text{(Adjoint rep.)} \tag{101}$$

$$\varphi \to \varphi' = g\varphi\,,\ \text{(Fundamental rep.)}, \tag{102}$$

and the lagrangian

$$\mathcal{L} = (D_\mu\varphi)^\dagger(D_\mu\varphi) - m^2\varphi^\dagger\varphi - \lambda(\varphi^\dagger\varphi)^2, \tag{103}$$

with $D_\mu = \partial_\mu + A_\mu$. $D_\mu : \Omega^p(U,V) \to \Omega^{p+1}(U,V)$, with V the vector space in which G acts via some representation. If g is global, exists a difference between the Abelian and non-Abelian cases.

$$A_\mu \to A'_\mu = gA_\mu g^{-1}. \tag{104}$$

We know from the Abelian case that $F_{\mu\nu} = F'_{\mu\nu} = gF_{\mu\nu}g^{-1}$. We would like to extend this transformation to the local non-Abelian case. This implies that

$$F_{\mu\nu} = \partial_{[\mu}A_{\nu]} + [A_\mu, A_\nu]. \tag{105}$$

Explicitly for $\mathfrak{su}(2)$ algebra, we have that

$$A_\mu = ig\frac{1}{2}\sigma_a A^a_\mu(x), \tag{106}$$

where the generators are the Pauli matrices σ_a. The are given by

$$\sigma_1 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad , \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad , \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{107}$$

They satisfy the relations

$$[\sigma_a, \sigma_b] = 2i\varepsilon_{abc}\sigma_c. \tag{108}$$

We can notice that

$$F_{\mu\nu} = -ig\frac{\sigma_a}{2}F^a_{\mu\nu}, \tag{109}$$

where

$$\begin{aligned} F_{\mu\nu} &= \partial_{[\mu}A_{\nu]} + [A_\mu, A_\nu] \\ &= -ig\frac{\sigma_a}{2}\left(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu\right) + \frac{(ig)^2}{4}\left[\sigma_a, \sigma_b\right]A^a_\mu A^b_\nu \\ &= -ig\frac{\sigma_a}{2}\left[\partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\varepsilon^a_{bc}A^b_\mu A^c_\nu\right] \quad , \quad \varepsilon^a_{bc} = \delta^{ad}\varepsilon_{dbc}, \end{aligned}$$

with $Tr(\sigma_a\sigma_b) = 2\delta_{ab}$. This is the bilinear form of $\mathfrak{su}(2)$ (Killing-Cartán form). **Homework:** be $g = 1 + i\frac{\sigma_a}{2}\epsilon^a(x)$. Find A'_μ and $F'_{\mu\nu}$.

1.9 The Yang–Mills action and field equations

From the Abelian gauge theory we know that the action

$$S[A_\mu] = -\frac{1}{4}\int d^4x\, F_{\mu\nu}F^{\mu\nu}, \tag{110}$$

is Lorentz invariant and gauge invariant. However, for the non-Abelian case F transforms in the adjoint representation

$$F \longrightarrow F' = gFg^{-1}. \tag{111}$$

How to construct an invariant action? we notice that

$$F'_{\mu\nu}F'^{\mu\nu} = gF_{\mu\nu}g^{-1}gF^{\mu\nu}g^{-1} = gF^2g^{-1}, \tag{112}$$

also transforms in the adjoint representation. **Homework :** Do this with internal indices $A_\mu \sim ig\sigma_a A^a_\mu$. But for the trace of this term

$$\mathrm{Tr}(F'^2) = \mathrm{Tr}(F^2). \tag{113}$$

Explicitly for $\mathbf{SU}(2)$

$$\mathrm{Tr}\left(F_{\mu\nu}F^{\mu\nu}\right) = \left(\frac{ig}{2}\right)^2 F^a_{\mu\nu}F^{b\mu\nu}\mathrm{Tr}(\sigma_a\sigma_b) = \frac{-g^2}{2}F^a_{\mu\nu}F^{a\mu\nu}, \tag{114}$$

where $\mathrm{Tr}(\sigma_a\sigma_b) = 2\delta_{ab}$. Now we can define the Yang-Mills action

$$S_{ym}[A_\mu] = \frac{1}{2g^2}\int d^4x\,\mathrm{Tr}\left(F_{\mu\nu}F^{\mu\nu}\right) = \frac{-1}{4}\int d^4x\, F^a_{\mu\nu}F^{a\mu\nu}. \tag{115}$$

Homework:try perturbations $A \to A + \delta A$ in this theory. The Yang-Mills theory os non-linear because in this action exists auto interaction terms (Tr(∂A^3),Tr(A^4)). The generalization of Yang-Mills theory to other groups is direct. Consider a field A which is evaluated in some Lie algebra $\mathfrak{g}$

$$A_\mu(x) = \alpha T_a A^a_\mu \quad , [T_a, T_b] = C^c_{ab} T_c. \tag{116}$$

The field strength take values on the algebra also

$$F_{\mu\nu} = \beta F^a_{\mu\nu} T_a. \tag{117}$$

The gauge transformation have the following form. For finite transformations

$$\begin{aligned} A_\mu \to A'_\mu &= g A_\mu g^{-1} + g\partial_\mu g^{-1}, \\ F'_{\mu\nu} \to F'_{\mu\nu} &= g F_{\mu\nu} g^{-1}, \end{aligned} \tag{118}$$

and for infinitesimal transformations ($g = 1 + \epsilon ...$)

$$\begin{aligned} A'_\mu - A_\mu = \delta A_\mu &= -\left(\partial_\mu \epsilon + [A_\mu, \epsilon]\right) = -D_\mu \epsilon, \\ F'_{\mu\nu} - F_{\mu\nu} = \delta F_{\mu\nu} &= [\epsilon, F_{\mu\nu}]. \end{aligned} \tag{119}$$

Observation : the definition of covariant derivative depends on the representation of the Lie group. If a field A is in the fundamental representation

$$D_\mu \varphi = \partial_\mu \varphi + A_\mu \varphi \quad , D_\mu \to D'_\mu = g D_\mu. \tag{120}$$

For A in the adjoint representation

$$D_\mu \epsilon = \partial_\mu \epsilon + [A_\mu, \epsilon] \quad , D_\mu \to D'_\mu = g D_\mu g^{-1}, \tag{121}$$

with $\epsilon = \epsilon^a T_a$. Finally the Yang-Mills action is given by

$$S = \frac{1}{2g^2} \int d^4x \mathrm{Tr}\left(F_{\mu\nu} F^{\mu\nu}\right), \tag{122}$$

where

$$F_{\mu\nu} = g F^a_{\mu\nu} T_a = g\left(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g C^{abc} A^b_\mu A^c_\nu\right) T_a, \tag{123}$$

and also we have that $Tr(T_a T_b) \sim \delta_{ab}$, if the Lie group is compact. **Homework:** Show that the EOM's for Yang-Mills action are

$$D_\mu F^{a\mu\nu} = 0 \quad , \quad D_\mu = \partial_\mu + g[A_\mu, A_\nu].$$

To find EOM's of Yang-Mills action, we vary w.r.t the fields

$$\delta S_{YM} = -\frac{1}{4} \int d^4x \left(\delta F^a_{\mu\nu} F^{\mu\nu}_a + F^a_{\mu\nu} \delta F^{\mu\nu}_a\right) = -\frac{1}{2} \int d^4x F^{a\mu\nu} \delta F_{a\mu\nu}, \tag{124}$$

with the consideration that

$$F_{\mu\nu} = g F^a_{\mu\nu} T_a = g\left(\partial_{[\mu} A^a_{\nu]} - g C^a_{bc} A^b_\mu A^c_\nu\right) T_a, \tag{125}$$

and if we have a compact group $\mathrm{Tr}(T_a T_b) = -\delta_{ab}$, $[T_a, T_b] = C^c_{ab} T_c$[1]. Explicitly the variation gives us

$$\delta F^a_{\mu\nu} = \partial_\mu(\delta A^a_\nu) - \partial_\nu(\delta A^a_\mu) + g\, C^a{}_{bc}\Big(\delta A^b_\mu\, A^c_\nu + A^b_\mu\, \delta A^c_\nu\Big), \tag{126}$$

$$= D_\mu(\delta A_\nu)^a - D_\nu(\delta A_\mu)^a, \tag{127}$$

with the covariant derivative defined as

$$(D_\mu\phi)^a \;=\; \partial_\mu\phi^a + g\, C^a{}_{bc}\, A^b_\mu\, \phi^c. \tag{128}$$

Without this covariant derivative the variation of the action loos like

$$\delta S_{YM} = -\int d^4x \left(\partial_\mu \delta A_{a\nu} + g C_{abc} A^b_\mu \delta A^c_\nu\right) = \int d^4x \left[\partial_\mu(F^{\mu\nu}_c) - g C_{abc} F^{a\mu\nu} A^b_\mu\right] \delta A^c_\nu = 0. \tag{129}$$

This, using the antisymmetry of the structure constant gives us

$$\delta S_{YM} = \int d^4x \left(\partial_\mu F^{\mu\nu}_c + g C_{cab} A^a_\mu F^{b\mu\nu}\right) \delta^c_\nu = 0, \tag{130}$$

then

$$\partial_\mu F^{\mu\nu}_c + g C_{cab} A^a_\mu F^{b\mu\nu} = 0. \tag{131}$$

With the covariant derivative

$$\delta S_{YM} = -\frac{1}{g^2}\int \mathrm{Tr}(F^{\mu\nu}\, \delta F_{\mu\nu})\, d^4x \tag{132}$$

$$= -\frac{2}{g^2}\int \mathrm{Tr}(F^{\mu\nu}\, D_\mu \delta A_\nu)\, d^4x \qquad \text{(Antisymmetry of } F^{\mu\nu}\text{)} \tag{133}$$

$$= \frac{2}{g^2}\int \mathrm{Tr}((D_\mu F^{\mu\nu})\, \delta A_\nu)\, d^4x \qquad \text{(Integration by parts)} \tag{134}$$

$$= -\frac{1}{g^2}\int (D_\mu F^{\mu\nu})^a\, \delta A^a_\nu\, d^4x \qquad (\mathrm{Tr}(T_a T_b) = -\tfrac{1}{2}\delta_{ab}) \tag{135}$$

Finally, EOM's of Yang-Mills action are given by

$$(D_\mu F^{\mu\nu})^a = 0. \tag{136}$$

1.10 Invariant bilinear forms and structure constants

We can tell that the structure constants are fully antisymmetric if exists an invariant bilinear form in the algebra. For this case we take as a invariant bilinear form δ_{ab}. Be $g_{ab} \equiv < T_a, T_b >$ a bilinear form $< \cdot, \cdot >: \mathfrak{g} \times \mathfrak{g} \to \mathbb{R}$. This is an invariant form if it doesn't change under the infinitesimal action (in the adjoint rep.) of $\mathfrak{g}$. How to show this? If $x, y, g \in \mathfrak{g} \to < x, y > = < Ad_g x, Ad_g y >$, with $Ad_x y = [x, y]$. Then we can compute the following

$$\begin{aligned}\langle x, y\rangle &= \left\langle gxg^{-1}, gyg^{-1}\right\rangle = \langle (1+\epsilon)x(1-\epsilon), (1+\epsilon)y(1-\epsilon)\rangle \\ &= \langle x + [\epsilon, x], y + [\epsilon, y]\rangle \\ &= \langle x, y\rangle + \langle x, [\epsilon, y]\rangle + \langle [\epsilon, x], y\rangle + o(\epsilon^2),\end{aligned} \tag{137}$$

[1] If it is not a bilinear form of $\mathfrak{g}$ the structure constant are not necessarily antisymmetric.

then the condition to be invariant is

$$\langle x, [\epsilon, y]\rangle + \langle [\epsilon, x], y\rangle = 0. \tag{138}$$

If $x = T_a, y = T_b, \epsilon = T_c$, then the condition is

$$\begin{aligned} \langle T_a, [T_c, T_b]\rangle + \langle [T_c, T_a], T_b\rangle = 0 \\ = \left\langle T_a, C^d_{bc} T_d \right\rangle + \left\langle C^d_{ca} T_d, T_b \right\rangle = 0 \\ g_{ad} C^d_{cb} + g_{db} C^d_{ca} = 0 \Rightarrow C_{acb} + C_{bca} = 0, \end{aligned} \tag{139}$$

permuting indices we have

$$\begin{aligned} C_{acb} - C_{bac} = 0 \\ \rightarrow C_{acb} - C_{bac} = 0 \\ \rightarrow -C_{abc} - C_{bac} = 0 \\ C_{abc} = -C_{bac}\,, \end{aligned} \tag{140}$$

and for this reason the structure constants are fully antisymmetric. **Homework:** take $\mathbf{SU}(2)$ and show that δ_{ab} is invariant.

1.11 Wolfram Mathematica coding class :

In this class we compute some codes to derive the meron solution of Yang-Mills equations for some Lie algebra. I will add here some mathematical preliminaries of the paper in which we have based our computation [21].

An exact non-Abelian solution: merons and the hedgehog black hole

Having obtained the Yang–Mills equation $(D_\mu F^{\mu\nu})^a = 0$ by varying the action, we now put it to work: we exhibit a closed-form, genuinely non-Abelian configuration — the *meron* — and show that, coupled to gravity, it sources an exact black hole. Throughout we keep the same conventions as above: a compact gauge group with anti-Hermitian generators $[T_a, T_b] = C^c{}_{ab} T_c$, fully antisymmetric structure constants C_{abc}, and $\mathrm{Tr}(T_a T_b) = -\frac{1}{2}\delta_{ab}$. For $SU(2)$ (the case of interest) a convenient realisation is

$$T_a = -\frac{i}{2}\,\sigma_a \quad \Longrightarrow \quad [T_a, T_b] = \varepsilon_{abc} T_c, \qquad C^c{}_{ab} = \varepsilon_{abc}, \qquad \mathrm{Tr}(T_a T_b) = -\tfrac{1}{2}\delta_{ab}, \tag{141}$$

with σ_a the Pauli matrices. The structure constants $C_{abc} = \varepsilon_{abc}$ are manifestly totally antisymmetric, in agreement with the invariant–bilinear–form argument of the previous section.[2]

[2] Indeed the Killing form of $\mathfrak{su}(2)$ is proportional to δ_{ab}, which is exactly the statement asked for in the homework there; the total antisymmetry of ε_{abc} is its immediate corollary.

The boundary term in the variation

The variation of $S_{\rm YM} = -\frac{1}{4}\int \mathrm{d}^4x\,\sqrt{-g}\,F^a_{\mu\nu}F^{a\,\mu\nu}$ carried out earlier discarded a total derivative. It is worth keeping, both because a well-posed variational principle requires it to vanish and because with boundaries (as in the black hole / holographic setting below) it is physical. Recall $\delta F^a_{\mu\nu} = D_\mu(\delta A_\nu)^a - D_\nu(\delta A_\mu)^a$, so that, using the antisymmetry of $F^{a\mu\nu}$,

$$\delta S_{\rm YM} = -\int \mathrm{d}^4x\,\sqrt{-g}\;F^{a\mu\nu}\,D_\mu(\delta A_\nu)^a. \tag{142}$$

Now integrate by parts *covariantly*. The key observation is that the singlet current $V^\mu \equiv F^{a\mu\nu}\delta A^a_\nu$ carries no free colour index, so its gauge-covariant divergence is an ordinary one, and

$$\frac{1}{\sqrt{-g}}\partial_\mu\big(\sqrt{-g}\,F^{a\mu\nu}\delta A^a_\nu\big) = (D_\mu F^{\mu\nu})^a\,\delta A^a_\nu + F^{a\mu\nu}\,D_\mu(\delta A_\nu)^a. \tag{143}$$

The cross terms in C cancel precisely because C_{abc} is totally antisymmetric:

$$gC^a{}_{bc}A^b_\mu F^{c\mu\nu}\delta A^a_\nu + gC^a{}_{bc}A^b_\mu F^{a\mu\nu}\delta A^c_\nu = g\,A^b_\mu F^{c\mu\nu}\delta A^a_\nu\big(C_{abc}+C_{cba}\big) = 0, \tag{144}$$

since $C_{cba} = -C_{abc}$. Feeding (143) into (142),

$$\boxed{\delta S_{\rm YM} = \underbrace{-\oint_{\partial M}\mathrm{d}\Sigma_\mu\,\sqrt{-g}\;F^{a\mu\nu}\,\delta A^a_\nu}_{\text{boundary term}} + \int \mathrm{d}^4x\,\sqrt{-g}\;(D_\mu F^{\mu\nu})^a\,\delta A^a_\nu} \tag{145}$$

with $\mathrm{d}\Sigma_\mu$ the outward-directed surface element. Demanding $\delta S_{\rm YM} = 0$ for arbitrary δA^a_ν that vanish on ∂M leaves the boundary piece inert and reproduces

$$(D_\mu F^{\mu\nu})^a = 0, \tag{146}$$

exactly as before. Note that the very same manipulation in trace form, $S_{\rm YM} = \frac{1}{2}\int\sqrt{-g}\;\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})$, gives $\delta S_{\rm YM} = 2\oint_{\partial M}\mathrm{d}\Sigma_\mu\sqrt{-g}\;\mathrm{Tr}(F^{\mu\nu}\delta A_\nu) - 2\int\sqrt{-g}\;\mathrm{Tr}\big((D_\mu F^{\mu\nu})\delta A_\nu\big)$, and projecting with $\mathrm{Tr}(T_aT_b) = -\frac{1}{2}\delta_{ab}$ returns (145). The boundary term is what one cancels with a Gibbons–Hawking–type / holographic counterterm when $\partial M \neq \varnothing$.

The meron ansatz and its field strength

For the remainder it is cleanest to absorb the coupling into the connection (equivalently set $g = 1$), and work with the Lie-algebra-valued one-form $A_\mu = A^a_\mu T_a$; the coupling can be restored at the end by $A \to gA$. Introduce a map $U(x) \in SU(2)$ and its *Maurer–Cartan form*

$$L_\mu \equiv U^{-1}\partial_\mu U \in \mathfrak{g}. \tag{147}$$

The **meron** is a fixed fraction λ of a pure gauge,

$$\boxed{A_\mu = \lambda\,L_\mu = \lambda\,U^{-1}\partial_\mu U, \qquad \lambda \neq 0, 1} \tag{148}$$

($\lambda = 0$ is trivial, $\lambda = 1$ is pure gauge with $F = 0$). Its curvature is remarkably simple. Using $\partial_\mu U^{-1} = -U^{-1}(\partial_\mu U)U^{-1}$ one proves the Maurer–Cartan identity

$$\partial_\mu L_\nu - \partial_\nu L_\mu = -[L_\mu, L_\nu], \tag{149}$$

because the symmetric second-derivative $U^{-1}\partial_\mu\partial_\nu U$ drops out of the antisymmetric combination. Hence

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] = \lambda(\partial_\mu L_\nu - \partial_\nu L_\mu) + \lambda^2[L_\mu, L_\nu] \\ &\overset{(149)}{=} \big(\lambda^2 - \lambda\big)[L_\mu, L_\nu] = \boxed{\lambda(\lambda-1)\,[\,U^{-1}\partial_\mu U,\ U^{-1}\partial_\nu U\,]}. \end{aligned} \tag{150}$$

The field strength is quadratic in the otherwise-flat Maurer–Cartan form and weighted by $\lambda(\lambda-1)$ — it vanishes at the two "pure" values and is maximised in between.

To make this explicit, parametrise the group element and read off the su(2) current:

$$U = Y^0\,\mathbf{1} + iY^i\sigma_i, \quad (Y^0)^2 + Y^iY_i = 1 \implies L_\mu = iP^k_\mu\sigma_k, \quad P^k_\mu = \varepsilon_{ijk}Y_i\partial_\mu Y_j + Y^0\partial_\mu Y^k - Y^k\partial_\mu Y^0. \tag{151}$$

A useful by-product (proved by the same ε-contraction $\varepsilon_{ijk}\varepsilon_{lmk} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ and the constraint $Y_a\partial_\mu Y_a = -Y^0\partial_\mu Y^0$) is that the bilinear of the current is the round metric on the group manifold S^3,

$$\delta_{mn}\,P^m_\mu P^n_\nu = G_{ij}\,\partial_\mu Y^i\partial_\nu Y^j, \qquad G_{ij} = \delta_{ij} + \frac{Y_iY_j}{1 - Y^kY_k}. \tag{152}$$

The hedgehog: a purely magnetic embedded monopole

Now specialise the map to be radial — the *hedgehog* ansatz,

$$U = \cos f(r)\,\mathbf{1} + i\,\hat n^i\sigma_i\,\sin f(r), \qquad Y^0 = \cos f, \quad Y^i = \hat n^i \sin f, \tag{153}$$

with $\hat n = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ the unit radial vector, $\hat n\cdot\hat n = 1$. Everything collapses at the maximal value $f = \pi/2$: then $Y^0 = 0$, $Y^i = \hat n^i$, so $\vec Y$ is a unit vector on the internal S^2 and only the first term of (151) survives, $P^k_\mu = \varepsilon_{ijk}Y_i\partial_\mu Y_j = (\vec Y\times\partial_\mu\vec Y)_k$.

Writing $F_{\mu\nu} = iF^m_{\mu\nu}\sigma_m$ with $F^m_{\mu\nu} = -2\lambda(\lambda-1)\varepsilon_{klm}P^k_\mu P^l_\nu$ and applying the quadruple cross-product identity $(\vec a\times\vec b)\times(\vec c\times\vec d) = [\vec a\cdot(\vec c\times\vec d)]\vec b - [\vec b\cdot(\vec c\times\vec d)]\vec a$ together with $\vec Y\cdot(\vec Y\times\cdots) = 0$, one finds the internal direction of F locks entirely onto the single unit vector Y^i:

$$\boxed{F^i_{\mu\nu} = Y^i\,\Pi_{\mu\nu}, \qquad \Pi_{\mu\nu} = 2\lambda(\lambda-1)\,\varepsilon_{mnq}\,Y^m\,\partial_\mu Y^q\,\partial_\nu Y^n} \tag{154}$$

so all the spacetime dependence sits in the *scalar* two-form $\Pi_{\mu\nu}$. Since $\Pi_{t\mu}\propto\partial_t Y = 0$, the configuration is purely magnetic. Promoted to a two-form and pulled back to the sphere,

$$\Pi = \lambda(\lambda-1)\,\varepsilon_{mnq}Y^m\,\mathrm{d}Y^q\wedge\mathrm{d}Y^n = -2\lambda(\lambda-1)\,\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi, \tag{155}$$

which is closed, $\mathrm{d}\Pi = 0$ (a top form on S^2), and is, up to the constant $-2\lambda(\lambda-1)$, a unit Dirac monopole. Thus the non-Abelian Bianchi identity and $\nabla^\nu\Pi_{\mu\nu} = 0$ both reduce to the two Maxwell equations of an embedded monopole.

Gravitational backreaction: an exact black hole

Because $Y^a Y^a = 1$, the colour indices in the stress tensor contract trivially, $F^a_{\mu\alpha}F^a_{\nu\beta} = \Pi_{\mu\alpha}\Pi_{\nu\beta}$, so the Yang–Mills source is precisely *twice* a Maxwell stress tensor,

$$T_{\mu\nu} = 2\Big(\Pi_{\mu\alpha}\Pi_{\nu}{}^{\alpha} - \tfrac{g_{\mu\nu}}{4}\Pi_{\lambda\tau}\Pi^{\lambda\tau}\Big), \tag{156}$$

the extra factor 2 being the fingerprint of the trace normalisation $\mathrm{Tr}(T_a T_b) = -\frac{1}{2}\delta_{ab}$. Solving $G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{\kappa}{e^2}T_{\mu\nu}$ on the static spherical ansatz $\mathrm{d}s^2 = -h\,\mathrm{d}t^2 + h^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$ gives a magnetic (A)dS–Reissner–Nordström metric,

$$\boxed{h(r) = 1 - \frac{\kappa M}{8\pi r} + \frac{4\kappa\,\lambda^2(\lambda-1)^2}{e^2\,r^2} - \frac{\Lambda r^2}{3}} \tag{157}$$

(with $\kappa = 16\pi G_N$, so the mass term is the Schwarzschild $2G_N M/r$). The $1/r^2$ charge coefficient is fixed entirely by λ, since $\Pi_{\theta\phi} = -2\lambda(\lambda-1)\sin\theta$ gives $T^t{}_t = -4\lambda^2(\lambda-1)^2/r^4$.

The field equation fixes $\lambda = \frac{1}{2}$

Unlike an ordinary integration constant, λ is *determined* by the Yang–Mills equation. With $F^i_{\mu\nu} = \Pi_{\mu\nu}Y^i$ and $\nabla^\nu\Pi_{\mu\nu} = 0$,

$$\nabla^\nu F^i_{\mu\nu} = \Pi_{\mu\nu}\nabla^\nu Y^i, \tag{158}$$

while the commutator piece evaluates, using $\varepsilon_{klm}\varepsilon_{abk} = \delta_{la}\delta_{mb} - \delta_{lb}\delta_{ma}$, $Y\cdot Y = 1$ and $Y\cdot\partial Y = 0$, to

$$[A^\nu, F_{\mu\nu}]^i = -2\lambda\,\Pi_{\mu\nu}\nabla^\nu Y^i. \tag{159}$$

Their sum is the Yang–Mills equation,

$$(D_\nu F^\nu{}_\mu)^i \;\propto\; (1-2\lambda)\,\Pi_{\mu\nu}\nabla^\nu Y^i = 0. \tag{160}$$

The tensor $\Pi_{\mu\nu}\nabla^\nu Y^i$ is not identically zero, so

$$\boxed{\lambda = \tfrac{1}{2}}. \tag{161}$$

The same value has a clean structural meaning. Writing $\sigma_r \equiv Y^i\sigma_i$ (the internal direction of F), a one-line computation with $[\vec{a}\cdot\vec{\sigma}, \vec{b}\cdot\vec{\sigma}] = 2i(\vec{a}\times\vec{b})\cdot\vec{\sigma}$ and BAC–CAB gives

$$\boxed{D_\mu\sigma_r = \partial_\mu\sigma_r + [A_\mu, \sigma_r] = (1-2\lambda)\,\partial_\mu\sigma_r} \tag{162}$$

so at $\lambda = \frac{1}{2}$ the field direction σ_r (equivalently Y^i) is *covariantly constant* — which is exactly the property one needs to define the conserved non-Abelian magnetic charge. The Yang–Mills equation, the existence of the charge, and the covariant constancy of Y^i are thus one and the same condition.

Remark 1. The solution is genuinely non-Abelian: σ_r is a hedgehog ($+\sigma_3$ at the north pole, $-\sigma_3$ at the south pole), so no globally smooth gauge rotation can align it with a fixed generator. The field $i\Pi\,\sigma_r$ is therefore *not* gauge-equivalent to an embedded Abelian monopole $i\Pi\,\sigma_3$, even though both have the same Π.

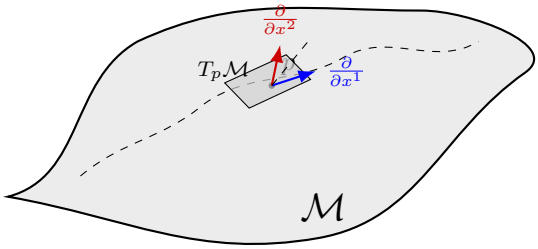
Homework. *(i)* Verify (149) directly from $\partial_\mu(U^{-1}U)=0$, and check that L_μ in (151) is traceless (i.e. lies in $\mathfrak{g}$) using the constraint $(Y^0)^2+Y^iY_i=1$. *(ii)* Prove the S^3 identity (152) by decomposing P^m_μ into its three pieces and using $\varepsilon_{ijk}\varepsilon_{lmk}=\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl}$. *(iii)* Starting from (162), show that the non-Abelian charge $Q\propto\oint_{S^2_\infty}\Pi$ is finite and fixed by λ, and evaluate it at $\lambda=\frac{1}{2}$.

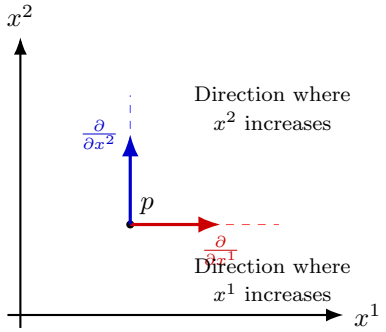
2 Differential Forms and Gravity in First Order Formalism

Introduction: Strong equivalence principle. Free falling reference frames are locally equivalent to a inertial reference frames. Given the equivalence principle, the spacetime is described by a manifold.

Spacetime as a manifold: coordinate bases and the metric

A *manifold* is a topological space which locally looks like $\mathbb{R}^n$ (a set of points that locally behaves as $\mathbb{R}^n$). Be the spacetime a manifold M of four dimensions, $\forall\, p\in M \quad \exists\ T_pM$ (tangent space for each point p) of Lorentzian signature $(-,+,+,+)$. We introduce a coordinate basis (Holonomic or contravariant)





Be a manifold of coordinates $x^\mu=(x^1,x^2,x^3,x^4)$ the associated coordinated basis is $\frac{\partial}{\partial x^\mu}$,i.e. on each point p the basis vectors are directional derivatives with respect to the coordinates

$$[\partial_\mu,\partial_\nu]=0 \quad ;\{\partial_\mu\}=\{\frac{\partial}{\partial x^\mu}\}\in T_pM. \tag{163}$$

We can introduce a dual basis (holonomic/covariant basis) $\forall\, p\in M$, we have ∂_μ. It is possible to define dx^μ such that $dx^\mu: T_pM\to\mathbb{R}$

$$\frac{\partial}{\partial x^\mu}\longrightarrow dx^\mu(\frac{\partial}{\partial x^\nu})=\delta^\mu_\nu, \tag{164}$$

extract the μ-th component of the coordinate basis (this is a linear application). A holonomic basis satisfy $ddx^\mu=0$. Therefore dx^μ is the basis of T^*_pM.

The metric g is a smooth tensor field with rank $(0,2)$ (the first slot for coordinate basis and the second for the dual basis) symmetric and non-degenerated. $g \in T_p^*M \otimes T_p^*M$, such that $\forall p \in M$

$$\begin{gathered} g_p : T_pM \otimes T_pM \Longrightarrow \mathbb{R} \\ g_p(\partial_\mu, \partial_\nu) = g_p(\partial_\nu, \partial_\mu) \\ \det(g) \neq 0. \end{gathered}$$

The inverse metric is $g^{-1} : T_p^*M \otimes T_p^*M \longrightarrow \mathbb{R}$. The metric $g = g_{\mu\nu} dx^\mu dx^\nu$ has smooth coefficients $g_{\mu\nu}$ in $U \in M$ with (U) a open smooth neighborhood. The components of the metric are defined as

$$g(\partial_a, \partial_b) = g_{\mu\nu} dx^\mu \otimes dx^\nu (\partial_\alpha \partial_\beta) = g_{\mu\nu} \delta^\mu_\alpha \delta^\nu_\beta = g_{\alpha\beta}. \tag{165}$$

In the same way

$$g^{-1}(dx^\mu, dx^\nu) = g^{\mu\nu}. \tag{166}$$

Orthonormal frames and the vielbein

Orthonormal basis: For a Lorentzian manifold M and T_pM it's tangent space with coordinates $z^a = \{z^1, ..z^4\}$. It posses an orthonormal basis if

$$g(e_a, e_b) = \eta_{ab} \quad , \eta_{ab} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{167}$$

Here η_{ab} is the Minkowski metric. Also we can define a dual orthonormal basis by

$$\begin{gathered} e^a : T_p(\mathbf{M}) \longrightarrow \mathbb{R}\,, \\ \longrightarrow e^a(e_b) = \delta^a_b\,. \end{gathered}$$

Also we have $g^{-1}(e^a, e^b) = \eta^{ab}$.

Observation : *The orthonormal basis doesn't are necessarily coordinated basis, in general they are not and satisfy being a holonomic basis :* $[e_a, e_b] = C^c_{ab} e_c$. An example of this claim is the polar coordinated basis in $\mathbb{R}^2$, i.e, $x^\mu = (r, \theta)$, where the tangent vectors to the curves (constant r and constant θ) are $\partial_\mu = (\partial_r, \partial_\theta)$. The satisfy

$$[\partial_\mu, \partial_\nu] = 0$$

so they are holonomic basis. Also we can see that

$$ds^2 = dr^2 + r^2 d\Omega^2 \to g(\partial_r, \partial_r) = 1\,;\, g(\partial_\theta, \partial_\theta) = r^2 \to |\partial_r| = 1\,;\, |\partial_\theta| = r\,.$$

If we consider $e_r = \partial_r, e_\theta = \frac{1}{r}\partial_\theta$ (a orthonormal basis), we can see from here that, if we have two vector fields (x, y), a function f and the Lie bracket, then

$$[x.fy] = x(fy) - (fy)x = (xf)y + fxy - fyx = (xf)y + f[x, y]\,,$$

then

$$[e_r, e_\theta] = [\partial_r, \frac{1}{r}\partial_\theta] = -\frac{1}{r^2}\partial_\theta = -r^{-1} e_\theta\,.$$

We see here how the "structure constant" appears for this non-commutative basis. The "anholonomy" measure the fail of the basis to be integrable as a coordinate system. [3]

Now we introduce the concept of **vielbein**. There exists a change of basis (isomorphism) between the coordinate basis $\{\partial_\mu\}$ of the coordinates x^μ and the *orthonormal basis* $\{e_a\}$ of coordinate Z^a of T_pM, which is given by the Jacobian matrix

$$e^a_\mu = \frac{\partial Z^a}{\partial x^\mu}\,;\, E^\mu_a = \frac{\partial x^\mu}{\partial Z^a}\,, \tag{168}$$

such that

$$e^a_\mu E^\nu_a = \delta^\mu_\nu\,, e^a_\mu E^\mu_b = \delta^a_b\,. \tag{169}$$

From here we can notice that

$$e_a = E^\mu_a \partial_\mu\,,\, \partial_\mu = e^a_\mu e_a \tag{170}$$

and the same for the dual basis

$$e^a = e^a_\mu dx^\mu\,,\, dx^\mu = E^\mu_a e^a\,. \tag{171}$$

If we consider $g(e_a, e_b) = \eta_{ab}$ then we can see that this relations are direct

$$g(E^\mu_a\partial_\mu, E^\nu_b\partial_\nu) = \eta_{ab} \to E^\mu_a E^\nu_b g_{\mu\nu} = \eta_{ab}\,, g^{-1}(e^a, e^b) = \eta^{ab} \leftrightarrow e^a_\mu e^b_\nu g^{\mu\nu} = \eta^{ab}\,. \tag{172}$$

Homework: Find the inverse relations $g \to \eta$ (one metric can induce the other).

We start from the observation that e^a_μ relates orthonormal basis and coordinates basis

$$e^a_\mu = \frac{\partial z^a}{\partial x^\mu},$$

but this only works in the case that we are working in holonomic basis. For example, consider that

$$\frac{\partial}{\partial x^\mu} f(z^a(x^\mu)) = \frac{\partial z^a}{\partial x^\mu}\frac{\partial f}{\partial z^a} \quad , \quad \therefore \frac{\partial}{\partial x^\mu} = \frac{\partial z^a}{\partial x^\mu}\frac{\partial}{\partial z^a}\,. \tag{173}$$

Here we can see that the piece corresponding to the holonomic basis is just $\frac{\partial}{\partial z^a}$, and we can try another examples like polar coordinates. Therefore, the relation $e^a_\mu = \frac{\partial z^a}{\partial x^\mu}$ is true only if the orthogonal basis is holonomic. And this imply that this definition of not the more general. The vielbein needs to be understood more broadly as a *local isomorphism* between the tangent bundle and an internal vector bundle with metric η_{ab}

$$e^a = e^a_\mu dx^\mu \quad , \quad e_a = e^\mu_a \partial_\mu\,. \tag{174}$$

We can notice that

$$g_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}\,, \tag{175}$$

in the left hand side we have 10 independent components because $g_{\mu\nu}$ is symmetric, and in the right hand side we have 16. The 6 redundant components corresponds to a gauge redundancy, specifically they are the 6 $\mathcal{SO}(3,1)$ generators (Lorentz group in D=4). If we consider a Lorentz transformation over the vielbeins

$$\bar{e}^a_\mu = \Lambda^a_b(x) e^b_\mu, \tag{176}$$

[3] The Maurer-Cartán forms are duals of the orthonormal basis.

then

$$g_{\mu\nu} = \bar{e}^a_\mu \bar{e}^b_\nu \eta_{ab} = \Lambda^a_c \Lambda^b_d e^c_\mu e^d_\nu \eta_{ab} = e^c_\mu e^d_\nu \eta_{cd}\,, \tag{177}$$

therefore the vielbein is a local Lorentz vector for the orthonormal basis. In addition to this, the vielbein transforms as a 1-form under coordinate transformations [4]

$$e'^a_\mu = \frac{\partial x^\nu}{\partial x'^\mu} e^a_\nu\,. \tag{178}$$

Fields of vectors and 1-forms

A vector field is an operator which maps real functions into real functions

$$X : C^\infty(\mathcal{M}) \longrightarrow C^\infty(\mathcal{M})\,. \tag{179}$$

If X represents a contravariant vector field, then $X(f) = g$ for a coordinate system $x^\mu = (x^1, x^2, ...x^n)$ this field is given by $X = X^\mu \partial_\mu$.

$$X(f) = X^\mu \partial_\mu f = g\,.$$

In general, $X : M \to T_pM$ such that $X_p \in T_pM$. From the dual coordinate basis it is possible to introduce a field of $1-$forms; $\omega_p : M \to T^*_pM$. This means that, on each point of M there is associated a $1-$form $\omega_p = \omega_\mu(p)dx^\mu$ (in orthonormal basis $\omega_p = \omega_a(p)e^a$). Moreover, we know that $\omega_p : T_pM \to \mathbb{R}$, where

$$\begin{aligned}\omega_p(f(p)X_p + g(p)Y_p) &= f(p)\omega_p(X_p) + g(p)\omega_p(X_p) \\ &= f(p)\omega_\mu(p)dx^\nu(p)dx^\mu(\partial_\nu) + g(p)\omega_\mu(p)Y^\nu(p)dx^\mu(\partial_\nu) \\ &= f(p)\omega_\mu(p)X^\mu(p) + g(p)\omega_\mu(p)X^\mu(p)\,.\end{aligned} \tag{180}$$

Using this $1-$forms it is possible to build a multi-linear map totally antisymmetric known as $p-$forms. This objects are invariant under general coordinate transformations by construction and the are very relevant in gauge theories.

Differential forms and the exterior algebra

Differential forms: They are a language that proportionate a description independent of the coordinates of a given physics theory. This language is constructed through the dual basis of the cotangent space T^*_pM, i.e, $dx^\mu : T_pM \to \mathbb{R}$, and considering multi-linear applications of it

$$\alpha_\sigma \,:\, T_\sigma M \times \cdots \times T_\sigma M \to \mathbb{R}\,,$$

where

$$\alpha \to dx^{\mu_1} \otimes dx^{\mu_2} \otimes \cdots \otimes dx^{\mu_p}\,.$$

We are interested in the subspace $\Lambda^p T_\sigma M \subset \otimes^p T_\sigma M$ (the totally antisymmetric space). We obtain this sub-space by applying the operator $\hat{A}$ of anti-symmetrization, which will generate a basis for this sub-space. This basis is given by

$$dx^\mu_1 \wedge \cdots \wedge dx^\mu_p = \sum_\sigma (-1)^{|\sigma|} dx^{\sigma(\mu_1)} \otimes dx^{\sigma(\mu_2)} \otimes \cdots \otimes dx^{\sigma(\mu_p)}\,, \tag{181}$$

Where σ denotes the permutation of the indices. For example :

[4] If we consider translations e^a_μ can be considered as a connection!.

- The most simple is with two indices

$$dx^\mu \wedge dx^\nu = dx^\mu \otimes dx^\nu - dx^\nu \otimes dx^\mu = -dx^\nu \wedge dx^\mu \,.$$

- With three indices we have

$$\begin{aligned} dx^\mu \wedge dx^\nu \wedge dx^\lambda = & \, dx^\mu \otimes dx^\nu \otimes dx^\lambda + dx^\nu \otimes dx^\lambda \otimes dx^\mu + dx^\lambda \otimes dx^\mu \otimes dx^\nu \\ & - dx^\nu \otimes dx^\mu \otimes dx^\lambda - dx^\mu \otimes dx^\lambda \otimes dx^\nu - dx^\lambda \otimes dx^\nu \otimes dx^\mu \,. \end{aligned}$$

Homework: compute the case of $dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_6}$. Finally, an element $\alpha \in \Lambda^p(M)$ can be written in the following way

$$\alpha = \frac{1}{p!} \alpha_{\mu_1 \cdots \mu_p} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_p} \,. \tag{182}$$

and we call this a $p-$form. Consider now $\alpha \in \Lambda^p(M)$, $\beta \in \Lambda^q(M)$ and $\delta \in \Lambda^r(M)$, then

- $dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_s} = 0$, if μ_s is repeated.
- $\alpha \wedge \alpha = 0$, if p is odd.
- $(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma)$.

The second one comes from

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha \,,$$

so is direct to see that

$$\beta = \alpha \implies \alpha \wedge \alpha = (-1)^{p^2} \alpha \wedge \alpha \implies (1 - (-1)^{p^2}) \alpha \wedge \alpha = 0 \,.$$

Exterior derivative : We define the exterior derivative as

$$d : \Lambda^p(M) \longrightarrow \Lambda^{p+1}(M) \,, \tag{183}$$

and acting in the following way

$$d\alpha = \frac{1}{p!} \partial_\lambda \alpha_{\mu_1 \cdots \mu_p} dx^\lambda \wedge dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_p} \,. \tag{184}$$

Let's see a example of how it acts on a $1-$form $A = A_\mu dx^\mu$. From this we can define F as in electrodynamics

$$\begin{aligned} F = dA &= \partial_\lambda A_\mu dx^\lambda \wedge dx^\mu \\ &= \frac{1}{2} \left(\partial_\lambda A_\mu dx^\lambda \wedge dx^\mu + \partial_\lambda A_\mu dx^\lambda \wedge dx^\mu \right) \\ &= \frac{1}{2} \left(\partial_\lambda A_\mu dx^\lambda \wedge dx^\mu + \partial_\mu A_\lambda dx^\mu \wedge dx^\lambda \right) \\ &= \frac{1}{2} \left(\partial_\lambda A_\mu - \partial_\mu A_\lambda \right) dx^\lambda \wedge dx^\mu \\ &= \frac{1}{2} F_{\lambda\mu} dx^\lambda \wedge dx^\mu \,, \end{aligned}$$

and we can see that in effect the exterior derivative gives us a $2-$ form when acts on a $1-$form. If $\alpha \in \Lambda^p(M), \beta \in \Lambda^q(M)$ and $\gamma \in \Lambda^D(M)$, with $D = dim(M)$, then:

- Nilpotency of exterior derivative $d(d)\alpha = 0$ for any smooth differential form on a smooth manifolds[5].
- $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$.
- $d\gamma = 0$.

Inner contraction: Let's us consider a vector field V, then we define $i_V : \Lambda^p(M) \to \Lambda^{p-1}(M)$

$$i_V(dx^\mu \wedge \cdots \wedge dx^{\mu_p}) = \sum_{k=1}^{p} (-1)^{k-1} dx^\mu \wedge \cdots \wedge d\hat{x}^{\mu_k} \wedge \cdots \wedge dx^{\mu_p} \,, \tag{185}$$

where the hat in $d\hat{x}^{\mu_k}$ means that term is omited. Then, if

$$i_{\partial_\nu}(dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_p}) = \delta^{\mu_1}_\nu dx^{\mu_2} \wedge \cdots \wedge dx^{\mu_p} - \delta^{\mu_2}_\nu dx^{\mu_1} \wedge dx^{\mu_3} \wedge \cdots \wedge dx^{\mu_p} + \cdots + (-1)^{k-1} \delta^{\mu_p}_\nu dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_{p-1}} \,. \tag{186}$$

With this we can apply the inner contraction operator on a $p-$form

$$\begin{aligned} i_V \alpha &= \frac{1}{p!} \alpha_{\mu_1 \cdots \mu_p} i_V(dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_p}) \\ &= \frac{1}{p!} \alpha_{\mu_1 \cdots \mu_p} p V^{\mu_1} dx^{\mu_2} \wedge \cdots \wedge dx^{\mu_p} \\ &= \frac{1}{(p-1)!} V^{\mu_1} \alpha_{\mu_1 \cdots \mu_p} dx^{\mu_2} \wedge \cdots \wedge dx^{\mu_p} \,. \end{aligned}$$

Some properties for $\alpha \in \Lambda^p$, $\beta \in \Lambda^q$ and $\phi \in \Lambda^0$ ($\phi : M \to \mathbb{R}$) are

1. $i_V i_V \alpha = 0$.
2. $i_V(\alpha + \beta) = i_V \alpha \wedge \beta + (-1)^p \alpha \wedge i_V \beta$.
3. $i_V \phi = 0$.

We know that we can change the basis between coordinate and orthonormal frames using the vielbein definition

$$\begin{aligned} T^{a_1 \cdots a_p}_{b_1 \cdots b_q} &= e^{a_1}_{\mu_1} \cdots e^{a_p}_{\mu_p} E^{\nu_1}_{b_1} \cdots E^{\nu_q}_{b_q} T^{\mu_1 \cdots \mu_p}_{\nu_1 \cdots \nu_q} \,, \\ T^{\mu_1 \cdots \mu_p}_{\nu_1 \cdots \nu_q} &= e^{b_1}_{\nu_1} \cdots e^{b_q}_{\nu_q} E^{\mu_1}_{a_1} \cdots E^{\mu_p}_{a_p} T^{a_1 \cdots a_p}_{b_1 \cdots b_q} \,, \end{aligned}$$

therefore

$$\begin{aligned} \alpha &= \frac{1}{p!} \alpha_{\mu_1 \cdots \mu_p} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_p} \\ &= \frac{1}{p!} \alpha_{a_1 \cdots a_p} e^{a_1} \wedge \cdots \wedge e^{a_p} \,. \end{aligned}$$

Now we can write the inner contraction operator in this basis E_a (E_a is for the tangent space and e^a is for the dual space)

$$i_a(e^{b_1} \wedge \cdots \wedge e^{b_p}) = \delta^{b_1}_a e^{b_2} \wedge \cdots \wedge e^{b_p} - \delta^{b_2} a e^{b_1} \wedge \cdots \wedge e^{b_p} + \cdots + (-1)^{p-1} \delta^{b_p}_a e^{b_1} \wedge \cdots \wedge e^{b_{p-1}} \,. \tag{187}$$

[5]This holds because the Clairaut's theorem and the nature of the wedge product.

As an example of the use we can see the following equation, which has the form of a typical equation in the Einstein-Cartán gravity context

$$Q_a + L_{ab} \wedge e^b = 0\,, \tag{188}$$

where $L_{ab} = L_{bc}e^c$ and $L_{ab} = -L_{ba}$. The goal is to find the expression for L_{ab}. If we apply the inner contraction in this equation

$$\begin{aligned} i_c(L_{ab} \wedge e^b) &= L_{abd} i_c(e^d) \wedge e^b - L_{ab} i_c(e^b) \\ &= L_{abd}\delta^d_c \wedge e^b - L_{ab}\delta^b_c \\ &= L_{abc} \wedge e^b - L_{ac}\,, \end{aligned}$$

we can notice that

$$i_a(L_{cb} \wedge e^b) = L_{cba} \wedge e^b - L_{ca} = L_{cba}e^b + L_{ac}\,,$$

and if we take this expression and subtract all the first equation that we have obtained, then

$$2i_{[a}L_{c]b}e^b = 2L_{ac} + (L_{cba} - L_{abc})e^b\,.$$

Finally we can apply the inner contraction twice on the first equation to find $(L_{cba} - L_{abc})$ as we will see here

$$\begin{aligned} i_d i_c(L_{ab} \wedge e^b) &= i_d(L_{abc}e^b - L_{acb}e^b) \\ &= L_{adc} - L_{acb} \\ &= L_{cad} - L_{dac}\,. \end{aligned}$$

Therefore, renaming indices, we will have this expression

$$2L_{ac} + i_a i_c(L_{bd}e^d)e^b = 2i_{[a}b_{c]b}e^b\,.$$

From the original equation we can see that all the terms can be written in terms of Q_a

$$\begin{aligned} 2L_{ac} - i_a i_c(Q_b)e^b &= -2i_{[a}Q_{c]} \\ \therefore L_{ac} &= -i_{[a}Q_{c]} + \frac{1}{2} i_a i_c(Q_b)e^b\,, \end{aligned}$$

so we conclude that

$$L_{ab} = -i_{[a}Q_{b]} + \frac{1}{2} i_a i_b(Q_c)e^c\,.$$

Volume form and Hodge duality

We want to understand the volumes in this new language of differential forms. Consider the following

$$e^{a_1} \wedge \cdots \wedge e^{a_d} = \frac{1}{d!}\delta^{a_1\cdots a_d}_{b_1\cdots b_d} e^{b_1} \wedge \cdots \wedge e^{b_d}\,, \tag{189}$$

so is direct to see that

$$\begin{aligned} e^1 \wedge \cdots \wedge e^d &= \frac{1}{d!}\delta^{1\cdots d}_{b_1\cdots b_d} e^{b_1} \wedge \cdots \wedge e^{b_d} \\ &= \frac{1}{d!}\epsilon_{b_1\cdots b_d} e^{b_1} \wedge \cdots \wedge e^{b_d}\,. \end{aligned}$$

Under Lorentz transformations $\bar{e}^a = \Lambda^a_b e^b$ we have that

$$\bar{e}^1 \wedge \cdots \wedge \bar{e}^d = \frac{1}{d!}\epsilon_{b_1\cdots b_d}\bar{e}^{b_1} \wedge \cdots \wedge \bar{e}^{b_d} = \frac{1}{d!}\epsilon_{b_1\cdots b_d}\Lambda^{b_1}_{a_1}\cdots\Lambda^{b_d}_{a_d}e^{a_1} \wedge \cdots \wedge e^{a_d}\,, \tag{190}$$

where we identify $\epsilon_{b_1\cdots b_d}\Lambda^{b_1}_{a_1}\cdots\Lambda^{b_d}_{a_d} = \det(\Lambda)\epsilon_{a_1\cdots a_d}$ [6].

The Lie groups invariant tensors allow us to build action principles which are invariant under that groups. In this case we can construct invariant actions for the Lorentz group.

For this case we have $e_1 \wedge \cdots \wedge e_d = \bar{e}_1 \wedge \cdots \wedge \bar{e}_d$. Finally we have find a $d-$form which is invariant under general coordinate transformations and under Lorentz transformations. Then, for any orientable manifold $M^{(d)}$ endowed with metricity which is codified by e^a

$$\mathrm{Vol}(M^{(d)}) = \int_{M^{(d)}} \tilde{v}^{(d)}\,, \quad \text{with}: \quad \tilde{v}^{(d)} = \frac{1}{d!}\epsilon_{a_1\cdots a_d}e^{a_1} \wedge \cdots \wedge e^{a_d}\,. \tag{191}$$

From here we can recover the known result in the coordinated basis [7]

$$\mathrm{Vol}(M^{(d)}) = \int_{M^{(d)}} \frac{1}{d!}\underbrace{\epsilon_{a_1\ldots a_d}\, e^{a_1}_{\mu_1}\cdots e^{a_d}_{\mu_d}}_{\det(e)\epsilon_{\mu_1\cdots\mu_d}}\mathrm{d}x^{\mu_1} \wedge \cdots \wedge \mathrm{d}x^{\mu_d}\,, \tag{192}$$

$$\begin{aligned}
\implies \mathrm{Vol}(M^{(d)}) &= \int_{M^{(d)}} \det(e)\frac{1}{d!}\epsilon_{\mu_1\cdots\mu_d}dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_d} \\
&= \int_{M^{(d)}} \sqrt{-g}dx^1 \wedge \cdots \wedge dx^d \\
&= \int_{M^{(d)}} \sqrt{-g}d^dx\,,
\end{aligned}$$

where we changed the indices of $\epsilon_{\mu_1\cdots}$ with Kronecker deltas. An important observation is because $\epsilon_{a_1\cdots a_d}$ is the Levi-Civita symbol, we need to consider

$$\eta^{a_1b_1}\eta^{a_2b_2}\cdots\eta^{a_db_d}\,\epsilon_{b_1\cdots b_d} = \det(\eta)\epsilon^{a_1\cdots a_d} = (-1)^{\eta_-}\epsilon^{a_1\cdots a_d}\,.$$

Hodge dual : The dimension of the spaces of $p-$forms is given by

$$\dim[\Lambda^p(M^{(d)})] = \frac{d!}{p!(d-p)!}\,, \tag{193}$$

and corresponds to the possible combinations without repetition of the $p-$elements over the total $d-$elements.

[6] Which $\epsilon_{a_1\cdots a_d}$is one of the invariant tensors of $\mathfrak{so}(3,1)$ when the determinant is $+1$. The other is $\eta_{cd} = \Lambda^a_c\Lambda^b_d\eta_{ab}$.

[7] Here we use the determinant coming from the equation $\eta_{ab}e^a_\mu e^b_\nu = g_{\mu\nu}$, which gives us $\det(e^a_\mu) = \sqrt{|\det g|} = \sqrt{-g}$.

Proof for the dimension

Let $M^{(d)}$ be a d-dimensional manifold. At any given point $x \in M^{(d)}$, the cotangent space T_x^*M is a vector space of dimension d. We can choose a coordinate basis of 1-forms for this space, which consists of d independent 1-forms:

$$\{dx^1, dx^2, \ldots, dx^d\}\,. \tag{194}$$

A p-form is an element of $\Lambda^p(T_x^*M)$, constructed by taking the wedge product of p of these basis 1-forms. A general basis element for the space of p-forms takes the form:

$$dx^{\mu_1} \wedge dx^{\mu_2} \wedge \cdots \wedge dx^{\mu_p}\,, \tag{195}$$

where each index $\mu_i \in \{1, \ldots, d\}$. To find the dimension of this space, we must determine the number of linearly independent basis elements that can be constructed.
The wedge product is totally antisymmetric, which imposes two critical constraints on the indices:

- **No repetitions:** The wedge product of any 1-form with itself vanishes,

$$dx^\mu \wedge dx^\mu = 0\,. \tag{196}$$

  Thus, for a basis element to be non-zero, all p indices must be distinct.

- **Antisymmetry under permutation:** Swapping any two indices only changes the overall sign of the form,

$$dx^\mu \wedge dx^\nu = -dx^\nu \wedge dx^\mu\,. \tag{197}$$

  Therefore, different orderings of the same set of indices do not yield linearly independent forms.

To systematically count the independent elements and avoid overcounting, we can restrict our basis vectors by imposing a strict ascending order on the indices:

$$\mu_1 < \mu_2 < \cdots < \mu_p\,. \tag{198}$$

The problem of finding the dimension of the space reduces to finding how many ways we can choose a subset of p distinct indices from a total set of d indices, without regard to the order of selection. This is exactly the definition of the binomial coefficient. Thus, the dimension of the space of p-forms is:

$$\begin{aligned}\dim[\Lambda^p(M^{(d)})] &= \binom{d}{p} \\ &= \frac{d!}{p!(d-p)!}\,.\end{aligned} \tag{199}$$

Given that $\binom{d}{p} = \binom{d}{d-p}$ [8], then $\Lambda^p(M^{(d)})$ and $\Lambda^{d-p}(M^{(d)})$ have the same dimension we can think about define an invertible map of the type $\star : \Lambda^p(M^{(d)}) \to \Lambda^{d-p}(M^{(d)})$. To define it is necessary to

[8] We just need to replace $d-p$ into the definition of the binomial coefficient: $\binom{d}{d-p} = \frac{d!}{(d-p)!\,[d-(d-p)]!} = \binom{d}{p}$.

introduce the concept of metricity in the manifold. Given a $p-$form α

$$\alpha = \frac{1}{p!}\alpha_{a_1\cdots a_p} e^{a_1} \wedge \cdots \wedge e^{a_p}\,,$$

the Hodge dual $\star\alpha$ of the $p-$form is defined as

$$\star\alpha = \frac{1}{p!(d-p)!}\alpha^{a_1\cdots a_p}\epsilon_{a_1\cdots a_p b_1\cdots b_{d-p}} e^{b_1} \wedge \cdots \wedge e^{b_{d-p}}\,. \tag{200}$$

We can write

$$(\star\alpha)_{b_1\cdots b_{d-p}} = \frac{1}{p!}\alpha^{a_1\cdots a_p}\epsilon_{a_1\cdots a_p b_1\cdots b_{d-p}}\,,$$

or more rigorously

$$\star\alpha = \frac{1}{(d-p)!}(\star\alpha)_{b_1\cdots b_{d-p}} e^{b_1} \wedge \cdots \wedge e^{b_{d-p}}\,.$$

Homework: It is possible to show that

$$\star\star\,\alpha = (-1)^{p(d-p)+\eta_-}\alpha\,. \tag{201}$$

Proof : Applying $\star$ twice yields a contraction of two Levi-Civita tensors

$$(\star\star\,\alpha)_{\rho_1\ldots\rho_p} = \frac{1}{p!(d-p)!}\alpha^{\mu_1\ldots\mu_p}\epsilon_{\mu_1\ldots\mu_p\nu_1\ldots\nu_{d-p}}\epsilon^{\nu_1\ldots\nu_{d-p}}{}_{\rho_1\ldots\rho_p}\,. \tag{202}$$

To properly contract the ν indices, we must first swap the p indices (μ) with the $(d-p)$ indices (ν) in the first tensor. Passing p indices through $(d-p)$ indices introduces a permutation sign:

$$\epsilon_{\mu_1\ldots\mu_p\nu_1\ldots\nu_{d-p}} = (-1)^{p(d-p)}\epsilon_{\nu_1\ldots\nu_{d-p}\mu_1\ldots\mu_p}\,. \tag{203}$$

Now, fully contracting the $(d-p)$ indices (ν) of the two tensors introduces a factor of $(d-p)!$, the antisymmetrization of the remaining p indices, and the sign of the metric determinant, $\mathrm{sgn}(g) = (-1)^{\eta_-}$:

$$\epsilon^{\nu_1\ldots\nu_{d-p}}{}_{\rho_1\ldots\rho_p}\epsilon_{\nu_1\ldots\nu_{d-p}\mu_1\ldots\mu_p} = (-1)^{\eta_-}(d-p)!\,p!\,\delta^{[\mu_1}_{\rho_1}\ldots\delta^{\mu_p]}_{\rho_p}\,. \tag{204}$$

Substituting this back into the first equation cancels the factorial denominators, leaving exactly the components of α multiplied by the accumulated signs:

$$\star\star\,\alpha = (-1)^{p(d-p)+\eta_-}\alpha\,. \quad \square \tag{205}$$

As an example, for $\alpha \in \Lambda^p(M^{(d)})$ we can compute

$$\begin{aligned}
\alpha\wedge\star\alpha &= \left(\frac{1}{p!}\alpha_{a_1\cdots a_p}e^{a_1}\wedge\cdots\wedge e^{a_p}\right)\wedge\left(\frac{1}{p!(d-p)!}\alpha^{b_1\cdots b_p}\,\epsilon_{b_1\cdots b_p c_1\cdots c_{d-p}}e^{c_1}\wedge\cdots\wedge e^{c_{d-p}}\right)\\
&= \frac{1}{p!p!(d-p)!}\alpha_{a_1\cdots a_p}\alpha^{b_1\cdots b_p}\epsilon_{b_1\cdots b_p c_1\cdots c_{d-p}}e^{a_1}\wedge\cdots\wedge e^{a_p}\wedge e^{c_1}\wedge\cdots\wedge e^{c_{d-p}}\\
&= \frac{1}{p!p!(d-p)!}\alpha_{a_1\cdots a_p}\alpha^{b_1\cdots b_p}\epsilon_{b_1\cdots b_p c_1\cdots c_{d-p}}\left(-\det(e)\epsilon^{a_1\cdots c_{d-p}}d^dx\right)\\
&= -\frac{1}{p!p!(d-p)!}\alpha_{a_1\cdots a_p}\alpha^{b_1\cdots b_p}\left(-p!(d-p)!\delta^{a_1}_{[b_1}\cdots\delta^{a_p}_{b_p]}\right)\det(e)d^dx\\
&= \frac{1}{p!}\alpha_{a_1\cdots a_p}\alpha^{b_1\cdots b_p}\frac{1}{p!}\delta^{a_1\cdots a_p}_{[b_1\cdots b_p}\det(e)d^dx\\
&= \frac{1}{p!}\alpha_{a_1\cdots a_p}\frac{1}{p!}\left(p!\alpha^{a_1\cdots a_p}\right)\det(e)d^dx\\
&= \frac{1}{p!}\alpha_{a_1\cdots a_p}\alpha^{a_1\cdots a_p}\det(e)d^dx = \frac{1}{p!}\alpha_{a_1\cdots a_p}\alpha^{a_1\cdots a_p}\tilde{v}(d)\,.
\end{aligned}$$

The Levi-Civita symbol and the Levi-Civita tensor

For the previous calculation we forget to make explicit the difference between the Levi-Civita *simbol* ϵ and the Levi-Civita *tensor* ε. The principal difference is that the symbol doesn't transform like a tensor, so we cannot raise or lower indices with the metric freely. The properties of the totally anti-symmetric objects that relates the Levi-Civita symbol and the Kronecker delta can be promoted to the tensor, but with the careful with respect to the spacetime signature. In the coordinated basis (for $d = 4$) the symbol is given by the following convention

$$\epsilon_{0123} = \epsilon^{0123} = 1\,,$$

or, in general,

$$\epsilon_{\mu\nu\lambda\rho} = \epsilon^{\mu\nu\lambda\rho} = \begin{cases} +1 & \text{even permutations of } (0123)\,, \\ -1 & \text{odd permutations of } (0123)\,, \\ \;\;0 & \text{repeated indices}\,. \end{cases}$$

Note that the two objects above are defined by the *same* numerical array: the position of the indices is here only a label, it does not indicate any contraction with the metric. With this convention, and without assuming any metricity, the symbol satisfies

$$\epsilon_{\mu\nu\lambda\rho}\,\epsilon^{\alpha\beta\gamma\delta} = \delta^{\alpha\beta\gamma\delta}_{\mu\nu\lambda\rho}\,,$$

where the generalized Kronecker delta is defined by

$$\delta^{\alpha\beta\gamma\delta}_{\mu\nu\lambda\rho} = 4!\,\delta^{[\alpha}_{\mu}\delta^{\beta}_{\nu}\delta^{\gamma}_{\lambda}\delta^{\delta]}_{\rho}\,.$$

The Levi-Civita *tensor* is then defined as

$$\varepsilon_{\mu\nu\lambda\rho} = \sqrt{-g}\,\epsilon_{\mu\nu\lambda\rho}\,.$$

To define this object we had to introduce the metric. This is a genuine tensor, so we can raise and lower its indices with the metric, and doing so we obtain

$$\varepsilon^{\mu\nu\lambda\rho} = \frac{\mathrm{sgn}(g)}{\sqrt{-g}}\,\epsilon^{\mu\nu\lambda\rho}\,, \qquad \text{with} \qquad \mathrm{sgn}(g) = \frac{\det(g)}{|\det(g)|}\,,$$

from which we note that

$$\varepsilon_{\mu\nu\lambda\rho}\,\varepsilon^{\alpha\beta\gamma\delta} = \mathrm{sgn}(g)\,\delta^{\alpha\beta\gamma\delta}_{\mu\nu\lambda\rho}\,.$$

If the metric has signature $(-+++)$, then $\mathrm{sgn}(g) = -1$ and we can write

$$\varepsilon_{0123} = \sqrt{-g}\,, \qquad \varepsilon^{0123} = -\frac{1}{\sqrt{-g}}\,.$$

The orthonormal basis (here the confusion begins)

We know that in the orthonormal basis the Levi-Civita symbol with covariant indices is an invariant tensor of $SO(3,1)$, which is precisely why it is tempting to use the symbol and the tensor interchangeably. Keeping the same convention as before,

$$\epsilon_{0123} = \epsilon^{0123} = 1\,,$$

we have

$$\epsilon_{abcd}\,\epsilon^{efgh} = \delta^{efgh}_{abcd}\,.$$

However, when we define the Levi-Civita tensor in this basis, given that $\sqrt{-\det(\eta)} = 1$, we obtain

$$\varepsilon_{abcd} = 1 \cdot \epsilon_{abcd}\,,$$

but the components with contravariant indices are

$$\varepsilon^{abcd} = \eta^{ae}\eta^{bf}\eta^{cg}\eta^{dh}\,\varepsilon_{efgh} = (-1)^t\,\epsilon^{abcd} = \mathrm{sgn}(\eta)\,\epsilon^{abcd}\,,$$

where t is the number of timelike (negative) directions of the signature. Therefore, if $\mathrm{sgn}(\eta) = (-+++)$, we have

$$\varepsilon_{abcd}\,\varepsilon^{efgh} = (-1)\,\delta^{efgh}_{abcd}\,,$$

and from here we can note that

$$\varepsilon_{0123} = -\varepsilon^{0123} = 1\,,$$

which is the convention we adopt. Indeed, this is nothing but

$$\varepsilon^{0123} = \eta^{00}\eta^{11}\eta^{22}\eta^{33}\,\varepsilon_{0123} = -\varepsilon_{0123}\,.$$

Observation: in the orthonormal basis it is possible to work either with the symbol or with the tensor, since their covariant components coincide. The subtlety is that they differ by a sign once the indices are raised, and this sign propagates through every contraction. To make this explicit we compute the volume form in both conventions.

Case 1: working with the symbol, $\epsilon_{0123} = \epsilon^{0123} = 1$. Then

$$\begin{aligned}
e^a \wedge e^b \wedge e^c \wedge e^d &= \frac{1}{4!}\,\delta^{abcd}_{efgh}\; e^e \wedge e^f \wedge e^g \wedge e^h \\
&= \frac{1}{4!}\,\epsilon_{efgh}\,\epsilon^{abcd}\; e^e \wedge e^f \wedge e^g \wedge e^h \\
&= \frac{1}{4!}\,\epsilon_{efgh}\,\epsilon^{abcd}\; e^e{}_\mu e^f{}_\nu e^g{}_\lambda e^h{}_\rho\; dx^\mu \wedge dx^\nu \wedge dx^\lambda \wedge dx^\rho \\
&= \frac{1}{4!}\,\epsilon^{abcd}\,\det(e)\,\epsilon_{\mu\nu\lambda\rho}\; dx^\mu \wedge dx^\nu \wedge dx^\lambda \wedge dx^\rho \\
&= \epsilon^{abcd}\,\det(e)\,d^4x\,,
\end{aligned}$$

where in the last line we used $\epsilon_{\mu\nu\lambda\rho}\,dx^\mu \wedge dx^\nu \wedge dx^\lambda \wedge dx^\rho = 4!\,d^4x$. Hence

$$e^0 \wedge e^1 \wedge e^2 \wedge e^3 = \epsilon^{0123}\,\det(e)\,d^4x = \det(e)\,d^4x\,.$$

Case 2: working with the tensor, with the convention $\varepsilon_{0123} = -\varepsilon^{0123} = 1$. Now $\delta^{abcd}_{efgh} = -\varepsilon^{abcd}\varepsilon_{efgh}$, so

$$\begin{aligned}
e^a \wedge e^b \wedge e^c \wedge e^d &= \frac{1}{4!}\,\delta^{abcd}_{efgh}\; e^e \wedge e^f \wedge e^g \wedge e^h \\
&= -\frac{1}{4!}\,\varepsilon^{abcd}\,\varepsilon_{efgh}\; e^e \wedge e^f \wedge e^g \wedge e^h \\
&= -\frac{1}{4!}\,\varepsilon^{abcd}\,\varepsilon_{efgh}\; e^e{}_\mu e^f{}_\nu e^g{}_\lambda e^h{}_\rho\; dx^\mu \wedge dx^\nu \wedge dx^\lambda \wedge dx^\rho \\
&= -\frac{1}{4!}\,\varepsilon^{abcd}\,\det(e)\,\epsilon_{\mu\nu\lambda\rho}\; dx^\mu \wedge dx^\nu \wedge dx^\lambda \wedge dx^\rho \\
&= -\varepsilon^{abcd}\,\det(e)\,d^4x\,,
\end{aligned}$$

and therefore

$$e^0 \wedge e^1 \wedge e^2 \wedge e^3 = -\varepsilon^{0123} \det(e)\, d^4x = \det(e)\, d^4x\,,$$

in agreement with Case 1, as it must be: the volume form is a well defined object and cannot depend on whether we choose to write it in terms of the symbol or of the tensor. We can write some properties that will be usefull

$$\begin{aligned}
\delta^{a_1\cdots a_p c_1\cdots c_r}_{b_1\cdots b_p c_1\cdots c_r} &= \frac{(d-p)!}{(d-r)!}\delta^{a_1\cdots a_p}_{b_1\cdots b_p} \\
\epsilon^{a_1\cdots a_d}\epsilon_{b_1\cdots b_d} &= -\delta^{a_1\cdots a_d}_{b_1\cdots b_d} \\
\delta^{a_1\cdots a_p}_{b_1\cdots b_p} &= -\frac{1}{(d-p)!}\epsilon^{a_1\cdots a_p c_{p+1}\cdots c_d}\epsilon_{b_1\cdots b_p c_{p+1}\cdots c_d} \\
\epsilon^{a_1\cdots a_p c_{p+1}\cdots c_d}\epsilon_{b_1\cdots b_p c_{p+1}\cdots c_d} &= -p!(d-p)!\delta^{a_1}_{[b_1}\cdots\delta^{a_p}_{b_p]}\,.
\end{aligned}$$

We can define the Levi-Civita tensor of rank (r,p)

$$\varepsilon^{a_1\cdots a_r}_{b_1\cdots b_p} = \eta^{a_1 c_1}\cdots\eta^{a_r c_r}\varepsilon_{c_1\cdots c_r b_1\cdots b_p}\,, \tag{206}$$

and with $r+p=d$

$$\varepsilon^{a_1\cdots a_r}{}_{b_{r+1}\cdots b_d}\varepsilon^{c_1\cdots c_r b_{r+1}\cdots b_d} = -r!(d-r)!\eta^{a_1[c-1}\cdots\eta^{a_r c_r]}\,,$$

where

$$\eta^{a_1[c-1}\cdots\eta^{a_r c_r]} = \frac{1}{r!}\sum_\sigma \mathrm{Sgn}(\sigma)\eta^{a_1 r_{\sigma(1)}}\cdots\eta^{a_p c_{\sigma(p)}}\,.$$

3 Connections, Torsion and Curvature

To motivate our approach, we begin by recalling the distinction between metric and affine geometry in the familiar setting of $\mathbb{R}^n$. Geometry can be decomposed into two conceptually independent layers: metric and affine. Metric properties govern measurable quantities such as lengths, angles, and volumes. In contrast, affine properties are characterized by invariance under affine transformations ($x \to Ax + B$), encompassing notions such as parallelism, line intersections, and the ratios of collinear segments. Notably, defining a straight line requires only affine structure, without any reference to a metric.

This is the principal hypothesis of the first order gravity, the independence of this two geometric characteristics. The affine properties are independent of the metric ones, however it is possible to obtain affine properties from the metric ones (for example, a straight line is the shortest distance between two points). In general, when the affine properties are not independent from the metric ones in curved geometries, we are working in a Riemannian geometry (Levi-Civita connection, no torsion).

Affine Connections and the Koszul Connection

The usual Christoffel symbols provide a first motivation for the geometrical notion of a connection. Under a change of coordinates $x^\mu \to \bar{x}^\mu$, the coefficients of an affine connection transform according

to

$$\bar{\Gamma}^{\rho}{}_{\mu\nu}(\bar{x}) = \frac{\partial \bar{x}^{\rho}}{\partial x^{\sigma}} \frac{\partial x^{\alpha}}{\partial \bar{x}^{\mu}} \frac{\partial x^{\beta}}{\partial \bar{x}^{\nu}} \Gamma^{\sigma}{}_{\alpha\beta}(x) + \frac{\partial \bar{x}^{\rho}}{\partial x^{\sigma}} \frac{\partial^2 x^{\sigma}}{\partial \bar{x}^{\mu} \partial \bar{x}^{\nu}}. \tag{207}$$

The second term is inhomogeneous. Consequently, $\Gamma^{\rho}{}_{\mu\nu}$ does *not* transform as a tensor.

For the Levi–Civita connection one can always choose locally inertial coordinates around a point p such that

$$\Gamma^{\rho}{}_{\mu\nu}(p) = 0. \tag{208}$$

At that point, the geodesic equation

$$\frac{d^2 x^{\mu}}{d\tau^2} + \Gamma^{\mu}{}_{\nu\rho} \frac{dx^{\nu}}{d\tau} \frac{dx^{\rho}}{d\tau} = 0 \tag{209}$$

reduces to

$$\left. \frac{d^2 x^{\mu}}{d\tau^2} \right|_p = 0. \tag{210}$$

The Levi–Civita connection is characterized by the two conditions

$$\nabla_{\rho} g_{\mu\nu} = 0, \qquad T^{\rho}{}_{\mu\nu} = 0, \tag{211}$$

namely metric compatibility and vanishing torsion.

Why introduce a more general connection?

In first-order gravity the metric need not be regarded as the fundamental variable. Instead, one introduces a coframe (vielbein) e^a and an independent connection $\omega^a{}_b$.
Therefore, it is useful to define a connection without imposing *a priori*

$$\nabla g = 0, \qquad T = 0.$$

The appropriate intrinsic definition is the *Koszul connection*.

Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields on a differentiable manifold M. Their Lie bracket is another vector field,

$$[X, Y] \in \mathfrak{X}(M), \tag{212}$$

defined through its action on a smooth function $f \in C^{\infty}(M)$:

$$[X, Y] f = X(Yf) - Y(Xf). \tag{213}$$

Writing

$$X = X^{\mu} \partial_{\mu}, \qquad Y = Y^{\mu} \partial_{\mu}, \tag{214}$$

one obtains

$$[X, Y] = \left(X^{\mu} \partial_{\mu} Y^{\nu} - Y^{\mu} \partial_{\mu} X^{\nu} \right) \partial_{\nu}. \tag{215}$$

The Lie bracket satisfies, for $f, g \in C^{\infty}(M)$,

$$[fX, Y] = f[X, Y] - Y(f) X\,, \tag{216}$$

$$[X, gY] = g[X, Y] + X(g) Y\,. \tag{217}$$

Koszul connection

Let M be a differentiable manifold and let $\mathfrak{X}(M)$ denote the space of smooth vector fields on M. A Koszul connection is a map

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \longrightarrow \mathfrak{X}(M), \qquad (X, Y) \longmapsto \nabla_X Y, \tag{218}$$

satisfying, for $X, Y, Z \in \mathfrak{X}(M)$, $f, g \in C^\infty(M)$ and $a, b \in \mathbb{R}$,

$$\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z, \tag{219}$$

$$\nabla_X (aY + bZ) = a\nabla_X Y + b\nabla_X Z, \tag{220}$$

$$\nabla_X (fY) = X(f)Y + f\nabla_X Y. \tag{221}$$

The last relation is the Leibniz rule. Geometrically, the connection provides a prescription for differentiating a vector field along another vector field. In particular, it provides the structure required to compare vectors at infinitesimally nearby points.

Consider a coordinate system x^μ and its associated basis $\{\partial_\mu\}$. The connection coefficients are defined by

$$\nabla_{\partial_\mu} \partial_\nu = \Gamma^\rho{}_{\mu\nu} \partial_\rho. \tag{222}$$

Let

$$X = X^\mu \partial_\mu, \qquad Y = Y^\nu \partial_\nu. \tag{223}$$

Using the defining properties of the Koszul connection,

$$\begin{aligned} \nabla_X Y &= \nabla_{X^\mu \partial_\mu} \left(Y^\nu \partial_\nu\right) \\ &= X^\mu \left[\partial_\mu Y^\nu \, \partial_\nu + Y^\nu \nabla_{\partial_\mu} \partial_\nu\right] \\ &= X^\mu \left(\partial_\mu Y^\rho + \Gamma^\rho{}_{\mu\nu} Y^\nu\right) \partial_\rho. \end{aligned} \tag{224}$$

Therefore,

$$\nabla_X Y = X^\mu (\nabla_\mu Y^\rho) \partial_\rho, \tag{225}$$

where

$$\boxed{\nabla_\mu Y^\rho = \partial_\mu Y^\rho + \Gamma^\rho{}_{\mu\nu} Y^\nu}. \tag{226}$$

The definition of a connection does not require a coordinate basis. Consider instead a local frame $\{e_a\}$ which is not necessarily of the form $e_a = \partial_a$.

We define the connection coefficients in this basis by

$$\nabla_{e_a} e_b = \Gamma^c{}_{ab} e_c. \tag{227}$$

For

$$X = X^a e_a, \qquad Y = Y^b e_b, \tag{228}$$

we obtain

$$\begin{aligned} \nabla_X Y &= X^a \nabla_{e_a} (Y^b e_b) \\ &= X^a \left[e_a(Y^c) + \Gamma^c{}_{ab} Y^b\right] e_c. \end{aligned} \tag{229}$$

Hence,

$$\boxed{\nabla_a Y^c = e_a(Y^c) + \Gamma^c{}_{ab} Y^b}. \tag{230}$$

The Kozul connection acting on 1-forms can be obtained from the duality relation. Let $\{e^a\}$ denote the coframe dual to $\{e_a\}$

$$e^a e_b = \delta^a{}_b. \tag{231}$$

Since

$$\nabla_{e_c} \delta^a{}_b = 0, \tag{232}$$

we have

$$\begin{aligned} 0 &= \nabla_{e_c} \left[e^a(e_b) \right] \\ &= (\nabla_{e_c} e^a)(e_b) + e^a(\nabla_{e_c} e_b). \end{aligned} \tag{233}$$

Using

$$\nabla_{e_c} e_b = \Gamma^d{}_{cb} e_d, \tag{234}$$

one finds

$$(\nabla_{e_c} e^a)(e_b) = -\Gamma^a{}_{cb}. \tag{235}$$

Therefore,

$$\boxed{\nabla_{e_c} e^a = -\Gamma^a{}_{cb} e^b}. \tag{236}$$

This is the familiar origin of the relative minus sign when the covariant derivative acts on a lower versus an upper tensor index.

Torsion and the Connection One-Forms

Torsion of a connection

Given a Koszul connection ∇, its torsion is the map

$$T : \mathfrak{X}(M) \times \mathfrak{X}(M) \longrightarrow \mathfrak{X}(M) \tag{237}$$

defined by

$$\boxed{T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]}. \tag{238}$$

It immediately follows that

$$T(X,Y) = -T(Y,X). \tag{239}$$

Moreover,

$$T(fX, Y) = fT(X,Y), \tag{240}$$

$$T(X, fY) = fT(X,Y), \tag{241}$$

so the torsion is $C^\infty(M)$-linear in both arguments. Hence torsion is a genuine tensor,

$$T \in \Gamma\big(TM \otimes \Lambda^2 T^* M\big)\,, \tag{242}$$

i.e. a $(1,2)$ tensor. For a coordinate basis,

$$[\partial_\mu, \partial_\nu] = 0. \tag{243}$$

Thus

$$\begin{aligned} T(\partial_\mu, \partial_\nu) &= \nabla_{\partial_\mu}\partial_\nu - \nabla_{\partial_\nu}\partial_\mu \\ &= (\Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu})\,\partial_\rho. \end{aligned} \tag{244}$$

Consequently,

$$\boxed{T^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu} = 2\Gamma^\rho{}_{[\mu\nu]}}. \tag{245}$$

For a general frame the basis vectors need not commute. Define the anholonomy coefficients $C^c{}_{ab}$ by

$$[e_a, e_b] = C^c{}_{ab} e_c. \tag{246}$$

Equation (238) then gives

$$\begin{aligned} T(e_a, e_b) &= \nabla_{e_a} e_b - \nabla_{e_b} e_a - [e_a, e_b] \\ &= (\Gamma^c{}_{ab} - \Gamma^c{}_{ba} - C^c{}_{ab})\, e_c. \end{aligned} \tag{247}$$

Hence

$$\boxed{T^c{}_{ab} = \Gamma^c{}_{ab} - \Gamma^c{}_{ba} - C^c{}_{ab}}. \tag{248}$$

Coordinate versus non-coordinate frames

The formula

$$T^\rho{}_{\mu\nu} = 2\Gamma^\rho{}_{[\mu\nu]}$$

is valid in a coordinate basis because $[\partial_\mu, \partial_\nu] = 0$.
For a non-holonomic frame, the additional term $-C^c{}_{ab}$ in (248) is essential.

The connection coefficients in the frame $\{e_a\}$ can be collected into a matrix of one-forms called the *connection one-forms*,

$$\omega^a{}_b = \Gamma^a{}_{cb}\, e^c. \tag{249}$$

Equivalently,

$$\nabla_X e_b = \omega^a{}_b(X) e_a. \tag{250}$$

The torsion can now be written as a vector-valued two-form

$$T^a = \frac{1}{2} T^a{}_{bc}\, e^b \wedge e^c. \tag{251}$$

Using the definition above one obtains Cartan's first structure equation,

$$\boxed{T^a = de^a + \omega^a{}_b \wedge e^b}. \tag{252}$$

Similarly, the curvature of the connection is encoded in the curvature two-form

$$\boxed{R^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b}. \tag{253}$$

Equations (252) and (253) are the first and second Cartan structure equations, respectively.

Introduce a vielbein $e^a{}_\mu$ such that

$$g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu e^b{}_\nu, \tag{254}$$

or, equivalently,

$$g = \eta_{ab}\, e^a \otimes e^b. \tag{255}$$

In an orthonormal frame,

$$g(e_a, e_b) = \eta_{ab}. \tag{256}$$

Metric compatibility,

$$\nabla g = 0, \tag{257}$$

then implies

$$\omega_{ab} + \omega_{ba} = 0, \tag{258}$$

so that

$$\boxed{\omega_{ab} = -\omega_{ba}}. \tag{259}$$

Thus a metric-compatible connection in an orthonormal frame takes values in the Lorentz algebra $\mathfrak{so}(1,3)$.

Why this language is natural in first-order gravity

In the first-order formulation of gravity the fundamental variables are taken to be e^a, ω^{ab} rather than the metric alone. The tetrad determines the spacetime metric,

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu,$$

while ω^{ab} determines parallel transport.
At the beginning of the first-order formulation, e^a and ω^{ab} are regarded as *independent fields.*

A First Look at the Einstein–Cartan Action

In four dimensions, the Einstein–Cartan or Palatini action can be written as

$$S[e,\omega] = \frac{1}{4\kappa}\int_M \epsilon_{abcd}\, e^a \wedge e^b \wedge R^{cd}[\omega], \tag{260}$$

where

$$R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}. \tag{261}$$

The two fields are varied independently. Since

$$\delta R^{ab} = D(\delta\omega^{ab}), \tag{262}$$

the ω-variation of the action gives, after integration by parts,

$$\epsilon_{abcd}\, T^a \wedge e^b = 0. \tag{263}$$

For an invertible tetrad and in the absence of matter carrying spin, this implies

$$\boxed{T^a = 0}. \tag{264}$$

Using Cartan's first structure equation,

$$de^a + \omega^a{}_b \wedge e^b = 0. \tag{265}$$

This equation determines the spin connection in terms of the tetrad,

$$\omega^{ab} = \omega^{ab}(e). \tag{266}$$

It is precisely the spin-connection version of the Levi–Civita connection.

The variation with respect to e^a gives

$$\boxed{\epsilon_{abcd}\, e^b \wedge R^{cd} = 0}. \tag{267}$$

After imposing the solution $\omega = \omega(e)$ of the torsion equation, this equation is equivalent to the vacuum Einstein equations.

There is an important distinction between the Levi–Civita case and a general affine connection. From (207), the inhomogeneous part of the coordinate transformation is symmetric in μ, ν. Therefore a coordinate transformation can modify $\Gamma^\rho{}_{(\mu\nu)}$ at a point, but not the antisymmetric part

$$\Gamma^\rho{}_{[\mu\nu]} = \frac{1}{2} T^\rho{}_{\mu\nu}. \tag{268}$$

Thus, when torsion is nonzero, one cannot choose a coordinate system in which all connection coefficients vanish at the point.

This is consistent with the fact that torsion is a tensor: a genuine nonzero tensor cannot be eliminated by a change of coordinates.

Geometric Interpretation of Torsion and the Lorentz Connection

In the previous class we introduced the general notion of a Koszul connection and the definition of torsion. We now want to develop a more geometric interpretation of torsion and connect this discussion with the vielbein formalism and the Lorentz connection, which will be the natural language for first-order gravity.

From the previous class we know that, in a coordinate basis $\{\partial_\mu\}$, the connection is written as

$$\nabla_{\partial_\mu} \partial_\nu = \Gamma^\lambda{}_{\mu\nu}\, \partial_\lambda, \qquad [\partial_\mu, \partial_\nu] = 0. \tag{269}$$

In an orthonormal basis, or more generally in a non-coordinate basis $\{e_a\}$, we write

$$\nabla_{e_a} e_b = \Gamma^c{}_{ab}\, e_c, \qquad [e_a, e_b] = C^c{}_{ab}\, e_c. \tag{270}$$

The torsion of a connection is defined by

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]. \tag{271}$$

Therefore, its components are:

- in a coordinate basis,
$$T^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\mu\nu} - \Gamma^{\lambda}{}_{\nu\mu}, \tag{272}$$
- in a non-holonomic basis,
$$T^{a}{}_{bc} = \Gamma^{a}{}_{bc} - \Gamma^{a}{}_{cb} - C^{a}{}_{bc}. \tag{273}$$

Geometric intuition

Torsion measures the failure of an infinitesimal parallelogram, constructed using parallel transport, to close. This gives a useful geometric interpretation of the tensor T.

The notion of parallel transport can be extracted directly from the definition of the covariant derivative. For a vector field $A^{\nu}(x)$, we have

$$\nabla_{\mu} A^{\nu} = \partial_{\mu} A^{\nu} + \Gamma^{\nu}{}_{\mu\lambda} A^{\lambda}. \tag{274}$$

We may rewrite this as

$$\begin{aligned} \nabla_{\mu} A^{\nu} &= \lim_{\Delta x^{\mu} \to 0} \frac{A^{\nu}(x + \Delta x) - A^{\nu}(x)}{\Delta x^{\mu}} + \Gamma^{\nu}{}_{\mu\lambda} A^{\lambda} \\ &= \lim_{\Delta x^{\mu} \to 0} \frac{A^{\nu}(x + \Delta x) - A^{\nu}_{T}(x + \Delta x)}{\Delta x^{\mu}}, \end{aligned} \tag{275}$$

where the vector transported parallelly to first order is defined by

$$A^{\nu}_{T}(x + \Delta x) = A^{\nu}(x) - \Gamma^{\nu}{}_{\mu\lambda}(x) A^{\lambda}(x) \Delta x^{\mu}. \tag{276}$$

Thus the connection tells us how a vector must change when transported from one tangent space to a neighboring one. Consider a point P together with two infinitesimal displacements dx^{μ}_{1} and dx^{μ}_{2}. Starting from P, define

$$Q = P + dx_2, \qquad S = P + dx_1. \tag{277}$$

There are now two possible ways of completing the infinitesimal parallelogram:

- transport dx_1 from Q,
- transport dx_2 from S.

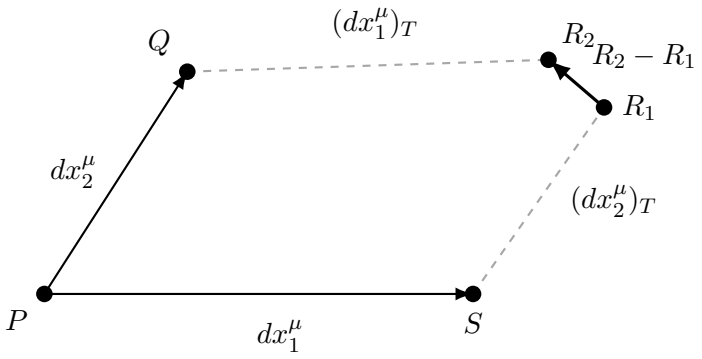


Figure 1: The infinitesimal parallelogram fails to close in the presence of torsion.

To first order in the connection and to second order in the infinitesimal displacements,

$$(dx_1^\mu)_T = dx_1^\mu - \Gamma^\mu{}_{\lambda\nu} dx_1^\nu dx_2^\lambda, \tag{278}$$

while

$$(dx_2^\mu)_T = dx_2^\mu - \Gamma^\mu{}_{\lambda\nu} dx_2^\nu dx_1^\lambda. \tag{279}$$

The construction is represented schematically by

The endpoint obtained by first moving along dx_1 and then transporting dx_2 is

$$\begin{aligned} x^\mu(R_1) &= x^\mu(S) + (dx_2^\mu)_T \\ &= x^\mu(P) + dx_1^\mu + dx_2^\mu - \Gamma^\mu{}_{\lambda\nu} dx_2^\nu dx_1^\lambda. \end{aligned} \tag{280}$$

Similarly,

$$\begin{aligned} x^\mu(R_2) &= x^\mu(Q) + (dx_1^\mu)_T \\ &= x^\mu(P) + dx_2^\mu + dx_1^\mu - \Gamma^\mu{}_{\lambda\nu} dx_1^\nu dx_2^\lambda. \end{aligned} \tag{281}$$

Subtracting,

$$\begin{aligned} x^\mu(R_2) - x^\mu(R_1) &= -\Gamma^\mu{}_{\lambda\nu} dx_1^\nu dx_2^\lambda + \Gamma^\mu{}_{\lambda\nu} dx_2^\nu dx_1^\lambda \\ &= \left(\Gamma^\mu{}_{\lambda\nu} - \Gamma^\mu{}_{\nu\lambda}\right) dx_2^\nu dx_1^\lambda \\ &= T^\mu{}_{\lambda\nu} dx_2^\nu dx_1^\lambda. \end{aligned} \tag{282}$$

Equation (282) shows that torsion measures the failure of an infinitesimal parallelogram constructed through parallel transport to close. Hence,

$$T^\mu{}_{\lambda\nu} \neq 0$$

implies that the two infinitesimal paths terminate at different points.

We now move to an orthonormal frame $\{e_a\}$ satisfying

$$g(e_a, e_b) = \eta_{ab}. \tag{283}$$

Since lengths and angles are preserved in an orthonormal frame, two neighboring orthonormal frames are related by Lorentz transformations. The connection in this basis is defined by

$$\nabla_{e_a} e_b = \Gamma^c{}_{ab}\, e_c \equiv \omega^c{}_b(e_a)\, e_c, \tag{284}$$

where $\omega^c{}_b$ is a one-form called the *Lorentz connection* or *spin connection*.

It can be expanded either in the coordinate coframe or in the orthonormal coframe:

$$\omega^a{}_b = \omega^a{}_{b\mu}\, dx^\mu, \tag{285}$$

or

$$\omega^a{}_b = \omega^a{}_{bc}\, e^c. \tag{286}$$

Suppose that the connection is metric compatible,

$$\nabla g = 0. \tag{287}$$

Then, for any vector field X,

$$0 = (\nabla_X g)(e_a, e_b). \tag{288}$$

Expanding this relation,

$$0 = X\big(g(e_a, e_b)\big) - g(\nabla_X e_a, e_b) - g(e_a, \nabla_X e_b). \tag{289}$$

Because $g(e_a, e_b) = \eta_{ab}$, and η_{ab} is constant,

$$X\big(g(e_a, e_b)\big) = 0. \tag{290}$$

Using

$$\nabla_X e_a = \omega^c{}_a(X) e_c, \tag{291}$$

we find

$$\begin{aligned} 0 &= -\omega^c{}_a(X)\eta_{cb} - \omega^c{}_b(X)\eta_{ac} \\ &= -\omega_{ba}(X) - \omega_{ab}(X). \end{aligned} \tag{292}$$

Therefore,

$$\boxed{\omega_{ab} = -\omega_{ba}.} \tag{293}$$

Lorentz-algebra-valued connection

The antisymmetry condition

$$\omega_{ab} = -\omega_{ba}$$

means that the spin connection takes values in the Lorentz Lie algebra $\mathfrak{so}(d-1,1)$. This is why ω^{ab} is called the Lorentz connection.

Let $V = V^b e_b$ be a vector expressed in an orthonormal frame. Then

$$\begin{aligned} \nabla_a(V^b e_b) &= \nabla_a(V^b) e_b + V^b \nabla_a e_b \\ &= \partial_a(V^b) e_b + V^b \omega^c{}_b(e_a) e_c \\ &= \Big[\partial_a(V^c) + \omega^c{}_b(e_a) V^b\Big]\, e_c. \end{aligned} \tag{294}$$

Hence,

$$\nabla_a V^b = \partial_a(V^b) + \omega^b{}_c(e_a)V^c. \tag{295}$$

Multiplying by the coframe e^a, we obtain

$$\begin{aligned} e^a \nabla_a V^b &= e^a \partial_a V^b + e^a \omega^b{}_{ca} V^c \\ &= dV^b + \omega^b_c V^c \,. \end{aligned}$$

This motivates the definition of the exterior covariant derivative

$$\boxed{DV^b = dV^b + \omega^b{}_c V^c.} \tag{296}$$

More generally, for a Lorentz tensor

$$M^{a_1 \cdots a_n}{}_{b_1 \cdots b_m}, \tag{297}$$

the exterior covariant derivative acts with one connection term for each Lorentz index:

$$\begin{aligned} DM^{a_1 \cdots a_n}{}_{b_1 \cdots b_m} = {} & dM^{a_1 \cdots a_n}{}_{b_1 \cdots b_m} \\ & + \omega^{a_1}{}_c M^{c a_2 \cdots a_n}{}_{b_1 \cdots b_m} + \cdots + \omega^{a_n}{}_c M^{a_1 \cdots a_{n-1} c}{}_{b_1 \cdots b_m} \\ & - \omega^c{}_{b_1} M^{a_1 \cdots a_n}{}_{c b_2 \cdots b_m} - \cdots - \omega^c{}_{b_m} M^{a_1 \cdots a_n}{}_{b_1 \cdots b_{m-1} c}. \end{aligned} \tag{298}$$

Upper Lorentz indices therefore contribute with a plus sign, while lower Lorentz indices contribute with a minus sign.

Under a local Lorentz transformation,

$$V'^a = \Lambda^a{}_b(x) V^b. \tag{299}$$

We require the covariant derivative to transform covariantly:

$$DV'^a = \Lambda^a{}_b DV^b. \tag{300}$$

This condition determines the transformation law of the spin connection:

$$\omega' = \Lambda \omega \Lambda^{-1} + \Lambda d\Lambda^{-1}. \tag{301}$$

In components,

$$\omega'^a{}_b = \Lambda^a{}_c \omega^c{}_d (\Lambda^{-1})^d{}_b + \Lambda^a{}_c d(\Lambda^{-1})^c{}_b \,, \tag{302}$$

and the inhomogeneous term is characteristic of a gauge connection.

Analogy with gauge theory

The Lorentz connection behaves exactly as a gauge field for local Lorentz transformations. Schematically,

$$\omega' = \Lambda \omega \Lambda^{-1} + \Lambda d\Lambda^{-1}.$$

This will later make the gauge-theoretic structure of first-order gravity particularly transparent.

We now want to express the torsion entirely in terms of the coframe e^a and the Lorentz connection $\omega^a{}_b$. For a one-form α,

$$d\alpha(X,Y) = X\big(\alpha(Y)\big) - Y\big(\alpha(X)\big) - \alpha([X,Y]). \tag{303}$$

For two one-forms α and β,

$$(\alpha \wedge \beta)(X,Y) = \alpha(X)\beta(Y) - \alpha(Y)\beta(X). \tag{304}$$

Let

$$T(X,Y) = T^a(X,Y)e_a. \tag{305}$$

Then

$$T^a(X,Y) = e^a(T(X,Y)). \tag{306}$$

Using the definition of torsion,

$$T^a(X,Y) = e^a\left(\nabla_X Y - \nabla_Y X - [X,Y]\right). \tag{307}$$

Now use the identity

$$X\big(e^a(Y)\big) = (\nabla_X e^a)(Y) + e^a(\nabla_X Y). \tag{308}$$

Since the connection acts on the dual frame as

$$\nabla_X e^a = -\omega^a{}_b(X)e^b, \tag{309}$$

we obtain

$$e^a(\nabla_X Y) = X\big(e^a(Y)\big) + \omega^a{}_b(X)e^b(Y). \tag{310}$$

Similarly,

$$e^a(\nabla_Y X) = Y\big(e^a(X)\big) + \omega^a{}_b(Y)e^b(X). \tag{311}$$

Therefore,

$$\begin{aligned} T^a(X,Y) = {} & X\big(e^a(Y)\big) - Y\big(e^a(X)\big) - e^a([X,Y]) \\ & + \omega^a{}_b(X)e^b(Y) - \omega^a{}_b(Y)e^b(X). \end{aligned} \tag{312}$$

Using (303),

$$X\big(e^a(Y)\big) - Y\big(e^a(X)\big) - e^a([X,Y]) = de^a(X,Y), \tag{313}$$

while from (304),

$$\omega^a{}_b(X)e^b(Y) - \omega^a{}_b(Y)e^b(X) = (\omega^a{}_b \wedge e^b)(X,Y). \tag{314}$$

Hence,

$$T^a(X,Y) = \left(de^a + \omega^a{}_b \wedge e^b\right)(X,Y). \tag{315}$$

Since this holds for arbitrary vector fields X and Y, we conclude that

$$\boxed{T^a = de^a + \omega^a{}_b \wedge e^b.} \tag{316}$$

First Cartan structure equation

The equation

$$\boxed{T^a = de^a + \omega^a{}_b \wedge e^b}$$

is Cartan's first structure equation.
It expresses the torsion two-form in terms of the vielbein and the Lorentz connection.

This equation is fundamental in the first-order formulation of gravity. The fundamental variables are the vielbein and the spin connection, which are initially treated as independent fields.

If the torsion vanishes $T^a = 0$, then Cartan's first structure equation becomes

$$de^a + \omega^a{}_b \wedge e^b = 0. \tag{317}$$

For an invertible vielbein, this equation determines the torsion-free spin connection $\omega^{ab}(e)$.

Thus the Levi–Civita connection can be recovered from the first-order formalism rather than being assumed from the beginning.

We begin by recalling the definition of torsion introduced in the previous class. Given a Koszul connection ∇, the torsion is defined by

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]. \tag{318}$$

Since $T(X,Y)$ is a vector field, it can be expanded in an orthonormal frame $\{e_a\}$ as

$$T(X,Y) = T^a(X,Y)e_a. \tag{319}$$

Acting with the dual coframe e^a, we obtain

$$T^a(X,Y) = e^a\big(T(X,Y)\big). \tag{320}$$

The torsion is therefore a vector-valued 2-form,

$$T \in \Omega^2(M, TM), \tag{321}$$

or equivalently a tensor of type $(1,2)$.

Let's consider two vector fields $X = X^a e_a\,, Y = Y^a e_a$.Then

$$T^a(X,Y) = e^a\left(\nabla_X Y - \nabla_Y X - [X,Y]\right). \tag{322}$$

Using

$$\nabla_X e_b = \omega^c{}_b(X) e_c, \tag{323}$$

we find

$$\begin{aligned} e^a(\nabla_X Y) &= e^a\left[\nabla_X (Y^b e_b)\right] \\ &= X(Y^a) + \omega^a{}_b(X) Y^b, \end{aligned} \tag{324}$$

and similarly

$$e^a(\nabla_Y X) = Y(X^a) + \omega^a{}_b(Y)X^b. \tag{325}$$

Therefore,

$$\begin{aligned} T^a(X,Y) = X(Y^a) - Y(X^a) - e^a([X,Y]) \\ + \omega^a{}_b(X)Y^b - \omega^a{}_b(Y)X^b. \end{aligned} \tag{326}$$

On the other hand, from the definition of the exterior derivative of a one-form,

$$\begin{aligned} de^a(X,Y) &= X\big(e^a(Y)\big) - Y\big(e^a(X)\big) - e^a([X,Y]) \\ &= X(Y^a) - Y(X^a) - e^a([X,Y]). \end{aligned} \tag{327}$$

Similarly,

$$\begin{aligned} (\omega^a{}_b \wedge e^b)(X,Y) &= \omega^a{}_b(X)e^b(Y) - \omega^a{}_b(Y)e^b(X) \\ &= \omega^a{}_b(X)Y^b - \omega^a{}_b(Y)X^b. \end{aligned} \tag{328}$$

Comparing with (326), we obtain

$$T^a(X,Y) = \left(de^a + \omega^a{}_b \wedge e^b\right)(X,Y). \tag{329}$$

Since this relation holds for arbitrary X and Y,

$$\boxed{T^a = de^a + \omega^a{}_b \wedge e^b = De^a.} \tag{330}$$

Thus the torsion is naturally a Lorentz-vector-valued 2-form. It may be expanded either in a coordinate basis,

$$T^a = \frac{1}{2} T^a{}_{\mu\nu}\, dx^\mu \wedge dx^\nu, \tag{331}$$

or in an orthonormal basis,

$$T^a = \frac{1}{2} T^a{}_{bc}\, e^b \wedge e^c. \tag{332}$$

The two sets of components are related by

$$T^a{}_{\mu\nu} = e^b{}_\mu e^c{}_\nu T^a{}_{bc}. \tag{333}$$

The Vielbein Postulate

Before introducing curvature, it is useful to establish explicitly the relation between the connection coefficients acting on coordinate indices and the Lorentz connection acting on internal indices.

Recall that

$$\nabla_{\partial_\mu}\partial_\nu = \Gamma^\rho{}_{\mu\nu}\partial_\rho, \tag{334}$$

while in the orthonormal frame

$$\nabla_\mu e_a = \omega^b{}_{a\mu} e_b. \tag{335}$$

A vector can be written either as

$$V = V^\mu \partial_\mu = V^a e_a, \tag{336}$$

with

$$V^a = e^a{}_\lambda V^\lambda. \tag{337}$$

We want the covariant derivative to represent the same geometrical object independently of the basis used. We therefore impose

$$\boxed{D_\mu V^a = e^a{}_\lambda \nabla_\mu V^\lambda.} \tag{338}$$

Let us examine this relation explicitly. The Lorentz-covariant derivative is

$$D_\mu V^a = \partial_\mu V^a + \omega^a{}_{b\mu} V^b. \tag{339}$$

Using $V^a = e^a{}_\lambda V^\lambda$,

$$\begin{aligned} D_\mu V^a &= \partial_\mu \left(e^a{}_\lambda V^\lambda \right) + \omega^a{}_{b\mu} e^b{}_\lambda V^\lambda \\ &= e^a{}_\lambda \partial_\mu V^\lambda + \left(\partial_\mu e^a{}_\lambda + \omega^a{}_{b\mu} e^b{}_\lambda \right) V^\lambda. \end{aligned} \tag{340}$$

On the other hand,

$$\begin{aligned} e^a{}_\rho \nabla_\mu V^\rho &= e^a{}_\rho \left(\partial_\mu V^\rho + \Gamma^\rho{}_{\mu\lambda} V^\lambda \right) \\ &= e^a{}_\rho \partial_\mu V^\rho + e^a{}_\rho \Gamma^\rho{}_{\mu\lambda} V^\lambda. \end{aligned} \tag{341}$$

Equating (340) and (341) for an arbitrary vector V^λ gives

$$\partial_\mu e^a{}_\lambda + \omega^a{}_{b\mu} e^b{}_\lambda - \Gamma^\rho{}_{\mu\lambda} e^a{}_\rho = 0. \tag{342}$$

This equation is usually written compactly as

$$\boxed{D_\mu e^a{}_\lambda = 0.} \tag{343}$$

Vielbein postulate

The relation

$$D_\mu e^a{}_\nu = \partial_\mu e^a{}_\nu + \omega^a{}_{b\mu} e^b{}_\nu - \Gamma^\rho{}_{\mu\nu} e^a{}_\rho = 0$$

is known as the *vielbein postulate* or *tetrad postulate*. It expresses the compatibility between the description using spacetime indices $\mu, \nu, \ldots$ and the description using Lorentz indices $a, b, \ldots$.

The tetrad postulate allows us to translate between the affine connection $\Gamma^\rho{}_{\mu\nu}$ and the Lorentz connection $\omega^a{}_{b\mu}$.

We can see that, multiplying (342) by the inverse vielbein $e_a{}^\rho$, we obtain

$$\Gamma^\rho{}_{\mu\nu} = e_a{}^\rho \left(\partial_\mu e^a{}_\nu + \omega^a{}_{b\mu} e^b{}_\nu \right). \tag{344}$$

Conversely,

$$\omega^a{}_{b\mu} = e^a{}_\rho \left(\partial_\mu e_b{}^\rho + \Gamma^\rho{}_{\mu\nu} e_b{}^\nu \right). \tag{345}$$

Two descriptions of the same connection

The quantities

$$\Gamma^\rho{}_{\mu\nu} \qquad \text{and} \qquad \omega^a{}_{b\mu}$$

are not two unrelated connections. Once a vielbein is chosen, they are two representations of the same geometrical connection: one acts on coordinate indices and the other on Lorentz indices.

Let us now verify explicitly that Cartan's first structure equation reproduces the familiar coordinate expression for torsion. Starting from

$$T^a = de^a + \omega^a{}_b \wedge e^b, \tag{346}$$

and writing $e^a = e^a{}_\mu dx^\mu \, , \omega^a{}_b = \omega^a{}_{b\mu} dx^\mu$ we find

$$\begin{aligned} de^a &= \partial_\mu e^a{}_\nu \, dx^\mu \wedge dx^\nu \\ &= \frac{1}{2} \left(\partial_\mu e^a{}_\nu - \partial_\nu e^a{}_\mu \right) dx^\mu \wedge dx^\nu, \end{aligned} \tag{347}$$

and

$$\begin{aligned} \omega^a{}_b \wedge e^b &= \omega^a{}_{b\mu} e^b{}_\nu dx^\mu \wedge dx^\nu \\ &= \frac{1}{2} \left(\omega^a{}_{b\mu} e^b{}_\nu - \omega^a{}_{b\nu} e^b{}_\mu \right) dx^\mu \wedge dx^\nu. \end{aligned} \tag{348}$$

Therefore,

$$T^a = \frac{1}{2} T^a{}_{\mu\nu} dx^\mu \wedge dx^\nu, \tag{349}$$

where

$$T^a{}_{\mu\nu} = 2\partial_{[\mu} e^a{}_{\nu]} + 2\omega^a{}_{b[\mu} e^b{}_{\nu]}. \tag{350}$$

Using the vielbein postulate,

$$\partial_\mu e^a{}_\nu + \omega^a{}_{b\mu} e^b{}_\nu = \Gamma^\rho{}_{\mu\nu} e^a{}_\rho, \tag{351}$$

we immediately obtain

$$\begin{aligned} T^a{}_{\mu\nu} &= e^a{}_\rho \left(\Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu} \right) \\ &= e^a{}_\rho T^\rho{}_{\mu\nu}. \end{aligned} \tag{352}$$

Thus

$$\boxed{T^a = \frac{1}{2} e^a{}_\rho T^\rho{}_{\mu\nu} \, dx^\mu \wedge dx^\nu.} \tag{353}$$

For a coordinate basis,

$$T^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu}. \tag{354}$$

This verifies explicitly the equivalence between the coordinate and vielbein descriptions of torsion.

Curvature and Cartan's Second Structure Equation

We now introduce curvature at the level of a general Koszul connection.

Riemann curvature

The curvature associated with a connection ∇ is the map

$$R : \mathfrak{X}(M) \times \mathfrak{X}(M) \times \mathfrak{X}(M) \longrightarrow \mathfrak{X}(M),$$

defined by

$$\boxed{R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.} \tag{355}$$

Equivalently,

$$R(X,Y) = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}. \tag{356}$$

The subtraction of the term $\nabla_{[X,Y]}$ is essential: it removes the contribution associated purely with the non-commutativity of the vector fields X and Y.

The map $R(X,Y)Z$ is $C^\infty(M)$-linear in each of its three vector arguments. Hence it defines a tensor of type $(1,3)$,

$$R \in \Gamma\left(TM \otimes T^*M \otimes T^*M \otimes T^*M\right). \tag{357}$$

Furthermore,

$$R(X,Y)Z = -R(Y,X)Z\,. \tag{358}$$

Thus the curvature tensor is antisymmetric in its first two vector arguments. Consider

$$\nabla_{e_a} e_b = \Gamma^c{}_{ab} e_c\,, \tag{359}$$

and let the frame be possibly non-holonomic,

$$[e_a, e_b] = C^c{}_{ab} e_c\,. \tag{360}$$

We define the components of the curvature by

$$R(e_a, e_b)e_c = R^d{}_{cab}\, e_d. \tag{361}$$

Using (355),

$$\begin{aligned} \nabla_{e_a} \nabla_{e_b} e_c &= \nabla_{e_a} \left(\Gamma^d{}_{bc} e_d\right) \\ &= e_a(\Gamma^d{}_{bc}) e_d + \Gamma^d{}_{bc} \Gamma^m{}_{ad} e_m, \end{aligned} \tag{362}$$

while

$$\nabla_{e_b} \nabla_{e_a} e_c = e_b(\Gamma^d{}_{ac}) e_d + \Gamma^d{}_{ac} \Gamma^m{}_{bd} e_m. \tag{363}$$

Finally,

$$\begin{aligned} \nabla_{[e_a, e_b]} e_c &= C^m{}_{ab} \nabla_{e_m} e_c \\ &= C^m{}_{ab} \Gamma^d{}_{mc} e_d. \end{aligned} \tag{364}$$

Collecting all terms gives

$$\begin{aligned} R^d{}_{cab} = e_a(\Gamma^d{}_{bc}) - e_b(\Gamma^d{}_{ac}) \\ + \Gamma^m{}_{bc}\Gamma^d{}_{am} - \Gamma^m{}_{ac}\Gamma^d{}_{bm} - C^m{}_{ab}\Gamma^d{}_{mc}. \end{aligned} \tag{365}$$

The final term is present because the basis need not be holonomic. For a coordinate basis,

$$C^\rho{}_{\mu\nu} = 0, \tag{366}$$

and the familiar expression is recovered:

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho{}_{\nu\sigma} - \partial_\nu\Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}. \tag{367}$$

Just as torsion can be described as a vector-valued 2-form, the curvature of the Lorentz connection can be encoded in the *curvature two-form*

$$R^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b\,, \tag{368}$$

and this is Cartan's second structure equation. The curvature two-form can be expanded in either basis:

$$R^a{}_b = \frac{1}{2}R^a{}_{b\mu\nu}\, dx^\mu \wedge dx^\nu, \tag{369}$$

or

$$R^a{}_b = \frac{1}{2}R^a{}_{bcd}\, e^c \wedge e^d. \tag{370}$$

The coordinate and Lorentz components are related through the vielbein

$$R^a{}_{b\mu\nu} = e^a{}_\rho e_b{}^\sigma R^\rho{}_{\sigma\mu\nu}. \tag{371}$$

Cartan's two structure equations

At this point we have obtained the two fundamental field strengths associated with the first-order geometrical variables:

$$\boxed{T^a = de^a + \omega^a{}_b \wedge e^b}$$

and

$$\boxed{R^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b.}$$

The first describes torsion, while the second describes curvature.

Bianchi Identities

The torsion and curvature are not independent. Their definitions imply differential identities known as the *Bianchi identities*. Let us begin with

$$T^a = de^a + \omega^a{}_b \wedge e^b. \tag{372}$$

Taking the exterior covariant derivative,

$$DT^a = dT^a + \omega^a{}_b \wedge T^b. \tag{373}$$

Substituting the first Cartan structure equation,

$$DT^a = d\left(de^a + \omega^a{}_b \wedge e^b\right) + \omega^a{}_b \wedge \left(de^b + \omega^b{}_c \wedge e^c\right). \tag{374}$$

Using $d^2 = 0$ and the graded Leibniz rule,

$$\begin{aligned} DT^a = {} & d\omega^a{}_b \wedge e^b - \omega^a{}_b \wedge de^b + \omega^a{}_b \wedge de^b \\ & + \omega^a{}_b \wedge \omega^b{}_c \wedge e^c. \end{aligned} \tag{375}$$

The middle terms cancel, giving

$$DT^a = (d\omega^a{}_b \omega^a{}_c \wedge \omega^c{}_b) \wedge e^b. \tag{376}$$

Recognizing the curvature two-form,

$$DT^a = R^a{}_b \wedge e^b. \tag{377}$$

This is the *first Bianchi identity* in Cartan form. Suppose now that $T^a = 0$, then the first Bianchi identity implies

$$R^a{}_b \wedge e^b = 0. \tag{378}$$

Using

$$R^a{}_b = \frac{1}{2} R^a{}_{bcd} e^c \wedge e^d, \tag{379}$$

we obtain

$$\frac{1}{2} R^a{}_{bcd} e^c \wedge e^d \wedge e^b = 0. \tag{380}$$

Since the wedge product is totally antisymmetric,

$$R^a{}_{[bcd]} = 0. \tag{381}$$

In a coordinate basis this becomes

$$R^\rho{}_{[\sigma\mu\nu]} = 0, \tag{382}$$

or equivalently,

$$R^\rho{}_{\sigma\mu\nu} + R^\rho{}_{\mu\nu\sigma} + R^\rho{}_{\nu\sigma\mu} = 0. \tag{383}$$

This is the usual algebraic or cyclic Bianchi identity.

Role of torsion

The relation

$$R^\rho{}_{[\sigma\mu\nu]} = 0$$

is the torsion-free form of the first Bianchi identity. The more general identity is

$$DT^a = R^a{}_b \wedge e^b,$$

which remains valid when torsion is nonzero.

There is a second identity involving only the curvature:

$$DR^a{}_b = 0. \tag{384}$$

To see this, it is convenient to regard ω and R as matrix-valued differential forms. Then

$$R = d\omega + \omega \wedge \omega. \tag{385}$$

The exterior covariant derivative of the curvature is

$$DR = dR + \omega \wedge R - R \wedge \omega. \tag{386}$$

Now,

$$\begin{aligned} dR &= d(d\omega + \omega \wedge \omega) \\ &= d\omega \wedge \omega - \omega \wedge d\omega, \end{aligned} \tag{387}$$

where we used $d^2 = 0$. Furthermore,

$$\begin{aligned} \omega \wedge R - R \wedge \omega = \omega \wedge d\omega + \omega \wedge \omega \wedge \omega \\ - d\omega \wedge \omega - \omega \wedge \omega \wedge \omega. \end{aligned} \tag{388}$$

Therefore all terms cancel and

$$DR = 0. \tag{389}$$

In Lorentz indices,

$$DR^a{}_b = dR^a{}_b + \omega^a{}_c \wedge R^c{}_b - \omega^c{}_b \wedge R^a{}_c = 0. \tag{390}$$

In the usual coordinate notation this is equivalent, in the torsion-free case, to

$$\nabla_{[\lambda} R^{\rho}{}_{|\sigma|\mu\nu]} = 0. \tag{391}$$

Contorsion and the Torsion-Free Spin Connection

We have seen that a metric-compatible Lorentz connection satisfies

$$D\eta_{ab} = 0, \tag{392}$$

or equivalently,

$$\omega_{ab} = -\omega_{ba}. \tag{393}$$

Metric compatibility, however, does not require the connection to be torsion-free. It is therefore useful to separate a general metric-compatible spin connection into a torsion-free part and a torsional contribution.

Let ω^{ab} be a general metric-compatible spin connection. We decompose it as

$$\omega^{ab} = \mathring{\omega}^{ab} + K^{ab}, \tag{394}$$

where $\mathring{\omega}^{ab}$ denotes the torsion-free spin connection, defined by

$$de^a + \mathring{\omega}^a{}_b \wedge e^b = 0, \tag{395}$$

and K^{ab} is called the *contorsion one-form*.

Since both ω^{ab} and $\mathring{\omega}^{ab}$ are metric-compatible,

$$\omega_{ab} = -\omega_{ba}, \qquad \mathring{\omega}_{ab} = -\mathring{\omega}_{ba}, \tag{396}$$

and therefore

$$K_{ab} = -K_{ba}. \tag{397}$$

The contorsion can be expanded either in the coordinate basis or in the orthonormal frame:

$$K^{ab} = K^{ab}{}_\mu \, dx^\mu = K^{ab}{}_c \, e^c. \tag{398}$$

Why the contorsion is a tensor

A connection transforms inhomogeneously under a local Lorentz transformation. Since K is the difference of two connections,

$$K = \omega - \mathring{\omega},$$

the inhomogeneous pieces cancel. Consequently,

$$K' = \Lambda K \Lambda^{-1}.$$

Thus the contorsion is a Lorentz-tensor-valued one-form rather than a connection.

Now, we can write the torsion in terms of the contorstion. Starting from Cartan's first structure equation,

$$T^a = de^a + \omega^a{}_b \wedge e^b, \tag{399}$$

and substituting

$$\omega^a{}_b = \mathring{\omega}^a{}_b + K^a{}_b, \tag{400}$$

we obtain

$$T^a = de^a + \mathring{\omega}^a{}_b \wedge e^b + K^a{}_b \wedge e^b. \tag{401}$$

The first two terms vanish by definition of the torsion-free connection, so

$$T^a = K^a{}_b \wedge e^b. \tag{402}$$

Thus all the torsion of the general connection is encoded in the contorsion.

To invert (402), it is useful to introduce the interior product

$$\iota_a \equiv \iota_{e_a}. \tag{403}$$

We first recall a useful algebraic identity. Suppose that Q_a is a set of two-forms and $L_{ab} = -L_{ba}$ is a set of one-forms satisfying

$$Q_a + L_{ab} \wedge e^b = 0. \tag{404}$$

Then

$$L_{ab} = -\iota_{[a} Q_{b]} + \frac{1}{2} \left(\iota_a \iota_b Q_c \right) e^c. \tag{405}$$

Here

$$\iota_{[a} Q_{b]} = \frac{1}{2} \left(\iota_a Q_b - \iota_b Q_a \right). \tag{406}$$

Applying this identity to

$$T_a - K_{ab} \wedge e^b = 0, \tag{407}$$

we identify

$$Q_a = T_a, \qquad L_{ab} = -K_{ab}, \tag{408}$$

and therefore obtain

$$K_{ab} = \iota_{[a} T_{b]} - \frac{1}{2} \left(\iota_a \iota_b T_c \right) e^c. \tag{409}$$

This gives the contorsion directly in terms of the torsion two-form. Now we write the torsion as

$$T_a = \frac{1}{2} T_{abc}\, e^b \wedge e^c, \qquad T_{abc} = -T_{acb}. \tag{410}$$

Then

$$\iota_a T_b = T_{bac} e^c. \tag{411}$$

Furthermore,

$$\iota_a \iota_b T_c = T_{cba}. \tag{412}$$

Hence (409) becomes

$$K_{ab} = \frac{1}{2} \left(T_{bac} - T_{abc} - T_{cba} \right) e^c. \tag{413}$$

Using the antisymmetry of torsion in its final two indices,

$$T_{cba} = -T_{cab}, \tag{414}$$

we may equivalently write

$$K_{abc} = \frac{1}{2} \left(T_{bac} - T_{abc} + T_{cab} \right). \tag{415}$$

Contorsion

For a metric-compatible connection,

$$\boxed{\omega^{ab} = \mathring{\omega}^{ab} + K^{ab}}$$

separates the connection into a torsion-free part and a tensorial torsional part. Moreover,

$$T^a = K^a{}_b \wedge e^b.$$

Thus K^{ab} contains precisely the information carried by torsion.

We can apply essentially the same argument directly to the torsion-free equation

$$de^a + \mathring{\omega}^a{}_b \wedge e^b = 0. \tag{416}$$

Introduce the coefficients of anholonomy $\Omega^a{}_{bc}$ through

$$de^a = \frac{1}{2}\Omega^a{}_{bc}\, e^b \wedge e^c, \tag{417}$$

where

$$\Omega^a{}_{bc} = -\Omega^a{}_{cb}. \tag{418}$$

Using (405), with

$$Q_a = de_a, \qquad L_{ab} = \mathring{\omega}_{ab}, \tag{419}$$

we obtain

$$\mathring{\omega}_{ab} = -\iota_{[a} de_{b]} + \frac{1}{2}\left(\iota_a \iota_b de_c\right) e^c. \tag{420}$$

In terms of the anholonomy coefficients,

$$\mathring{\omega}_{abc} = \frac{1}{2}\left(\Omega_{abc} + \Omega_{bca} - \Omega_{cab}\right). \tag{421}$$

$$\mathring{\omega}_{ab} = \frac{1}{2}\left(\Omega_{abc} + \Omega_{bca} - \Omega_{cab}\right) e^c. \tag{422}$$

The precise signs in this expression depend on the convention chosen in (417). Here we have defined $de^a = +\frac{1}{2}\Omega^a{}_{bc} e^b \wedge e^c$. In order to recover the Levi-Civita connection, we can now use the vielbein postulate,

$$D_\mu e^a{}_\nu = \partial_\mu e^a{}_\nu + \mathring{\omega}^a{}_{b\mu} e^b{}_\nu - \mathring{\Gamma}^\rho{}_{\mu\nu} e^a{}_\rho = 0. \tag{423}$$

Let $E_a{}^\mu$ denote the inverse vielbein,

$$E_a{}^\mu e^a{}_\nu = \delta^\mu{}_\nu, \qquad E_a{}^\mu e^b{}_\mu = \delta_a{}^b. \tag{424}$$

Multiplying the vielbein postulate by $E_a{}^\rho$ gives

$$\mathring{\Gamma}^\rho{}_{\mu\nu} = E_a{}^\rho \left(\partial_\mu e^a{}_\nu + \mathring{\omega}^a{}_{b\mu} e^b{}_\nu\right). \tag{425}$$

Since $\mathring{\omega}^{ab}$ is both metric-compatible and torsion-free, the resulting affine connection is precisely the Levi–Civita connection:

$$\mathring{\Gamma}^\lambda{}_{\mu\nu} = \frac{1}{2} g^{\lambda\rho}\left(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\right). \tag{426}$$

An important consistency check

Starting from the torsion-free Cartan equation

$$de^a + \mathring{\omega}^a{}_b \wedge e^b = 0$$

and using the vielbein postulate reproduces the ordinary Christoffel symbols.
Thus the metric and vielbein descriptions of the Levi–Civita connection are completely equivalent.

Curvature in the Presence of Torsion

The curvature two-form of the full connection is

$$R^{ab}(\omega) = d\omega^{ab} + \omega^{a}{}_{c} \wedge \omega^{cb}. \tag{427}$$

Substituting

$$\omega^{ab} = \mathring{\omega}^{ab} + K^{ab}, \tag{428}$$

gives

$$\begin{aligned} R^{ab} = {} & d\mathring{\omega}^{ab} + dK^{ab} \\ & + (\mathring{\omega}^{a}{}_{c} + K^{a}{}_{c}) \wedge \left(\mathring{\omega}^{cb} + K^{cb}\right). \end{aligned} \tag{429}$$

Expanding,

$$\begin{aligned} R^{ab} = {} & d\mathring{\omega}^{ab} + \mathring{\omega}^{a}{}_{c} \wedge \mathring{\omega}^{cb} \\ & + dK^{ab} + \mathring{\omega}^{a}{}_{c} \wedge K^{cb} + K^{a}{}_{c} \wedge \mathring{\omega}^{cb} \\ & + K^{a}{}_{c} \wedge K^{cb}. \end{aligned} \tag{430}$$

The first line is the curvature of the torsion-free connection,

$$\mathring{R}^{ab} = d\mathring{\omega}^{ab} + \mathring{\omega}^{a}{}_{c} \wedge \mathring{\omega}^{cb}. \tag{431}$$

The middle terms define the exterior covariant derivative with respect to $\mathring{\omega}$,

$$\mathring{D}K^{ab} = dK^{ab} + \mathring{\omega}^{a}{}_{c} \wedge K^{cb} + \mathring{\omega}^{b}{}_{c} \wedge K^{ac}. \tag{432}$$

Consequently,

$$R^{ab} = \mathring{R}^{ab} + \mathring{D}K^{ab} + K^{a}{}_{c} \wedge K^{cb}. \tag{433}$$

Curvature decomposition

The decomposition

$$\omega = \mathring{\omega} + K$$

induces

$$\boxed{R(\omega) = R(\mathring{\omega}) + \mathring{D}K + K \wedge K.}$$

Thus torsion modifies the Riemannian curvature through both a term linear in the contorsion and a term quadratic in the contorsion.

If the contorsion is small, for example

$$K^{ab} = \varepsilon K^{ab}_{(1)}, \tag{434}$$

then

$$R^{ab} = \mathring{R}^{ab} + \varepsilon\, \mathring{D} K^{ab}_{(1)} + \varepsilon^2 K^{a}_{(1)c} \wedge K^{cb}_{(1)}. \tag{435}$$

Hence, to linear order,

$$\delta R^{ab} = \mathring{D} K^{ab}. \tag{436}$$

This is analogous to the familiar decomposition of a metric into a background plus a perturbation,

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \varepsilon h_{\mu\nu}. \tag{437}$$

The curvature two-form may be expanded as

$$\mathring{R}^{ab} = \frac{1}{2} \mathring{R}^{ab}{}_{\mu\nu} dx^\mu \wedge dx^\nu. \tag{438}$$

The Lorentz components are related to the usual Riemann tensor by

$$\mathring{R}^{ab}{}_{\mu\nu} = e^a{}_\rho E^{b\sigma} \mathring{R}^\rho{}_{\sigma\mu\nu}. \tag{439}$$

$$\mathring{R}^{ab} = \frac{1}{2} e^a{}_\rho E^{b\sigma} \mathring{R}^\rho{}_{\sigma\mu\nu} dx^\mu \wedge dx^\nu. \tag{440}$$

In terms of the torsion-free spin connection,

$$\begin{aligned} \mathring{R}^{ab}{}_{\mu\nu} = {} & \partial_\mu \mathring{\omega}^{ab}{}_\nu - \partial_\nu \mathring{\omega}^{ab}{}_\mu \\ & + \mathring{\omega}^a{}_{c\mu} \mathring{\omega}^{cb}{}_\nu - \mathring{\omega}^a{}_{c\nu} \mathring{\omega}^{cb}{}_\mu. \end{aligned} \tag{441}$$

Invariant Tensors of the Lorentz Algebra

We now introduce a concept that will be particularly useful when constructing Lorentz-invariant gravitational actions. Let $\mathfrak{g}$ be a Lie algebra with generators X_A satisfying

$$[X_A, X_B] = C^C{}_{AB} X_C, \tag{442}$$

where $C^C{}_{AB}$ are the structure constants. A rank-n tensor

$$\Delta_{A_1 \cdots A_n} \tag{443}$$

is said to be an *invariant tensor* of the Lie algebra if it satisfies

$$\sum_{i=1}^{n} C^F{}_{BA_i} \Delta_{A_1 \cdots A_{i-1} F A_{i+1} \cdots A_n} = 0. \tag{444}$$

Explicitly,

$$\begin{aligned} 0 = {} & C^F{}_{BA_1} \Delta_{F A_2 \cdots A_n} + C^F{}_{BA_2} \Delta_{A_1 F \cdots A_n} + \cdots \\ & + C^F{}_{BA_n} \Delta_{A_1 \cdots A_{n-1} F}. \end{aligned} \tag{445}$$

This is simply the infinitesimal statement that Δ is unchanged under the action of the Lie group.

For the Lorentz group $SO(d-1,1)$, two particularly important invariant tensors are η_{ab} and $\epsilon_{a_1\cdots a_d}$. The first is the Minkowski metric in the tangent space, while the second is the totally antisymmetric Levi–Civita symbol.

These are precisely the tensors that allow Lorentz indices to be contracted when constructing gravitational actions. For example, in four dimensions the Einstein–Cartan action involves

$$\epsilon_{abcd}\, e^a \wedge e^b \wedge R^{cd}. \tag{446}$$

Another important invariant bilinear form of a Lie algebra is the Killing form,

$$\kappa_{AB} = C^C{}_{AD} C^D{}_{BC}. \tag{447}$$

For the Lorentz algebra, whose generators may be written as J_{ab}, the Killing form is proportional to

$$\kappa_{ab,cd} \propto \eta_{ac}\eta_{bd} - \eta_{ad}\eta_{bc}. \tag{448}$$

Thus the tangent-space Minkowski metric ultimately provides the invariant quadratic structure on the Lorentz algebra.

A useful property of invariant tensors is that they are covariantly constant with respect to a connection valued in the corresponding Lie algebra.

For the Lorentz metric,

$$D\eta_{ab} = d\eta_{ab} - \omega^c{}_a \eta_{cb} - \omega^c{}_b \eta_{ac}. \tag{449}$$

Since η_{ab} has constant components,

$$d\eta_{ab} = 0, \tag{450}$$

and therefore

$$\begin{aligned} D\eta_{ab} &= -\omega_{ba} - \omega_{ab} \\ &= 0, \end{aligned} \tag{451}$$

where metric compatibility was used. Hence,

$$D\eta_{ab} = 0. \tag{452}$$

The same property holds for the Levi–Civita tensor:

$$D\epsilon_{a_1\cdots a_d} = 0. \tag{453}$$

Let us verify this explicitly.

Since the components of $\epsilon_{a_1\cdots a_d}$ are constant,

$$d\epsilon_{a_1\cdots a_d} = 0. \tag{454}$$

The Lorentz-covariant derivative therefore gives

$$\begin{aligned} D\epsilon_{a_1\cdots a_d} = & -\omega^{b}{}_{a_1}\epsilon_{ba_2\cdots a_d} - \omega^{b}{}_{a_2}\epsilon_{a_1 ba_3\cdots a_d} \\ & - \cdots - \omega^{b}{}_{a_d}\epsilon_{a_1\cdots a_{d-1}b}. \end{aligned} \tag{455}$$

Lorentz invariance of the Levi–Civita tensor implies

$$\omega^{b}{}_{a_1}\epsilon_{ba_2\cdots a_d} + \omega^{b}{}_{a_2}\epsilon_{a_1 ba_3\cdots a_d} + \cdots + \omega^{b}{}_{a_d}\epsilon_{a_1\cdots a_{d-1}b} = 0. \tag{456}$$

Consequently,

$$D\epsilon_{a_1\cdots a_d} = 0. \tag{457}$$

Why this identity is useful

The identity

$$D\epsilon_{a_1\cdots a_d} = 0$$

is extremely useful in the first-order formulation of gravity.
For example, when varying the Einstein–Cartan or Palatini action, it allows the exterior covariant derivative to pass through the Levi–Civita tensor without generating additional terms.

4 First-Order Formulations of General Relativity

The Palatini Formulation in a Coordinate Basis

We now turn to the first-order formulation of gravity in a coordinate basis. In the metric formulation of General Relativity, the affine connection is assumed from the beginning to be the Levi–Civita connection,

$$\Gamma^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\mu\nu}(g). \tag{458}$$

Therefore, the metric is the only independent gravitational degree of freedom. In the Palatini formulation, by contrast, the metric and the affine connection are regarded as independent variables:

$$\boxed{g_{\mu\nu}, \qquad \Gamma^{\lambda}{}_{\mu\nu}.} \tag{459}$$

We work in a coordinate basis, so $[\partial_\mu, \partial_\nu] = 0$. The Palatini action is

$$\boxed{S[g,\Gamma] = \frac{1}{2\kappa}\int d^4x\,\sqrt{-g}\,g^{\mu\nu}R_{\mu\nu}(\Gamma).} \tag{460}$$

Here the Ricci tensor is constructed entirely from the independent connection $\Gamma^{\lambda}{}_{\mu\nu}$. Since $g_{\mu\nu}$ and $\Gamma^{\lambda}{}_{\mu\nu}$ are independent, varying the metric does not vary the Ricci tensor:

$$\delta_g R_{\mu\nu}(\Gamma) = 0. \tag{461}$$

Therefore,

$$\delta_g S = \frac{1}{2\kappa}\int d^4x\ \left[\delta(\sqrt{-g})\, g^{\mu\nu}R_{\mu\nu} + \sqrt{-g}\,\delta g^{\mu\nu}R_{\mu\nu}\right]. \tag{462}$$

Recall the standard identity

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu}. \tag{463}$$

Hence

$$\delta_g S = \frac{1}{2\kappa}\int d^4x\,\sqrt{-g}\left[R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R\right]\delta g^{\mu\nu}. \tag{464}$$

Since $\delta g^{\mu\nu}$ is symmetric, strictly speaking only the symmetric part of the Ricci tensor contributes. Thus the metric field equation is

$$R_{(\mu\nu)}(\Gamma) - \frac{1}{2}g_{\mu\nu}R(\Gamma) = 0. \tag{465}$$

If the Ricci tensor is symmetric, this reduces to the usual expression

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0. \tag{466}$$

We now keep the metric fixed and vary the affine connection. From (460),

$$\delta_\Gamma S = \frac{1}{2\kappa}\int d^4x\,\sqrt{-g}\, g^{\mu\nu}\delta_\Gamma R_{\mu\nu}. \tag{467}$$

To evaluate this variation, we first need the variation of the Riemann tensor. For a general connection,

$$R^\lambda{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\lambda{}_{\nu\sigma} - \partial_\nu\Gamma^\lambda{}_{\mu\sigma} + \Gamma^\lambda{}_{\mu\rho}\Gamma^\rho{}_{\nu\sigma} - \Gamma^\lambda{}_{\nu\rho}\Gamma^\rho{}_{\mu\sigma}. \tag{468}$$

Varying gives us

$$\begin{aligned}\delta R^\lambda{}_{\sigma\mu\nu} = {}& \partial_\mu\delta\Gamma^\lambda{}_{\nu\sigma} - \partial_\nu\delta\Gamma^\lambda{}_{\mu\sigma} \\ & + \delta\Gamma^\lambda{}_{\mu\rho}\Gamma^\rho{}_{\nu\sigma} + \Gamma^\lambda{}_{\mu\rho}\delta\Gamma^\rho{}_{\nu\sigma} \\ & - \delta\Gamma^\lambda{}_{\nu\rho}\Gamma^\rho{}_{\mu\sigma} - \Gamma^\lambda{}_{\nu\rho}\delta\Gamma^\rho{}_{\mu\sigma}.\end{aligned} \tag{469}$$

The difference between two connections transforms tensorially. Therefore,

$$\delta\Gamma^\lambda{}_{\mu\nu} \tag{470}$$

is a $(1,2)$ tensor, and it is useful to rewrite the previous expression using covariant derivatives.

With

$$T^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu}, \tag{471}$$

one obtains the generalized Palatini identity

$$\delta R^\lambda{}_{\sigma\mu\nu} = \nabla_\mu\delta\Gamma^\lambda{}_{\nu\sigma} - \nabla_\nu\delta\Gamma^\lambda{}_{\mu\sigma} + T^\rho{}_{\mu\nu}\delta\Gamma^\lambda{}_{\rho\sigma}. \tag{472}$$

Why does the torsion term appear?

If the connection is symmetric,

$$T^{\rho}{}_{\mu\nu} = 0,$$

the familiar Palatini identity is recovered:

$$\delta R^{\lambda}{}_{\sigma\mu\nu} = \nabla_{\mu}\delta\Gamma^{\lambda}{}_{\nu\sigma} - \nabla_{\nu}\delta\Gamma^{\lambda}{}_{\mu\sigma}.$$

For a connection with torsion, the antisymmetric part of $\Gamma^{\rho}{}_{\mu\nu}$ produces the additional term in (472).

Using the definition $R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}$ we obtain

$$\delta R_{\mu\nu} = \nabla_{\lambda}\delta\Gamma^{\lambda}{}_{\nu\mu} - \nabla_{\nu}\delta\Gamma^{\lambda}{}_{\lambda\mu} + T^{\rho}{}_{\lambda\nu}\delta\Gamma^{\lambda}{}_{\rho\mu}. \tag{473}$$

Substituting this into the variation of the action gives

$$\delta_{\Gamma} S = \frac{1}{2\kappa}\int d^4x\,\sqrt{-g}\,g^{\mu\nu}\Big(\nabla_{\lambda}\delta\Gamma^{\lambda}{}_{\nu\mu} - \nabla_{\nu}\delta\Gamma^{\lambda}{}_{\lambda\mu} + T^{\rho}{}_{\lambda\nu}\delta\Gamma^{\lambda}{}_{\rho\mu}\Big). \tag{474}$$

The first two terms contain derivatives of the variation. We therefore need an integration-by-parts formula appropriate for a connection with torsion. We can introduce

$$\mathfrak{g}^{\mu\nu} \equiv \sqrt{-g}\,g^{\mu\nu}. \tag{475}$$

This is not an ordinary tensor: it is a contravariant tensor density of weight $+1$.

Its covariant derivative is therefore

$$\begin{aligned}\nabla_{\lambda}\mathfrak{g}^{\mu\nu} = \partial_{\lambda}\mathfrak{g}^{\mu\nu} &+ \Gamma^{\mu}{}_{\lambda\rho}\mathfrak{g}^{\rho\nu} + \Gamma^{\nu}{}_{\lambda\rho}\mathfrak{g}^{\mu\rho} \\ &- \Gamma^{\rho}{}_{\lambda\rho}\mathfrak{g}^{\mu\nu}.\end{aligned} \tag{476}$$

The last term is characteristic of a density of weight $+1$. Let $\mathfrak{V}^{\lambda}$ be a vector density of weight $+1$. Its covariant divergence is

$$\nabla_{\lambda}\mathfrak{V}^{\lambda} = \partial_{\lambda}\mathfrak{V}^{\lambda} + \Gamma^{\lambda}{}_{\lambda\rho}\mathfrak{V}^{\rho} - \Gamma^{\rho}{}_{\lambda\rho}\mathfrak{V}^{\lambda}. \tag{477}$$

Renaming dummy indices in the second term,

$$\nabla_{\lambda}\mathfrak{V}^{\lambda} = \partial_{\lambda}\mathfrak{V}^{\lambda} + (\Gamma^{\rho}{}_{\rho\lambda} - \Gamma^{\rho}{}_{\lambda\rho})\,\mathfrak{V}^{\lambda}. \tag{478}$$

Define the torsion trace

$$T_{\lambda} \equiv T^{\rho}{}_{\lambda\rho} = \Gamma^{\rho}{}_{\lambda\rho} - \Gamma^{\rho}{}_{\rho\lambda}. \tag{479}$$

Therefore,

$$\nabla_{\lambda}\mathfrak{V}^{\lambda} = \partial_{\lambda}\mathfrak{V}^{\lambda} - T_{\lambda}\mathfrak{V}^{\lambda}. \tag{480}$$

Equivalently,

$$\partial_\lambda \mathfrak{V}^\lambda = \nabla_\lambda \mathfrak{V}^\lambda + T_\lambda \mathfrak{V}^\lambda. \tag{481}$$

This is the origin of the additional torsion contribution in the integration-by-parts formula. Let's consider the following

$$I_1 = \int d^4x\, \mathfrak{g}^{\mu\nu} \nabla_\lambda \delta\Gamma^\lambda{}_{\nu\mu}. \tag{482}$$

Using the Leibniz rule,

$$\nabla_\lambda \left(\mathfrak{g}^{\mu\nu} \delta\Gamma^\lambda{}_{\nu\mu} \right) = \left(\nabla_\lambda \mathfrak{g}^{\mu\nu} \right) \delta\Gamma^\lambda{}_{\nu\mu} + \mathfrak{g}^{\mu\nu} \nabla_\lambda \delta\Gamma^\lambda{}_{\nu\mu}. \tag{483}$$

Hence

$$\mathfrak{g}^{\mu\nu} \nabla_\lambda \delta\Gamma^\lambda{}_{\nu\mu} = \nabla_\lambda \left(\mathfrak{g}^{\mu\nu} \delta\Gamma^\lambda{}_{\nu\mu} \right) - \left(\nabla_\lambda \mathfrak{g}^{\mu\nu} \right) \delta\Gamma^\lambda{}_{\nu\mu}. \tag{484}$$

The first term is the covariant divergence of a vector density. Using (481),

$$\nabla_\lambda \left(\mathfrak{g}^{\mu\nu} \delta\Gamma^\lambda{}_{\nu\mu} \right) = \partial_\lambda \left(\mathfrak{g}^{\mu\nu} \delta\Gamma^\lambda{}_{\nu\mu} \right) - T_\lambda \mathfrak{g}^{\mu\nu} \delta\Gamma^\lambda{}_{\nu\mu}. \tag{485}$$

Therefore,

$$I_1 = \int d^4x\, \partial_\lambda \left(\mathfrak{g}^{\mu\nu} \delta\Gamma^\lambda{}_{\nu\mu} \right) - \int d^4x\, \left[\nabla_\lambda \mathfrak{g}^{\mu\nu} + T_\lambda \mathfrak{g}^{\mu\nu} \right] \delta\Gamma^\lambda{}_{\nu\mu}. \tag{486}$$

Assuming the variation vanishes sufficiently rapidly at the boundary, the ordinary divergence term can be discarded. Thus

$$\boxed{I_1 \doteq - \int d^4x\, \left[\nabla_\lambda (\sqrt{-g}\, g^{\mu\nu}) + T_\lambda \sqrt{-g}\, g^{\mu\nu} \right] \delta\Gamma^\lambda{}_{\nu\mu},} \tag{487}$$

where $\doteq$ denotes equality up to a boundary term.

Important subtlety

With torsion, the covariant divergence of a weight-1 vector density is not simply an ordinary total derivative:

$$\nabla_\mu \mathfrak{V}^\mu \neq \partial_\mu \mathfrak{V}^\mu.$$

Instead,

$$\nabla_\mu \mathfrak{V}^\mu = \partial_\mu \mathfrak{V}^\mu - T_\mu \mathfrak{V}^\mu.$$

This is precisely why the torsion trace appears in the connection variation.

Now consider

$$I_2 = - \int d^4x\, \mathfrak{g}^{\mu\nu} \nabla_\nu \delta\Gamma^\lambda{}_{\lambda\mu}. \tag{488}$$

Proceeding in exactly the same way,

$$I_2 \doteq \int d^4x\, \left[\nabla_\nu \mathfrak{g}^{\mu\nu} + T_\nu \mathfrak{g}^{\mu\nu} \right] \delta\Gamma^\lambda{}_{\lambda\mu}. \tag{489}$$

To combine this with the other terms, write

$$\delta\Gamma^{\rho}{}_{\rho\mu} = \delta^{\nu}{}_{\lambda}\delta\Gamma^{\lambda}{}_{\nu\mu}. \tag{490}$$

Combining (487), (489) and the explicit torsion term in (474), we obtain

$$\delta_{\Gamma}S \doteq \frac{1}{2\kappa}\int d^4x \left\{ -\nabla_{\lambda}\mathfrak{g}^{\mu\nu} - T_{\lambda}\mathfrak{g}^{\mu\nu} + \delta^{\nu}{}_{\lambda}\left(\nabla_{\rho}\mathfrak{g}^{\mu\rho} + T_{\rho}\mathfrak{g}^{\mu\rho}\right) + \mathfrak{g}^{\mu\rho}T^{\nu}{}_{\lambda\rho} \right\}\delta\Gamma^{\lambda}{}_{\nu\mu}. \tag{491}$$

Since $\delta\Gamma^{\lambda}{}_{\nu\mu}$ is arbitrary, the connection equation of motion is

$$\boxed{-\nabla_{\lambda}\mathfrak{g}^{\mu\nu} - T_{\lambda}\mathfrak{g}^{\mu\nu} + \delta^{\nu}{}_{\lambda}\left(\nabla_{\rho}\mathfrak{g}^{\mu\rho} + T_{\rho}\mathfrak{g}^{\mu\rho}\right) + \mathfrak{g}^{\mu\rho}T^{\nu}{}_{\lambda\rho} = 0,} \tag{492}$$

with

$$\mathfrak{g}^{\mu\nu} = \sqrt{-g}\, g^{\mu\nu}. \tag{493}$$

A particularly important special case is obtained by imposing a symmetric connection,

$$T^{\lambda}{}_{\mu\nu} = 0. \tag{494}$$

Then the connection equation reduces to

$$-\nabla_{\lambda}\mathfrak{g}^{\mu\nu} + \delta^{\nu}{}_{\lambda}\nabla_{\rho}\mathfrak{g}^{\mu\rho} = 0. \tag{495}$$

Contracting λ with ν in $n > 1$ dimensions gives

$$\nabla_{\rho}\mathfrak{g}^{\mu\rho} = 0, \tag{496}$$

and therefore

$$\boxed{\nabla_{\lambda}\left(\sqrt{-g}\, g^{\mu\nu}\right) = 0.} \tag{497}$$

For a torsion-free connection this implies

$$\nabla_{\lambda}g_{\mu\nu} = 0, \tag{498}$$

so that the independent connection reduces dynamically to the Levi–Civita connection,

$$\Gamma^{\lambda}{}_{\mu\nu} = \left\{ {\lambda \atop \mu\nu} \right\}. \tag{499}$$

Thus the usual metric formulation of General Relativity is recovered from the Palatini theory.

If neither torsion-freeness nor metric compatibility is imposed before the variation, the Einstein–Hilbert–Palatini action possesses a projective invariance. In our index convention,

$$\Gamma^{\lambda}{}_{\mu\nu} \longrightarrow \Gamma^{\lambda}{}_{\mu\nu} + \delta^{\lambda}{}_{\nu}A_{\mu} \tag{500}$$

changes the Ricci tensor only by an antisymmetric contribution and therefore leaves $g^{\mu\nu}R_{\mu\nu}$ unchanged.

Consequently, for a completely unrestricted affine connection, the connection is determined only up to this projective mode. If one requires a torsion-free or metric-compatible connection, this freedom is removed and the Levi–Civita connection is recovered uniquely. Here A_{μ} is an arbitrary $1-$form. For this reason it is not possible determine all the components from the equations of motion.

We consider the projective transformation on the self-parallels equation

$$\frac{d^2x^\rho}{d\lambda^2} + \Gamma^\rho_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0\,.$$

Under the transformation we have

$$\Gamma'^\rho_{\mu\nu} = \Gamma^\rho_{\mu\nu}\dot{x}^\mu\dot{x}\nu + \dot{x}^\mu A_\mu\dot{x}^\rho\,.$$

Hence we have the new equation

$$\frac{d^2x^\rho}{d\lambda^2} + \Gamma^\rho_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -(A_\mu\dot{x}^\mu)\dot{x}^\rho\,.$$

The last term can be absorbed by reparametrization of the curve, so the projective transformation is such that preserves the non-parametrized trajectories.

Field Equations of the Einstein–Cartan Action

We continue with the Einstein–Cartan action introduced in the previous class,

$$S[e,\omega] = \int_M \left[\alpha\,\epsilon_{abcd}\,R^{ab}\wedge e^c\wedge e^d + \beta\,\epsilon_{abcd}\,e^a\wedge e^b\wedge e^c\wedge e^d\right], \tag{501}$$

where the vielbein e^a and the Lorentz connection ω^{ab} are regarded as independent fields.

From the variation with respect to the vielbein we found

$$\epsilon_{abcd}\left(R^{ab} + \frac{2\beta}{\alpha}e^a\wedge e^b\right)\wedge e^c = 0. \tag{502}$$

We now study the independent variation with respect to the Lorentz connection. Since the cosmological term does not depend explicitly on ω^{ab},

$$\delta_\omega S = \alpha\int_M \epsilon_{abcd}\,\delta_\omega R^{ab}\wedge e^c\wedge e^d. \tag{503}$$

Recall that

$$R^{ab} = d\omega^{ab} + \omega^a{}_c\wedge\omega^{cb}. \tag{504}$$

Its variation is therefore

$$\delta R^{ab} = d(\delta\omega^{ab}) + \delta\omega^a{}_c\wedge\omega^{cb} + \omega^a{}_c\wedge\delta\omega^{cb}. \tag{505}$$

This is precisely the exterior Lorentz-covariant derivative of the variation:

$$\delta R^{ab} = D(\delta\omega^{ab}). \tag{506}$$

Hence

$$\delta_\omega S = \alpha \int_M \epsilon_{abcd}\, D(\delta\omega^{ab}) \wedge e^c \wedge e^d. \tag{507}$$

Because $\delta\omega^{ab}$ is a one-form, the graded Leibniz rule gives

$$D\left(\delta\omega^{ab} \wedge e^c \wedge e^d\right) = D(\delta\omega^{ab}) \wedge e^c \wedge e^d - \delta\omega^{ab} \wedge D(e^c \wedge e^d). \tag{508}$$

Therefore,

$$D(\delta\omega^{ab}) \wedge e^c \wedge e^d = D\left(\delta\omega^{ab} \wedge e^c \wedge e^d\right) + \delta\omega^{ab} \wedge D(e^c \wedge e^d). \tag{509}$$

Since the Levi–Civita tensor is Lorentz invariant,

$$D\epsilon_{abcd} = 0. \tag{510}$$

Thus

$$\delta_\omega S = \alpha \int_M D\left(\epsilon_{abcd}\delta\omega^{ab} \wedge e^c \wedge e^d\right) + \alpha \int_M \epsilon_{abcd}\delta\omega^{ab} \wedge D(e^c \wedge e^d). \tag{511}$$

The quantity inside the first covariant derivative is a Lorentz scalar. Therefore, on it, $D = d$. Consequently, the first term is an ordinary exterior derivative,

$$\int_M d\left(\epsilon_{abcd}\delta\omega^{ab} \wedge e^c \wedge e^d\right), \tag{512}$$

and hence becomes a boundary term by Stokes' theorem. We are therefore left with

$$\delta_\omega S \doteq \alpha \int_M \epsilon_{abcd}\, \delta\omega^{ab} \wedge D(e^c \wedge e^d), \tag{513}$$

where $\doteq$ denotes equality up to a boundary term. Using the graded (for differential forms the degree of the form somethimes change a sign) Leibniz rule,

$$\begin{aligned} D(e^c \wedge e^d) &= De^c \wedge e^d - e^c \wedge De^d \\ &= T^c \wedge e^d - e^c \wedge T^d. \end{aligned} \tag{514}$$

Therefore,

$$\delta_\omega S \doteq \alpha \int_M \epsilon_{abcd}\delta\omega^{ab} \wedge \left(T^c \wedge e^d - e^c \wedge T^d\right). \tag{515}$$

Since T^d is a two-form,

$$e^c \wedge T^d = T^d \wedge e^c. \tag{516}$$

Relabeling $c \leftrightarrow d$ in the second term and using the antisymmetry of ϵ_{abcd} gives

$$\delta_\omega S \doteq 2\alpha \int_M \epsilon_{abcd}\, \delta\omega^{ab} \wedge T^c \wedge e^d. \tag{517}$$

Since $\delta\omega^{ab}$ is arbitrary,

$$\epsilon_{abcd}\, T^c \wedge e^d = 0. \tag{518}$$

For an invertible vielbein, this equation implies

$$T^a = 0. \tag{519}$$

Thus the independent spin connection is dynamically forced to be the torsion-free spin connection,

$$\omega^{ab} = \mathring{\omega}^{ab}(e). \tag{520}$$

This is an important difference between the formulation and the final dynamics: the connection is treated as independent when constructing the action, but its own equation of motion determines it algebraically.

Normalization and the Cosmological Constant

We now determine the constants α and β by requiring that, after imposing the torsion equation, the Einstein–Cartan action agrees with the Einstein–Hilbert action with cosmological constant,

$$S_{\mathrm{EH}} = \frac{1}{2\kappa}\int_M d^4x\,\sqrt{-g}\,(R - 2\Lambda)\,. \tag{521}$$

We use the conventions

$$R^{ab} = \frac{1}{2} R^{ab}{}_{ef} e^e \wedge e^f, \tag{522}$$

and

$$\mathrm{vol} = \frac{1}{4!}\epsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d = \sqrt{-g}\, d^4x. \tag{523}$$

Let's consider first

$$I_R = \alpha \int_M \epsilon_{abcd} R^{ab} \wedge e^c \wedge e^d. \tag{524}$$

Substituting the component expansion of the curvature,

$$I_R = \frac{\alpha}{2}\int_M \epsilon_{abcd} R^{ab}{}_{ef} e^e \wedge e^f \wedge e^c \wedge e^d. \tag{525}$$

Using the standard contraction of Levi–Civita tensors, this reduces to

$$\boxed{\epsilon_{abcd} R^{ab} \wedge e^c \wedge e^d = 2R\,\mathrm{vol}.} \tag{526}$$

Therefore,

$$I_R = 2\alpha \int_M d^4x\,\sqrt{-g}\,R. \tag{527}$$

Comparison with the Einstein–Hilbert coefficient,

$$\frac{1}{2\kappa}\int_M d^4x\,\sqrt{-g}\,R, \tag{528}$$

gives

$$\alpha = \frac{1}{4\kappa}. \tag{529}$$

Normalization convention

The factor of 1/4 follows from the convention

$$R^{ab} = \frac{1}{2}R^{ab}{}_{cd}e^c \wedge e^d.$$

With this convention, the standard Einstein–Cartan action reproduces the Einstein–Hilbert normalization $1/(2\kappa)$.

From (523),

$$\epsilon_{abcd}e^a \wedge e^b \wedge e^c \wedge e^d = 4!\,\sqrt{-g}\,d^4x. \tag{530}$$

Therefore,

$$I_\Lambda = \beta\int_M \epsilon_{abcd}e^a \wedge e^b \wedge e^c \wedge e^d = 4!\,\beta\int_M d^4x\,\sqrt{-g}. \tag{531}$$

The cosmological term in the metric (formulation of the) action is

$$-\frac{\Lambda}{\kappa}\int_M d^4x\,\sqrt{-g}. \tag{532}$$

Hence to match this factor exactly, the relation between the constant is determined

$$4!\,\beta = -\frac{\Lambda}{\kappa}, \tag{533}$$

and therefore

$$\beta = -\frac{\Lambda}{4!\,\kappa} = -\frac{\Lambda}{24\kappa}. \tag{534}$$

Substituting (529) and (534) into (501), we obtain

$$S[e,\omega] = \frac{1}{4\kappa}\int_M \epsilon_{abcd}\left[R^{ab} \wedge e^c \wedge e^d - \frac{\Lambda}{6}e^a \wedge e^b \wedge e^c \wedge e^d\right]. \tag{535}$$

Equivalently,

$$S[e,\omega] = \frac{1}{4\kappa}\int_M \epsilon_{abcd}\left(R^{ab} - \frac{\Lambda}{6}e^a \wedge e^b\right) \wedge e^c \wedge e^d. \tag{536}$$

The two independent equations of motion are therefore

$$\epsilon_{abcd}\left(R^{ab} - \frac{\Lambda}{3}e^a \wedge e^b\right) \wedge e^c = 0, \tag{537}$$

and

$$\epsilon_{abcd} T^c \wedge e^d = 0. \tag{538}$$

For an invertible vielbein, the second equation gives

$$T^a = 0. \tag{539}$$

The spin connection is then the Levi–Civita spin connection $\mathring{\omega}^{ab}(e)$, and substitution back into (535) reproduces the usual Einstein–Hilbert theory with cosmological constant.

5 Poincaré Gauge Theory of Gravity

Gravity as a Gauge Theory

We now want to understand how the geometrical structures introduced in the previous classes can be interpreted from the perspective of gauge theory if it is possible, which is a fact itself not trivial to prove. In a gauge theory, one starts with a Lie group G describing the gauge symmetry. Its Lie algebra $\mathfrak{g}$ is generated by elements T_A satisfying

$$[T_A, T_B] = C^C{}_{AB}\, T_C, \tag{540}$$

where $C^C{}_{AB}$ are the structure constants of the algebra. To formulate the gauge theory, we introduce a gauge field taking values in the Lie algebra,

$$A_\mu = A_\mu^A T_A. \tag{541}$$

Equivalently, in the language of differential forms, the gauge connection is the Lie-algebra-valued one-form

$$A = A^A T_A, \qquad A^A = A_\mu^A dx^\mu. \tag{542}$$

Thus,

$$A \in \Omega^1(M, \mathfrak{g}). \tag{543}$$

Associated with the gauge connection A is a Lie-algebra-valued two-form called the *field strength* or *curvature*,

$$F = dA + A \wedge A. \tag{544}$$

Since A is Lie-algebra valued, the product $A \wedge A$ contains both the wedge product of differential forms and the product of the Lie algebra generators.

Writing

$$A = A^A T_A, \tag{545}$$

we have

$$A \wedge A = A^A \wedge A^B\, T_A T_B. \tag{546}$$

Because

$$A^A \wedge A^B = -A^B \wedge A^A, \tag{547}$$

only the antisymmetric part of $T_A T_B$ contributes. Therefore,

$$\begin{aligned} A \wedge A &= \frac{1}{2} A^A \wedge A^B [T_A, T_B] \\ &= \frac{1}{2} C^C{}_{AB} A^A \wedge A^B T_C. \end{aligned} \tag{548}$$

Hence

$$F = F^A T_A, \tag{549}$$

with

$$F^A = dA^A + \frac{1}{2} C^A{}_{BC} A^B \wedge A^C. \tag{550}$$

Let ϕ be a field transforming in some representation of the gauge group. Its covariant exterior derivative is

$$D\phi = (d + A)\phi. \tag{551}$$

Applying the covariant derivative twice,

$$\begin{aligned} D^2\phi &= (d + A)(d + A)\phi \\ &= d(d\phi) + d(A\phi) + A \wedge d\phi + A \wedge A\,\phi. \end{aligned} \tag{552}$$

Using the fact that $d^2 = 0$ and the graded Leibiniz rule

$$d(A\phi) = dA\,\phi - A \wedge d\phi, \tag{553}$$

we obtain

$$D^2\phi = (dA + A \wedge A)\,\phi. \tag{554}$$

Therefore,

$$D^2\phi = F\phi. \tag{555}$$

Thus the curvature measures the failure of two covariant derivatives to commute. This is the gauge-theory counterpart of the geometrical interpretation of the Riemann tensor studied previously.

Suppose now that the gauge curvature vanishes,

$$F = 0. \tag{556}$$

Then

$$dA + A \wedge A = 0. \tag{557}$$

This is the Maurer–Cartan equation. Locally, a flat connection can be written as a pure gauge. A standard example is the left Maurer–Cartan form

$$\boxed{\Theta = g^{-1} dg,} \tag{558}$$

where $g(x)$ is a group-valued field. Let us verify that it satisfies the Maurer–Cartan equation. We first use

$$d(g^{-1}g) = 0, \tag{559}$$

which gives

$$d(g^{-1})\, g + g^{-1} dg = 0. \tag{560}$$

Therefore,

$$d(g^{-1}) = -g^{-1}(dg)g^{-1}. \tag{561}$$

Now,

$$\begin{aligned} d\Theta &= d(g^{-1}dg) \\ &= d(g^{-1}) \wedge dg + g^{-1}d^2 g \\ &= -g^{-1}dg\, g^{-1} \wedge dg. \end{aligned} \tag{562}$$

But

$$\Theta \wedge \Theta = g^{-1}dg \wedge g^{-1}dg. \tag{563}$$

Hence

$$\boxed{d\Theta + \Theta \wedge \Theta = 0.} \tag{564}$$

The Maurer–Cartan form therefore provides the canonical example of a flat, or pure-gauge, connection.

Flat versus trivial

The condition

$$F = 0$$

implies that a connection is locally pure gauge. Globally, however, a flat connection need not be gauge-equivalent to zero if the underlying space has nontrivial topology.

The curvature also satisfies a differential identity. Define the covariant derivative acting on a Lie-algebra-valued form by

$$D = d + [A, \cdot], \tag{565}$$

with the appropriate graded commutator. Starting from

$$F = dA + A \wedge A, \tag{566}$$

one obtains

$$DF = 0. \tag{567}$$

In matrix notation,

$$DF = dF + A \wedge F - F \wedge A = 0. \tag{568}$$

This is the gauge-theory form of the Bianchi identity.

Non-Abelian versus Abelian case

For a general non-Abelian gauge theory the Bianchi identity is

$$DF = 0,$$

rather than $dF = 0$.
In an Abelian theory, where the commutator terms vanish, this reduces to $dF = 0$.

The Poincaré Algebra and Its Gauge Connection

We now apply this general construction to gravity.The relevant symmetry group is the Poincaré group

$$ISO(3,1) = SO(3,1) \ltimes \mathbb{R}^{3,1}. \tag{569}$$

Its generators consist of translations P_a and Lorentz transformations J_{ab},

$$J_{ab} = -J_{ba}. \tag{570}$$

Collectively, we may write

$$T_A = \{P_a, J_{ab}\}. \tag{571}$$

The Poincaré algebra is given by the commutation relations

$$[P_a, P_b] = 0, \tag{572}$$
$$[J_{ab}, P_c] = \eta_{bc}P_a - \eta_{ac}P_b \tag{573}$$
$$[J_{ab}, J_{cd}] = \eta_{ad}J_{bc} + \eta_{bc}J_{ad} - \eta_{ac}J_{bd} - \eta_{bd}J_{ac}.. \tag{574}$$

These commutation relations determine the structure constants of the Poincaré algebra.

Since the algebra contains both translations and Lorentz transformations, the associated gauge connection naturally decomposes into two pieces

$$A = e^a P_a + \frac{1}{2}\omega^{ab}J_{ab}. \tag{575}$$

The translational part is identified with the vielbein one-form

$$e^a = e^a{}_\mu dx^\mu, \tag{576}$$

while the Lorentz part is the spin connection

$$\omega^{ab} = \omega^{ab}{}_\mu dx^\mu. \tag{577}$$

The factor of $1/2$ compensates for the antisymmetry

$$J_{ab} = -J_{ba}. \tag{578}$$

We now compute

$$F = dA + A \wedge A. \tag{579}$$

Using (575),

$$dA = de^a P_a + \frac{1}{2} d\omega^{ab} J_{ab}. \tag{580}$$

The quadratic term can be decomposed according to the Poincaré commutation relations. Since

$$[P_a, P_b] = 0, \tag{581}$$

there is no $e^a \wedge e^b$ contribution to the curvature of the Poincaré connection.

The mixed term involving J_{ab} and P_c gives the translational part of the curvature, while the J–J term gives its Lorentz part.

Thus the total curvature can be written as

$$F = T^a P_a + \frac{1}{2} R^{ab} J_{ab}, \tag{582}$$

where we will now recover explicitly

$$T^a = de^a + \omega^a{}_b \wedge e^b \tag{583}$$

and

$$R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}. \tag{584}$$

Consider the part of the curvature proportional to P_a. Using

$$[J_{ab}, P_c] = \eta_{bc} P_a - \eta_{ac} P_b, \tag{585}$$

the mixed contribution is

$$F\big|_P = de^a P_a + \frac{1}{2} \omega^{bc} \wedge e^d [J_{bc}, P_d]. \tag{586}$$

Substituting the commutator,

$$F\big|_P = de^a P_a + \frac{1}{2} \omega^{bc} \wedge e^d \left(\eta_{cd} P_b - \eta_{bd} P_c\right). \tag{587}$$

The two terms are equal after relabeling dummy indices and using

$$\omega^{bc} = -\omega^{cb}. \tag{588}$$

Therefore,

$$F\big|_P = \left(de^a + \omega^a{}_b \wedge e^b\right) P_a. \tag{589}$$

Here we rembember a known quantity

$$T^a = de^a + \omega^a{}_b \wedge e^b. \tag{590}$$

We recognize precisely Cartan's first structure equation. Thus, in the Poincaré gauge description, the torsion is the field strength associated with the translational sector.

We now consider the component proportional to J_{ab}. Since the translations commute, only the Lorentz connection contributes:

$$F\big|_J = \frac{1}{2} d\omega^{ab} J_{ab} + \frac{1}{8} \omega^{ab} \wedge \omega^{cd} [J_{ab}, J_{cd}]. \tag{591}$$

Using the Lorentz algebra (574) and simplifying the dummy indices, one obtains

$$F\big|_J = \frac{1}{2} \left(d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb} \right) J_{ab}. \tag{592}$$

Therefore,

$$R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}. \tag{593}$$

This is Cartan's second structure equation. Combining the translational and Lorentz sectors gives

$$F = \left(de^a + \omega^a{}_b \wedge e^b \right) P_a + \frac{1}{2} \left(d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb} \right) J_{ab}. \tag{594}$$

Equivalently,

$$F = T^a P_a + \frac{1}{2} R^{ab} J_{ab}. \tag{595}$$

An important advantage of the gauge formulation is that both Cartan Bianchi identities follow at once from the single gauge-theory identity

$$DF = 0. \tag{596}$$

Substituting

$$F = T^a P_a + \frac{1}{2} R^{ab} J_{ab}, \tag{597}$$

and separating the translational and Lorentz generators, one obtains

$$DT^a = R^a{}_b \wedge e^b, \tag{598}$$

together with

$$DR^{ab} = 0. \tag{599}$$

These are precisely the first and second Bianchi identities obtained previously from Cartan geometry.

The gauge-theory formulation therefore reorganizes the geometrical quantities of first-order gravity into a single Poincaré-valued connection,

$$A = e^a P_a + \frac{1}{2}\omega^{ab} J_{ab}, \tag{600}$$

and a single curvature,

$$F = T^a P_a + \frac{1}{2} R^{ab} J_{ab}. \tag{601}$$

The translational field strength is the torsion,

$$T^a = De^a, \tag{602}$$

while the Lorentz field strength is the curvature,

$$R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}. \tag{603}$$

This reproduces the Cartan structure equations directly from the Poincaré algebra.

A geometrical subtlety

The relation between gravity and ordinary Yang–Mills theory is very close but not completely literal. The vielbein e^a is not merely an ordinary internal gauge field: it also identifies the internal translation space with tangent directions on spacetime. This "soldering" property is what allows the gauge variables to describe spacetime geometry.
A more precise mathematical framework for this viewpoint is *Cartan geometry* or *Poincaré gauge theory*.

Gauge Transformations of the Gravitational Fields

We now study the gauge transformations of the fields introduced in the Poincaré gauge formulation of gravity. Recall that the Poincaré-valued connection is

$$A = e^a P_a + \frac{1}{2}\omega^{ab} J_{ab}, \tag{604}$$

where P_a are the generators of translations and J_{ab} are the generators of Lorentz transformations.

Let's consider

$$U \in G \tag{605}$$

be a gauge transformation. We choose the convention

$$U = e^{-\lambda}, \tag{606}$$

so that, infinitesimally,

$$U = 1 - \lambda + \mathcal{O}(\lambda^2), \qquad U^{-1} = 1 + \lambda + \mathcal{O}(\lambda^2). \tag{607}$$

The finite transformation law for a connection is

$$A' = UAU^{-1} + U\,dU^{-1}. \tag{608}$$

Expanding to first order,

$$\begin{aligned} A' &= (1-\lambda)A(1+\lambda) + (1-\lambda)d(1+\lambda) \\ &= A + A\lambda - \lambda A + d\lambda + \mathcal{O}(\lambda^2). \end{aligned} \tag{609}$$

Therefore,

$$\delta_\lambda A \equiv A' - A = d\lambda + [A, \lambda] = D\lambda. \tag{610}$$

Thus the infinitesimal transformation of a gauge connection is the covariant derivative of the gauge parameter.

For the Poincaré group, the gauge parameter has both a translational and a Lorentz part:

$$\lambda = \lambda^a P_a + \frac{1}{2}\lambda^{ab} J_{ab}, \tag{611}$$

with

$$\lambda^{ab} = -\lambda^{ba}. \tag{612}$$

Using

$$A = e^a P_a + \frac{1}{2}\omega^{ab} J_{ab}, \tag{613}$$

we obtain

$$\begin{aligned} \delta A &= \delta e^a P_a + \frac{1}{2}\delta\omega^{ab} J_{ab} \\ &= d\lambda^a P_a + \frac{1}{2} d\lambda^{ab} J_{ab} + \left[e^a P_a + \frac{1}{2}\omega^{ab} J_{ab},\ \lambda^c P_c + \frac{1}{2}\lambda^{cd} J_{cd} \right]. \end{aligned} \tag{614}$$

We now separate the translational and Lorentz parts. We use the Poincaré algebra

$$[P_a, P_b] = 0, \tag{615}$$

and

$$[J_{ab}, P_c] = \eta_{bc} P_a - \eta_{ac} P_b. \tag{616}$$

The translational part of $[A, \lambda]$ receives contributions from

$$\left[\frac{1}{2}\omega^{ab} J_{ab}, \lambda^c P_c \right], \quad \text{and} \quad \left[e^a P_a, \frac{1}{2}\lambda^{bc} J_{bc} \right]. \tag{617}$$

For the first term,

$$\frac{1}{2}\omega^{ab}\lambda^c [J_{ab}, P_c] = \frac{1}{2}\omega^{ab}\lambda^c \left(\eta_{bc} P_a - \eta_{ac} P_b\right) = \omega^a{}_b \lambda^b P_a. \tag{618}$$

For the second,

$$\frac{1}{2}e^a\lambda^{bc}[P_a, J_{bc}] = -\frac{1}{2}e^a\lambda^{bc}\left(\eta_{ca}P_b - \eta_{ba}P_c\right) = -\lambda^a{}_b e^b P_a. \tag{619}$$

Therefore,

$$\delta e^a = d\lambda^a + \omega^a{}_b\lambda^b - \lambda^a{}_b e^b. \tag{620}$$

Introducing the Lorentz-covariant derivative

$$D\lambda^a = d\lambda^a + \omega^a{}_b\lambda^b, \tag{621}$$

we may write

$$\delta e^a = D\lambda^a - \lambda^a{}_b e^b. \tag{622}$$

The Lorentz part follows from

$$[J_{ab}, J_{cd}] = \eta_{ad}J_{bc} + \eta_{bc}J_{ad} - \eta_{ac}J_{bd} - \eta_{bd}J_{ac}. \tag{623}$$

After collecting the coefficient multiplying J_{ab}, one finds

$$\delta\omega^{ab} = d\lambda^{ab} + \omega^a{}_c\lambda^{cb} + \omega^b{}_c\lambda^{ac}. \tag{624}$$

Equivalently,

$$\delta\omega^{ab} = D\lambda^{ab}. \tag{625}$$

Thus the complete infinitesimal Poincaré transformation is

$$\begin{aligned} \delta e^a &= D\lambda^a - \lambda^a{}_b e^b, \\ \delta\omega^{ab} &= D\lambda^{ab}. \end{aligned} \tag{626}$$

It is useful to distinguish the two sectors. For a pure translation,

$$\lambda^{ab} = 0, \tag{627}$$

and therefore

$$\delta_{\mathrm{T}} e^a = D\lambda^a, \qquad \delta_{\mathrm{T}}\omega^{ab} = 0. \tag{628}$$

For a pure Lorentz transformation,

$$\lambda^a = 0, \tag{629}$$

and

$$\delta_{\mathrm{L}} e^a = -\lambda^a{}_b e^b, \qquad \delta_{\mathrm{L}}\omega^{ab} = D\lambda^{ab}. \tag{630}$$

The Lorentz transformation of the vielbein is therefore the usual vector transformation in the tangent space. The curvature

$$F = dA + A \wedge A \tag{631}$$

transforms covariantly,

$$F' = UFU^{-1}. \tag{632}$$

Infinitesimally we have

$$\delta_\lambda F = [F, \lambda]. \tag{633}$$

Using

$$F = T^a P_a + \frac{1}{2} R^{ab} J_{ab}, \tag{634}$$

one obtains, in particular, for a pure translation,

$$\delta_{\mathrm{T}} R^{ab} = 0, \tag{635}$$

while

$$\delta_{\mathrm{T}} T^a = R^a{}_b \, \lambda^b. \tag{636}$$

The last relation can also be seen directly:

$$\begin{aligned} \delta_{\mathrm{T}} T^a &= \delta_{\mathrm{T}} (D e^a) \\ &= D(\delta_{\mathrm{T}} e^a) \\ &= D^2 \lambda^a \\ &= R^a{}_b \lambda^b. \end{aligned} \tag{637}$$

This observation will become important below.

Local Translations and the Einstein–Cartan Action

Consider, for simplicity, the Einstein–Cartan action without a cosmological term,

$$S_{\mathrm{EC}}[e, \omega] = \frac{1}{4\kappa} \int_M \epsilon_{abcd} R^{ab} \wedge e^c \wedge e^d. \tag{638}$$

This action is manifestly invariant under local Lorentz transformations. Indeed, all Lorentz indices are contracted with the invariant tensor ϵ_{abcd}.

It is also invariant under spacetime diffeomorphisms, since its integrand is a four-form. The more subtle question is whether it is invariant under the local translations (628).

Under a pure translation,

$$\delta_{\mathrm{T}} e^a = D\lambda^a, \qquad \delta_{\mathrm{T}} \omega^{ab} = 0. \tag{639}$$

Consequently,

$$\delta_{\mathrm{T}} R^{ab} = 0. \tag{640}$$

The variation of the action is therefore

$$\begin{aligned}\delta_{\mathrm{T}} S_{\mathrm{EC}} &= \frac{1}{4\kappa}\int_M \epsilon_{abcd} R^{ab} \wedge \delta_{\mathrm{T}}\left(e^c \wedge e^d\right) \\ &= \frac{1}{2\kappa}\int_M \epsilon_{abcd} R^{ab} \wedge D\lambda^c \wedge e^d. \end{aligned} \tag{641}$$

To integrate by parts covariantly, consider

$$D\left(R^{ab}\lambda^c e^d\right). \tag{642}$$

Since R^{ab} is a two-form and λ^c is a zero-form,

$$D\left(R^{ab}\lambda^c e^d\right) = DR^{ab}\,\lambda^c e^d + R^{ab} D\lambda^c\, e^d + R^{ab}\lambda^c De^d. \tag{643}$$

Using the Bianchi identity

$$DR^{ab} = 0 \tag{644}$$

and

$$De^d = T^d, \tag{645}$$

we find

$$R^{ab} D\lambda^c e^d = D\left(R^{ab}\lambda^c e^d\right) - R^{ab}\lambda^c T^d. \tag{646}$$

Therefore,

$$\begin{aligned}\delta_{\mathrm{T}} S_{\mathrm{EC}} = \frac{1}{2\kappa}\int_M D\left(\epsilon_{abcd} R^{ab}\lambda^c e^d\right) \\ - \frac{1}{2\kappa}\int_M \epsilon_{abcd} R^{ab}\lambda^c T^d. \end{aligned} \tag{647}$$

Up to a boundary term,

$$\delta_{\mathrm{T}} S_{\mathrm{EC}} \doteq -\frac{1}{2\kappa}\int_M \epsilon_{abcd} R^{ab} \wedge T^d\,\lambda^c. \tag{648}$$

Hence the Einstein–Cartan action is, in general, *not invariant off shell* under local Poincaré translations.

At first sight, one might try to impose

$$T^a = 0, \tag{649}$$

which would make (648) vanish. However, the torsion-free condition itself is not preserved under a generic local translation.

Indeed,

$$\begin{aligned}\delta_{\mathrm{T}} T^a &= D(\delta_{\mathrm{T}} e^a) \\ &= D(D\lambda^a) \\ &= D^2\lambda^a \\ &= R^a{}_b \lambda^b. \end{aligned} \tag{650}$$

Thus

$$\delta_{\mathrm{T}} T^a = R^a{}_b \lambda^b, \tag{651}$$

which is generically nonzero. Therefore even if we consider

$$T^a = 0 \tag{652}$$

initially, a local translation generally takes us outside the torsion-free sector whenever the curvature is nonvanishing. This shows that the ordinary Einstein–Cartan action does not possess local internal translations as an independent off-shell gauge symmetry.

At this point an important distinction must be made. A coordinate transformation, or more generally a diffeomorphism, is *not* the same transformation as an internal Poincaré translation.

Under an infinitesimal diffeomorphism generated by the vector field

$$\xi = \xi^\mu \partial_\mu, \tag{653}$$

a field transforms according to its Lie derivative. For the vielbein,

$$\delta_\xi e^a = \mathcal{L}_\xi e^a, \tag{654}$$

or in components,

$$\delta_\xi e^a{}_\mu = \xi^\lambda \partial_\lambda e^a{}_\mu + e^a{}_\lambda \partial_\mu \xi^\lambda. \tag{655}$$

Thus, with respect to its spacetime index, the vielbein transforms as a covariant vector. Similarly, the spin connection is a spacetime one-form, and therefore

$$\delta_\xi \omega^{ab} = \mathcal{L}_\xi \omega^{ab}, \tag{656}$$

or

$$\delta_\xi \omega^{ab}{}_\mu = \xi^\lambda \partial_\lambda \omega^{ab}{}_\mu + \omega^{ab}{}_\lambda \partial_\mu \xi^\lambda. \tag{657}$$

By contrast, under a pure internal Poincaré translation we found

$$\delta_{\mathrm{T}} e^a = D\lambda^a, \qquad \delta_{\mathrm{T}} \omega^{ab} = 0. \tag{658}$$

These are manifestly different transformation laws. The relation between the two can be made more explicit by using Cartan's formula

$$\mathcal{L}_\xi = \iota_\xi d + d\iota_\xi. \tag{659}$$

For the vielbein,

$$\mathcal{L}_\xi e^a = \iota_\xi de^a + d(\iota_\xi e^a). \tag{660}$$

Using

$$de^a = T^a - \omega^a{}_b \wedge e^b, \tag{661}$$

we obtain

$$\mathcal{L}_\xi e^a = D(\iota_\xi e^a) - (\iota_\xi \omega^a{}_b) e^b + \iota_\xi T^a. \tag{662}$$

Likewise, for the spin connection,

$$\mathcal{L}_\xi \omega^{ab} = D(\iota_\xi \omega^{ab}) + \iota_\xi R^{ab}. \tag{663}$$

Now define field-dependent gauge parameters

$$\lambda^a = \iota_\xi e^a, \qquad \lambda^{ab} = \iota_\xi \omega^{ab}. \tag{664}$$

The corresponding Poincaré gauge transformation gives

$$\delta_\lambda e^a = D(\iota_\xi e^a) - (\iota_\xi \omega^a{}_b) e^b, \tag{665}$$

so that

$$\mathcal{L}_\xi e^a = \delta_\lambda e^a + \iota_\xi T^a. \tag{666}$$

Similarly,

$$\mathcal{L}_\xi \omega^{ab} = \delta_\lambda \omega^{ab} + \iota_\xi R^{ab}. \tag{667}$$

Thus diffeomorphisms and internal Poincaré gauge transformations are closely related, but they are not identical in a generic curved, torsionful geometry.

Conceptual distinction

Local translations act in the internal Poincaré gauge space:

$$\delta_{\mathrm{T}} e^a = D\lambda^a, \qquad \delta_{\mathrm{T}} \omega^{ab} = 0.$$

Diffeomorphisms act on spacetime through the Lie derivative:

$$\delta_\xi e^a = \mathcal{L}_\xi e^a, \qquad \delta_\xi \omega^{ab} = \mathcal{L}_\xi \omega^{ab}.$$

Their difference is controlled precisely by torsion and curvature, as shown in (662) and (663).

Local Poincaré Translations and Diffeomorphisms

We have seen that local translations in the Poincaré gauge formulation are not, in general, identical to general coordinate transformations or diffeomorphisms.

Under a local Poincaré translation with parameter ρ^a, the first-order gravitational variables transform as

$$\delta_\rho e^a = D\rho^a, \qquad \delta_\rho \omega^{ab} = 0. \tag{668}$$

On the other hand, a spacetime diffeomorphism generated by a vector field

$$\xi = \xi^\mu \partial_\mu \tag{669}$$

acts through the Lie derivative. To compare these two transformations, it is convenient to derive a gauge-covariant expression for the Lie derivative of a connection.

Let A be a Lie-algebra-valued connection one-form. Cartan's formula for the Lie derivative gives

$$\mathcal{L}_\xi A = d(\iota_\xi A) + \iota_\xi(dA), \tag{670}$$

where ι_ξ denotes contraction with the vector field ξ.

Using

$$F = dA + A \wedge A, \tag{671}$$

we may write

$$\begin{aligned} \mathcal{L}_\xi A &= d(\iota_\xi A) + \iota_\xi \left(F - A \wedge A\right) \\ &= d(\iota_\xi A) + \iota_\xi F - \iota_\xi(A \wedge A). \end{aligned} \tag{672}$$

The contraction operator satisfies the graded Leibniz rule,

$$\iota_\xi(\alpha \wedge \beta) = (\iota_\xi \alpha) \wedge \beta + (-1)^p \alpha \wedge (\iota_\xi \beta), \tag{673}$$

when α is a p-form. Since A is a one-form,

$$\begin{aligned} \iota_\xi(A \wedge A) &= (\iota_\xi A)A - A(\iota_\xi A) \\ &= [\iota_\xi A, A]. \end{aligned} \tag{674}$$

Therefore,

$$\mathcal{L}_\xi A = d(\iota_\xi A) + [A, \iota_\xi A] + \iota_\xi F. \tag{675}$$

Recognizing the gauge-covariant derivative,

$$D(\iota_\xi A) = d(\iota_\xi A) + [A, \iota_\xi A], \tag{676}$$

we arrive at the important identity

$$\mathcal{L}_\xi A = D(\iota_\xi A) + \iota_\xi F. \tag{677}$$

This relation makes the connection between gauge transformations and diffeomorphisms particularly transparent. For the Lorentz connection,

$$F^{ab} = R^{ab}, \tag{678}$$

and therefore

$$\mathcal{L}_\xi \omega^{ab} = D(\iota_\xi \omega^{ab}) + \iota_\xi R^{ab}. \tag{679}$$

In components,

$$(\mathcal{L}_\xi \omega^{ab})_\mu = D_\mu \left(\xi^\nu \omega^{ab}{}_\nu\right) + \xi^\nu R^{ab}{}_{\nu\mu}. \tag{680}$$

The first term has precisely the form of a local Lorentz transformation with the field-dependent parameter

$$\lambda^{ab} = \iota_\xi \omega^{ab} = \xi^\mu \omega^{ab}{}_\mu. \tag{681}$$

The second term measures the difference between the diffeomorphism and this internal gauge transformation. For the vielbein, Cartan's first structure equation is

$$T^a = De^a. \tag{682}$$

Starting from the ordinary Lie derivative,

$$\mathcal{L}_\xi e^a = d(\iota_\xi e^a) + \iota_\xi (de^a), \tag{683}$$

and using

$$de^a = T^a - \omega^a{}_b \wedge e^b, \tag{684}$$

we obtain

$$\mathcal{L}_\xi e^a = D(\iota_\xi e^a) - (\iota_\xi \omega^a{}_b) e^b + \iota_\xi T^a. \tag{685}$$

Define the field-dependent parameters

$$\rho^a = \iota_\xi e^a, \qquad \lambda^{ab} = \iota_\xi \omega^{ab}. \tag{686}$$

Then

$$\mathcal{L}_\xi e^a = D\rho^a - \lambda^a{}_b e^b + \iota_\xi T^a. \tag{687}$$

Thus, when the torsion vanishes,

$$T^a = 0, \tag{688}$$

a diffeomorphism can locally be represented as a combination of

- a local translation with parameter $\rho^a = \iota_\xi e^a$, and
- a local Lorentz transformation with parameter $\lambda^{ab} = \iota_\xi \omega^{ab}$.

In this precise sense, internal translations and spacetime diffeomorphisms become closely related in the torsion-free sector. There is, however, an important subtlety.

Imposing

$$T^a = 0 \tag{689}$$

allows us to solve the spin connection as a function of the vielbein,

$$\omega^{ab} = \mathring{\omega}^{ab}(e). \tag{690}$$

Once this is done, e^a and ω^{ab} are no longer independent. Therefore a variation of the vielbein necessarily induces a variation of the spin connection,

$$\delta\omega^{ab} = \frac{\delta \mathring{\omega}^{ab}}{\delta e^c}\, \delta e^c, \tag{691}$$

which is generically nonzero.

Consequently, the transformation law

$$\delta_\rho \omega^{ab} = 0 \tag{692}$$

cannot be maintained after the torsion-free condition has been solved explicitly. This is why, in the usual second-order formulation of general relativity, the manifest gauge symmetries are diffeomorphisms for the metric formulation $S[g_{\mu\nu}]$, together with local Lorentz transformations when working with vielbeins.

Gauge translations versus diffeomorphisms

The identification

$$\rho^a = \iota_\xi e^a$$

is field-dependent. Therefore the relation between local translations and diffeomorphisms is not a simple identification of two independent Lie algebras. This field dependence is one reason why the symmetry structure of gravity is naturally described in terms of gauge algebroid-like structures rather than a finite-dimensional Lie algebra with constant structure coefficients.

6 Gravity in 2+1 Dimensions

Three-dimensional gravity has several remarkable properties that make it qualitatively different from gravity in four or more spacetime dimensions.

In particular, pure Einstein gravity in $2+1$ dimensions has no local propagating gravitational degrees of freedom. At the same time, its first-order formulation has a particularly elegant gauge-theoretic description.

The Einstein–Cartan Action in Three Dimensions

We begin with the Einstein–Cartan formulation. Let $\mathcal{M}_3$ be a three-dimensional spacetime. Since the vielbein e^a is a one-form and the curvature R^{ab} is a two-form, the combination

$$\epsilon_{abc}\, R^{ab} \wedge e^c \tag{693}$$

is already a three-form and can therefore be integrated over $\mathcal{M}_3$. The Einstein–Cartan action with vanishing cosmological constant may thus be written as

$$S[e,\omega] = k \int_{M_3} \epsilon_{abc} R^{ab} \wedge e^c, \tag{694}$$

where k is an overall normalization. Consider the local translational transformation

$$\delta_\rho e^a = D\rho^a, \qquad \delta_\rho \omega^{ab} = 0. \tag{695}$$

Since

$$\delta_\rho R^{ab} = 0, \tag{696}$$

the variation of the action is

$$\delta_\rho S = k \int_{M_3} \epsilon_{abc} R^{ab} \wedge D\rho^c. \tag{697}$$

Using the Leibniz rule,

$$D\left(\epsilon_{abc} R^{ab} \rho^c\right) = \epsilon_{abc} DR^{ab}\, \rho^c + \epsilon_{abc} R^{ab} \wedge D\rho^c. \tag{698}$$

The Bianchi identity gives

$$DR^{ab} = 0. \tag{699}$$

Therefore,

$$\delta_\rho S = k \int_{M_3} D\left(\epsilon_{abc} R^{ab} \rho^c\right). \tag{700}$$

Since the quantity inside D is a Lorentz scalar, D reduces to the ordinary exterior derivative,

$$\delta_\rho S = k \int_{M_3} d\left(\epsilon_{abc} R^{ab} \rho^c\right). \tag{701}$$

Thus

$$\delta_\rho S = k \int_{\partial M_3} \epsilon_{abc} R^{ab} \rho^c. \tag{702}$$

Hence the three-dimensional Einstein–Cartan action is invariant under local translations up to a boundary term. This is a special feature of three dimensions.

Cosmological Constant and the (A)dS Gauge Groups

The action with cosmological constant can be written as

$$S[e,\omega] = k \int_{M_3} \epsilon_{abc} \left(R^{ab} \wedge e^c - \frac{\Lambda}{3} e^a \wedge e^b \wedge e^c \right). \tag{703}$$

It is often useful to introduce a length scale ℓ such that

$$\Lambda = \begin{cases} -\ell^{-2}, & \mathrm{AdS}_3, \\ 0, & \text{Minkowski}, \\ +\ell^{-2}, & \mathrm{dS}_3. \end{cases} \tag{704}$$

Therefore,

$$-\frac{\Lambda}{3} e^a e^b e^c = \begin{cases} +\dfrac{1}{3\ell^2} e^a e^b e^c, & \Lambda < 0, \\ -\dfrac{1}{3\ell^2} e^a e^b e^c, & \Lambda > 0. \end{cases} \tag{705}$$

Here and below wedge products between forms may occasionally be left implicit. The corresponding isometry groups are

Λ	Gauge/isometry group
0	$ISO(2,1)$
$\Lambda < 0$	$SO(2,2)$
$\Lambda > 0$	$SO(3,1)$.

(706)

The first corresponds to Minkowski space, the second to AdS_3, and the third to dS_3.

A convenient convention for the $(A)dS$ algebra is

$$[P_a, P_b] = -\Lambda J_{ab}. \tag{707}$$

With this convention, an infinitesimal translational transformation is

$$\boxed{\delta_\rho e^a = D\rho^a,} \tag{708}$$

together with

$$\delta_\rho \omega^{ab} = -\Lambda \left(e^a \rho^b - e^b \rho^a \right). \tag{709}$$

For $\Lambda = 0$, this reduces to the Poincaré transformation

$$\delta_\rho e^a = D\rho^a, \qquad \delta_\rho \omega^{ab} = 0. \tag{710}$$

The Lorentz transformations remain

$$\delta_\lambda e^a = -\lambda^a{}_b e^b, \qquad \delta_\lambda \omega^{ab} = D\lambda^{ab}. \tag{711}$$

Thus, unlike the generic four-dimensional Einstein–Cartan action, the three-dimensional theory can be organized naturally as a gauge theory of the Poincaré, anti-de Sitter, or de Sitter group.

Metric Formulation in Three Dimensions

We may also eliminate the spin connection by imposing its equation of motion

$$T^a = 0. \tag{712}$$

For an invertible vielbein, this determines

$$\omega^{ab} = \mathring{\omega}^{ab}(e). \tag{713}$$

Using the vielbein postulate then leads to the usual metric formulation. With the conventional normalization, the action is

$$S_{\rm EH}[g] = \frac{1}{16\pi G} \int_{M_3} d^3x \, \sqrt{-g} \, (R - 2\Lambda) + S_{\rm boundary}, \tag{714}$$

where G is the three-dimensional Newton constant. The Einstein equations are

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 0. \tag{715}$$

Taking the trace in three dimensions,

$$0 = R - \frac{3}{2} R + 3\Lambda, \tag{716}$$

and hence

$$\boxed{R = 6\Lambda.} \tag{717}$$

Substituting this result back into the Einstein equation gives

$$R_{\mu\nu} - 3\Lambda g_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \tag{718}$$

so that

$$\boxed{R_{\mu\nu} = 2\Lambda g_{\mu\nu}.} \tag{719}$$

Therefore every vacuum solution of three-dimensional Einstein gravity is an Einstein manifold.

Why Gravity in Three Dimensions Has No Local Gravitons

The previous result becomes much stronger in three dimensions because the Weyl tensor vanishes identically. To understand this, recall the decomposition of the Riemann tensor in d dimensions:

$$R_{\alpha\beta\gamma\delta} = C_{\alpha\beta\gamma\delta} + \frac{2}{d-2} \left(g_{\alpha[\gamma} R_{\delta]\beta} - g_{\beta[\gamma} R_{\delta]\alpha} \right) - \frac{2R}{(d-1)(d-2)} g_{\alpha[\gamma} g_{\delta]\beta}. \tag{720}$$

In $d = 3$, this becomes

$$R_{\alpha\beta\gamma\delta} = C_{\alpha\beta\gamma\delta} + 2 \left(g_{\alpha[\gamma} R_{\delta]\beta} - g_{\beta[\gamma} R_{\delta]\alpha} \right) - R\, g_{\alpha[\gamma} g_{\delta]\beta}. \tag{721}$$

However,

$$C_{\alpha\beta\gamma\delta} = 0 \qquad \text{identically in } d = 3. \tag{722}$$

Consequently, the entire Riemann tensor is determined algebraically by the Ricci tensor and the Ricci scalar. For an Einstein solution,

$$R_{\mu\nu} = 2\Lambda g_{\mu\nu}, \qquad R = 6\Lambda. \tag{723}$$

Substitution into (721) yields

$$R_{\alpha\beta\gamma\delta} = \Lambda \left(g_{\alpha\gamma} g_{\beta\delta} - g_{\alpha\delta} g_{\beta\gamma} \right). \tag{724}$$

This is precisely the Riemann tensor of a constant-curvature spacetime. Therefore every vacuum solution is locally

$$\begin{cases} \mathrm{AdS}_3, & \Lambda < 0, \\ \mathbb{R}^{2,1}, & \Lambda = 0, \\ \mathrm{dS}_3, & \Lambda > 0. \end{cases} \tag{725}$$

This statement is *local*. Globally, different identifications and topologies can still lead to physically inequivalent geometries.

The absence of the Weyl tensor in three dimensions can also be understood by counting independent components. The Riemann tensor satisfies

$$R_{\mu\nu\rho\sigma} = -R_{\nu\mu\rho\sigma} = -R_{\mu\nu\sigma\rho} = R_{\rho\sigma\mu\nu}. \tag{726}$$

An antisymmetric pair $[\mu\nu]$ has

$$N = \frac{d(d-1)}{2} \tag{727}$$

independent values. Thus the pair symmetry allows us initially to regard the Riemann tensor as a symmetric $N \times N$ matrix, giving

$$\frac{N(N+1)}{2} \tag{728}$$

components. We must still impose the algebraic Bianchi identity

$$R_{[\mu\nu\rho\sigma]} = 0. \tag{729}$$

It removes

$$\binom{d}{4} = \frac{d(d-1)(d-2)(d-3)}{4!} \tag{730}$$

components. Therefore the number of independent components of the Riemann tensor is

$$\begin{aligned} N_{\mathrm{Riem}} &= \frac{1}{2}N(N+1) - \binom{d}{4} \\ &= \boxed{\frac{d^2(d^2-1)}{12}}. \end{aligned} \tag{731}$$

For $d = 3$,

$$N_{\mathrm{Riem}} = \frac{9(8)}{12} = 6. \tag{732}$$

The Ricci tensor is symmetric and therefore has

$$N_{\mathrm{Ricci}} = \frac{d(d+1)}{2} \tag{733}$$

independent components. In three dimensions,

$$N_{\mathrm{Ricci}} = \frac{3(4)}{2} = 6. \tag{734}$$

Thus

$$N_{\text{Riem}} = N_{\text{Ricci}} = 6 \qquad (d = 3). \tag{735}$$

There are no independent components left over for a Weyl tensor. More generally,

$$\begin{aligned} N_{\text{Weyl}} &= N_{\text{Riem}} - N_{\text{Ricci}} \\ &= \frac{d^2(d^2-1)}{12} - \frac{d(d+1)}{2} \\ &= \frac{d(d+1)(d+2)(d-3)}{12}. \end{aligned} \tag{736}$$

Indeed,

$$N_{\text{Weyl}} = 0 \qquad \text{for } d = 3. \tag{737}$$

The same conclusion can be obtained directly from the gravitational degrees of freedom. A symmetric metric $g_{\mu\nu}$ in d spacetime dimensions contains $\frac{d(d+1)}{2}$ independent components.

Diffeomorphism invariance removes d arbitrary functions. In the Hamiltonian formulation there are also d corresponding constraint equations. Thus the number of local configuration-space degrees of freedom is

$$\begin{aligned} N_{\text{phys}} &= \frac{d(d+1)}{2} - d - d \\ &= \frac{d(d-3)}{2}. \end{aligned} \tag{738}$$

For four-dimensional general relativity,

$$N_{\text{phys}}(d = 4) = 2, \tag{739}$$

corresponding to the two graviton polarizations. In three spacetime dimensions,

$$N_{\text{phys}}(d = 3) = 0. \tag{740}$$

Thus pure Einstein gravity in $2 + 1$ dimensions has no local propagating gravitational degrees of freedom.

Local triviality does not mean global triviality

The statement

$$N_{\text{phys}} = 0$$

means that there are no local graviton degrees of freedom.
It does *not* mean that three-dimensional gravity is physically empty. Global identifications, nontrivial topology, boundaries and asymptotic degrees of freedom can all carry nontrivial information. Indeed, this will be crucial when studying geometries such as the BTZ black hole and the asymptotic symmetry structure of AdS_3.

Gravity in $2+1$ Dimensions as a Gauge Theory

We now make more precise the gauge-theoretic formulation of three-dimensional gravity. For definiteness, we begin with a negative cosmological constant,

$$\Lambda = -\frac{1}{\ell^2}, \tag{741}$$

so that the maximally symmetric vacuum is AdS_3 and the corresponding isometry group is

$$SO(2,2). \tag{742}$$

The central idea is to combine the spin connection ω^{ab} and the vielbein e^a into a single $SO(2,2)$ gauge connection.nLet capital Latin indices

$$A, B, \ldots = 0, 1, 2, 3 \tag{743}$$

denote indices in the fundamental representation of $SO(2,2)$, while

$$a, b, \ldots = 0, 1, 2 \tag{744}$$

denote Lorentz indices of $SO(2,1)$. We introduce the $SO(2,2)$ connection

$$W^{AB} = \begin{pmatrix} \omega^{ab} & \dfrac{1}{\ell} e^a \\ -\dfrac{1}{\ell} e^b & 0 \end{pmatrix}. \tag{745}$$

The connection is antisymmetric,

$$W^{AB} = -W^{BA}, \tag{746}$$

as appropriate for an orthogonal group. The invariant metric of $SO(2,2)$ may be written as

$$\Pi_{AB} = \begin{pmatrix} \eta_{ab} & 0 \\ 0 & -1 \end{pmatrix}, \tag{747}$$

where η_{ab} is the Lorentz metric in three dimensions. The group $SO(2,2)$ possesses two important invariant tensors:

$$\Pi_{AB}, \qquad \epsilon_{ABCD}. \tag{748}$$

The three-dimensional Levi–Civita tensor is naturally obtained from the four-dimensional one by fixing the extra internal index:

$$\epsilon_{abc} \equiv \epsilon_{abc3}. \tag{749}$$

These invariant tensors will allow us to construct gauge-invariant polynomials from the curvature. The curvature two-form associated with W^{AB} is

$$F^{AB} = dW^{AB} + W^A{}_C \wedge W^{CB}, \tag{750}$$

where

$$W^A{}_C = W^{AD}\Pi_{DC}. \tag{751}$$

Let us calculate its components. For the purely Lorentz components,

$$\begin{aligned} F^{ab} &= d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb} + \frac{1}{\ell^2} e^a \wedge e^b \\ &= R^{ab} + \frac{1}{\ell^2} e^a \wedge e^b. \end{aligned} \tag{752}$$

For the mixed components,

$$\begin{aligned} F^{a3} &= \frac{1}{\ell} \left(de^a + \omega^a{}_b \wedge e^b \right) \\ &= \frac{1}{\ell} T^a. \end{aligned} \tag{753}$$

Therefore,

$$F^{AB} = \begin{pmatrix} R^{ab} + \dfrac{1}{\ell^2} e^a \wedge e^b & \dfrac{1}{\ell} T^a \\ -\dfrac{1}{\ell} T^b & 0 \end{pmatrix}. \tag{754}$$

This equation is one of the key results of the gauge formulation: the ordinary gravitational curvature and torsion are simply different components of a single $SO(2,2)$ field strength.

In particular,

$$F^{AB} = 0 \tag{755}$$

implies

$$T^a = 0, \qquad R^{ab} = -\frac{1}{\ell^2} e^a \wedge e^b, \tag{756}$$

which are precisely the vacuum Einstein equations for locally AdS_3 geometry.

Invariant Polynomials of the $SO(2,2)$ Curvature

Using the invariant tensors ϵ_{ABCD} and Π_{AB}, we can construct gauge-invariant four-forms from the curvature. Two particularly important examples are the Euler-type and Pontryagin-type densities.

Consider

$$E_4 = \frac{1}{2} \epsilon_{ABCD} F^{AB} \wedge F^{CD}. \tag{757}$$

Since ϵ_{ABCD} is completely antisymmetric, a nonzero contribution must contain the index 3. Hence,

$$E_4 = 2\epsilon_{abc} F^{a3} \wedge F^{bc}. \tag{758}$$

Using (754),

$$\begin{aligned} E_4 &= \frac{2}{\ell} \epsilon_{abc} T^a \wedge \left(R^{bc} + \frac{1}{\ell^2} e^b \wedge e^c \right) \\ &= \frac{2}{\ell} \epsilon_{abc} T^a \wedge R^{bc} + \frac{2}{\ell^3} \epsilon_{abc} T^a \wedge e^b \wedge e^c. \end{aligned} \tag{759}$$

Now recall

$$T^a = De^a, \qquad DR^{ab} = 0. \tag{760}$$

Therefore,

$$d\left(\epsilon_{abc} R^{ab} \wedge e^c\right) = \epsilon_{abc} R^{ab} \wedge T^c. \tag{761}$$

Similarly,

$$d\left(\epsilon_{abc} e^a \wedge e^b \wedge e^c\right) = 3\epsilon_{abc} T^a \wedge e^b \wedge e^c. \tag{762}$$

It follows that

$$E_4 = d\left[\frac{2}{\ell}\epsilon_{abc}\left(R^{ab} \wedge e^c + \frac{1}{3\ell^2} e^a \wedge e^b \wedge e^c\right)\right]. \tag{763}$$

This is already a remarkable result. The three-form inside the derivative is proportional to the three-dimensional Einstein–Cartan Lagrangian with negative cosmological constant. Indeed, defining

$$L^{(3)}_{\mathrm{EC}} = \epsilon_{abc}\left(R^{ab} \wedge e^c + \frac{1}{3\ell^2} e^a \wedge e^b \wedge e^c\right), \tag{764}$$

we have

$$E_4 = d\left(\frac{2}{\ell} L^{(3)}_{\mathrm{EC}}\right). \tag{765}$$

Thus the Einstein–Cartan Lagrangian in $2+1$ dimensions arises naturally as a Chern–Simons form associated with the Euler invariant of the $SO(2,2)$ curvature.

A second invariant is obtained using the invariant metric:

$$P_4 = F^A{}_B \wedge F^B{}_A. \tag{766}$$

Substituting the decomposition of the curvature gives

$$P_4 = R^a{}_b \wedge R^b{}_a + \frac{2}{\ell^2}\left(T^a \wedge T_a - R_{ab} \wedge e^a \wedge e^b\right). \tag{767}$$

The first term is the usual Lorentz Pontryagin density. The remaining combination satisfies the Nieh–Yan identity

$$d(e_a \wedge T^a) = T_a \wedge T^a - R_{ab} \wedge e^a \wedge e^b. \tag{768}$$

Moreover,

$$R^a{}_b \wedge R^b{}_a = d\left[\omega^a{}_b \wedge \left(R^b{}_a - \frac{1}{3}\omega^b{}_c \wedge \omega^c{}_a\right)\right]. \tag{769}$$

Hence

$$\boxed{P_4 = d\left[\omega^a{}_b \wedge \left(R^b{}_a - \frac{1}{3}\omega^b{}_c \wedge \omega^c{}_a\right) + \frac{2}{\ell^2} e_a \wedge T^a\right].} \tag{770}$$

The three-form

$$\omega^a{}_b \wedge \left(R^b{}_a - \frac{1}{3}\omega^b{}_c \wedge \omega^c{}_a \right) \tag{771}$$

is the Lorentz Chern–Simons form. Thus both of the natural four-dimensional invariant polynomials give rise to three-dimensional Chern–Simons-type Lagrangians.

At first sight it may appear strange to introduce four-forms when our physical spacetime is three-dimensional. The reason is that characteristic forms such as E_4 and P_4 are locally exact:

$$E_4 = dQ_E, \qquad P_4 = dQ_P. \tag{772}$$

The three-forms Q_E and Q_P can then be integrated directly over a three-dimensional manifold. The Einstein–Cartan action itself is therefore understood as the Chern–Simons form whose exterior derivative gives the Euler characteristic density associated with the enlarged $SO(2,2)$ connection. This observation motivates the general theory of Chern–Simons forms.

Invariant Polynomials and Chern–Simons Forms

Let A be a Lie-algebra-valued connection and

$$F = dA + A \wedge A \tag{773}$$

its curvature. Under a finite gauge transformation $g(x)$,

$$A \longrightarrow A^g = g^{-1}Ag + g^{-1}dg, \tag{774}$$

while the curvature transforms homogeneously,

$$F \longrightarrow F^g = g^{-1}Fg. \tag{775}$$

This difference is fundamental: A is a connection and transforms inhomogeneously, whereas F transforms covariantly. Let

$$\left\langle T_{A_1} \cdots T_{A_{n+1}} \right\rangle \tag{776}$$

denote an invariant symmetric multilinear form on the Lie algebra.

Using it, we construct the $(2n+2)$-form

$$P_{2n+2}(F) = \left\langle F^{n+1} \right\rangle . \tag{777}$$

The notation means

$$\left\langle F^{n+1} \right\rangle = \left\langle F \wedge F \wedge \cdots \wedge F \right\rangle , \tag{778}$$

with $n+1$ factors of F. Because F transforms in the adjoint representation,

$$F^g = g^{-1}Fg, \tag{779}$$

the invariance of the multilinear form implies

$$\left\langle (F^g)^{n+1} \right\rangle = \left\langle F^{n+1} \right\rangle . \tag{780}$$

Hence these forms are gauge invariant.

The Chern–Weil Theorem

The structure encountered above is a particular example of thenChern–Weil theorem. Let

$$P_{2n+2}(F) = \left\langle F^{n+1} \right\rangle \tag{781}$$

be an invariant polynomial of degree $n+1$ in the curvature. The theorem implies two crucial properties.

We have

$$d \left\langle F^{n+1} \right\rangle = 0. \tag{782}$$

To see this, note that because the complete expression is gauge invariant,

$$d \left\langle F^{n+1} \right\rangle = D \left\langle F^{n+1} \right\rangle. \tag{783}$$

Using the Leibniz rule and invariance of the bracket,

$$D \left\langle F^{n+1} \right\rangle = (n+1) \left\langle DF\, F^{n} \right\rangle. \tag{784}$$

The Bianchi identity

$$DF = 0 \tag{785}$$

therefore gives

$$d \left\langle F^{n+1} \right\rangle = 0. \tag{786}$$

Thus characteristic forms are closed. Consider two connections A and $\bar{A}$, with curvatures

$$F = dA + A^2, \qquad \bar{F} = d\bar{A} + \bar{A}^2. \tag{787}$$

The difference of their characteristic forms is exact:

$$\left\langle F^{n+1} \right\rangle - \left\langle \bar{F}^{\,n+1} \right\rangle = dQ_T^{(2n+1)}(A, \bar{A}). \tag{788}$$

The $(2n+1)$-form

$$Q_T^{(2n+1)}(A, \bar{A}) \tag{789}$$

is called the *transgression form*. To construct it, define

$$\Theta = A - \bar{A} \tag{790}$$

and interpolate between the two connections:

$$A_t = \bar{A} + t\Theta, \qquad 0 \le t \le 1. \tag{791}$$

Thus

$$A_0 = \bar{A}, \qquad A_1 = A. \tag{792}$$

The corresponding curvature is

$$F_t = dA_t + A_t \wedge A_t. \tag{793}$$

Differentiating with respect to t gives

$$\frac{dF_t}{dt} = D_t \Theta, \tag{794}$$

where

$$D_t = d + [A_t, \cdot]. \tag{795}$$

Therefore,

$$\begin{aligned} \frac{d}{dt} \left\langle F_t^{n+1} \right\rangle &= (n+1) \left\langle D_t \Theta \, F_t^n \right\rangle \\ &= (n+1) d \left\langle \Theta F_t^n \right\rangle, \end{aligned} \tag{796}$$

where the Bianchi identity $D_t F_t = 0$ has been used. Integrating from $t = 0$ to $t = 1$ yields

$$\left\langle F^{n+1} \right\rangle - \left\langle \bar{F}^{\,n+1} \right\rangle = (n+1) d \int_0^1 dt \, \left\langle \Theta F_t^n \right\rangle. \tag{797}$$

Hence

$$Q_T^{(2n+1)}(A, \bar{A}) = (n+1) \int_0^1 dt \, \left\langle \Theta F_t^n \right\rangle. \tag{798}$$

Locally, or after choosing a trivial reference connection, we may set

$$\bar{A} = 0. \tag{799}$$

Then

$$\Theta = A, \qquad A_t = tA. \tag{800}$$

Its curvature is

$$\begin{aligned} F_t &= d(tA) + (tA)^2 \\ &= t \, dA + t^2 A^2. \end{aligned} \tag{801}$$

Equation (788) becomes

$$\left\langle F^{n+1} \right\rangle = d Q_{\mathrm{CS}}^{(2n+1)}(A), \tag{802}$$

with

$$Q_{\mathrm{CS}}^{(2n+1)}(A) = (n+1) \int_0^1 dt \, \left\langle A \left(t \, dA + t^2 A^2 \right)^n \right\rangle. \tag{803}$$

This is the general Chern–Simons form. The relation

$$d Q_{\mathrm{CS}}^{(2n+1)} = \left\langle F^{n+1} \right\rangle \tag{804}$$

explains why a gauge-invariant characteristic class in even dimension naturally produces a Chern–Simons theory in one dimension lower.

The case relevant for three-dimensional gravity corresponds to $n = 1$. Then

$$\begin{aligned} Q^{(3)}_{\mathrm{CS}}(A) &= 2\int_0^1 dt\, \left\langle A\left(t\,dA + t^2 A^2\right)\right\rangle \\ &= 2\left[\frac{t^2}{2}\langle A\,dA\rangle + \frac{t^3}{3}\left\langle A^3\right\rangle\right]_0^1 . \end{aligned} \tag{805}$$

Therefore,

$$Q^{(3)}_{\mathrm{CS}}(A) = \left\langle A\wedge dA + \frac{2}{3}A\wedge A\wedge A\right\rangle . \tag{806}$$

Its exterior derivative is

$$dQ^{(3)}_{\mathrm{CS}}(A) = \langle F\wedge F\rangle . \tag{807}$$

Thus the familiar three-dimensional Chern–Simons Lagrangian is the transgression form associated with the four-dimensional invariant polynomial $\langle F\wedge F\rangle$.

We can now understand the previous calculation conceptually.

For the $SO(2,2)$ connection

$$W^{AB} = \begin{pmatrix} \omega^{ab} & e^a/\ell \\ -e^b/\ell & 0 \end{pmatrix}, \tag{808}$$

choose the invariant bilinear form defined by

$$\langle J_{AB}J_{CD}\rangle \propto \epsilon_{ABCD}. \tag{809}$$

Then

$$\langle F\wedge F\rangle \propto E_4, \tag{810}$$

and the corresponding Chern–Simons three-form is proportional to

$$\epsilon_{abc}\left(R^{ab}\wedge e^c + \frac{1}{3\ell^2}e^a\wedge e^b\wedge e^c\right). \tag{811}$$

But this is precisely the first-order Einstein–Hilbert action with negative cosmological constant.

Therefore, in three dimensions,

$$\boxed{\text{Einstein gravity with } \Lambda<0 \quad\longleftrightarrow\quad SO(2,2) \text{ Chern–Simons theory.}} \tag{812}$$

The equivalence will become even more transparent after decomposing

$$SO(2,2) \simeq SL(2,\mathbb{R})\times SL(2,\mathbb{R}), \tag{813}$$

which allows the gravitational theory to be written in terms of two independent $SL(2,\mathbb{R})$ Chern–Simons connections.

Chern–Simons Theory and $2+1$-Dimensional Gravity

We now study the Chern–Simons action itself and then show explicitly how three-dimensional Einstein gravity can be obtained from it.

The Chern–Simons Action

Let M be a three-dimensional manifold and let $\mathfrak{g}$ be a Lie algebra with generators T_a,

$$[T_a, T_b] = C_{ab}{}^c T_c. \tag{814}$$

A gauge connection is a Lie-algebra-valued one-form,

$$A = A_\mu{}^a \, dx^\mu \, T_a. \tag{815}$$

The Chern–Simons action is

$$S_{\mathrm{CS}}[A] = \frac{k}{4\pi} \int_M \left\langle A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right\rangle, \tag{816}$$

where k is called the *Chern–Simons level* and $\langle \cdots \rangle$ denotes an invariant symmetric bilinear form on the Lie algebra. The latter is defined by

$$\langle T_a, T_b \rangle = g_{ab}. \tag{817}$$

For a semisimple Lie algebra one can use the Killing form, although more generally Chern–Simons theory only requires a symmetric, invariant and non-degenerate bilinear form. Let us derive the equations of motion. Varying the first term gives

$$\delta \langle A \wedge dA \rangle = \langle \delta A \wedge dA + A \wedge d(\delta A) \rangle \, . \tag{818}$$

Using

$$d(A \wedge \delta A) = dA \wedge \delta A - A \wedge d(\delta A), \tag{819}$$

we obtain

$$A \wedge d(\delta A) = dA \wedge \delta A - d(A \wedge \delta A). \tag{820}$$

Since dA is a two-form and δA is a one-form,

$$dA \wedge \delta A = \delta A \wedge dA. \tag{821}$$

Therefore,

$$\delta \langle A \wedge dA \rangle = 2 \langle \delta A \wedge dA \rangle - d \langle A \wedge \delta A \rangle. \tag{822}$$

For the cubic term, cyclicity of the invariant form gives

$$\delta \langle A \wedge A \wedge A \rangle = 3 \langle \delta A \wedge A \wedge A \rangle \,. \tag{823}$$

Hence

$$\begin{aligned} \delta S_{\mathrm{CS}} = \frac{k}{2\pi} \int_M \langle \delta A \wedge (dA + A \wedge A) \rangle \\ - \frac{k}{4\pi} \int_{\partial M} \langle A \wedge \delta A \rangle \,. \end{aligned} \tag{824}$$

The curvature of A is

$$F = dA + A \wedge A. \tag{825}$$

Thus, if the boundary term vanishes because of appropriate boundary conditions, or after adding a suitable boundary term to the action, the bulk variation becomes

$$\delta S_{\mathrm{CS}} = \frac{k}{2\pi} \int_M \langle \delta A \wedge F \rangle. \tag{826}$$

The equations of motion therefore follow from

$$\langle \delta A \wedge F \rangle = 0. \tag{827}$$

Expanding

$$\delta A = \delta A^a T_a, \qquad F = F^b T_b, \tag{828}$$

we obtain

$$\langle \delta A \wedge F \rangle = \delta A^a \wedge F^b \langle T_a, T_b \rangle = \delta A^a \wedge F^b g_{ab}. \tag{829}$$

Therefore the equations of motion initially take the form

$$g_{ab} F^b = 0. \tag{830}$$

In order to conclude from this equation that

$$F^a = 0, \tag{831}$$

the matrix g_{ab} must be invertible:

$$\det g_{ab} \neq 0. \tag{832}$$

Thus the bilinear form used in the Chern–Simons action must be *non-degenerate*.

The field equation is consequently

$$F = 0. \tag{833}$$

This is the fundamental equation of motion of pure Chern–Simons theory. The bilinear form must also be invariant under the adjoint action of the Lie algebra. For any generators,

$$\langle [T_a, T_b], T_c \rangle + \langle T_b, [T_a, T_c] \rangle = 0. \tag{834}$$

Using

$$[T_a, T_b] = C_{ab}{}^d T_d, \tag{835}$$

this condition becomes

$$C_{ab}{}^d g_{dc} + C_{ac}{}^d g_{bd} = 0. \tag{836}$$

This is the algebraic condition that guarantees gauge invariance of the Chern–Simons action, up to possible boundary contributions.

Flat Connections and Local Degrees of Freedom

The equation

$$F = 0 \tag{837}$$

states that the gauge connection is flat.

Locally, on a simply connected region, every flat connection can be written as a pure gauge:

$$A = g^{-1} dg. \tag{838}$$

Therefore all local solutions are gauge equivalent.

This explains an important property of three-dimensional Chern–Simons theory: *Pure Chern–Simons theory has no local propagating bulk degrees of freedom.*

This does *not* mean that the theory is completely trivial. If the topology of the manifold is nontrivial, flat connections may possess nontrivial holonomies. Moreover, in the presence of a boundary, degrees of freedom that were gauge in the bulk may become physical boundary degrees of freedom.

This fact will become extremely important when studying three-dimensional gravity and, in particular, asymptotic symmetries.

Three-Dimensional Gravity as an $SO(2,2)$ Chern–Simons Theory

We now return to gravity with negative cosmological constant,

$$\Lambda = -\frac{1}{\ell^2}. \tag{839}$$

The first-order Einstein–Cartan action in three dimensions is

$$S_{\mathrm{EC}}[e, \omega] = \frac{1}{16\pi G} \int_M \epsilon_{abc} \left(R^{ab} \wedge e^c - \frac{\Lambda}{3} e^a \wedge e^b \wedge e^c \right). \tag{840}$$

For $\Lambda = -1/\ell^2$ this becomes

$$S_{\mathrm{EC}} = \frac{1}{16\pi G}\int_M \epsilon_{abc}\left(R^{ab}\wedge e^c + \frac{1}{3\ell^2}e^a\wedge e^b\wedge e^c\right). \tag{841}$$

An important simplification occurs only in three dimensions. An antisymmetric Lorentz tensor ω^{ab} contains

$$\frac{3(3-1)}{2} = 3 \tag{842}$$

independent components, exactly the same number as a Lorentz vector. We can therefore dualize it according to

$$\omega_a = \frac{1}{2}\epsilon_{abc}\omega^{bc}. \tag{843}$$

The inverse relation is

$$\omega^{ab} = -\epsilon^{abc}\omega_c, \tag{844}$$

with the sign depending on the convention used for raising the Levi–Civita tensor. Likewise, we define the dual curvature

$$R_a = \frac{1}{2}\epsilon_{abc}R^{bc}. \tag{845}$$

It follows that

$$\epsilon_{abc}R^{ab}\wedge e^c = 2e^a\wedge R_a. \tag{846}$$

Consequently, the Einstein–Cartan action can be written as

$$S_{\mathrm{EC}} = \frac{1}{16\pi G}\int_M \left[2e^a\wedge R_a - \frac{\Lambda}{3}\epsilon_{abc}e^a\wedge e^b\wedge e^c\right]. \tag{847}$$

The dual curvature is

$$R^a = d\omega^a + \frac{1}{2}\epsilon^a{}_{bc}\omega^b\wedge\omega^c. \tag{848}$$

The $\mathfrak{so}(2,2)$ Algebra

For $\Lambda < 0$, the relevant gauge algebra is $\mathfrak{so}(2,2)$. Using Lorentz generators J_a and AdS translation generators P_a, we write

$$[J_a, J_b] = \epsilon_{ab}{}^c J_c, \tag{849}$$
$$[J_a, P_b] = \epsilon_{ab}{}^c P_c, \tag{850}$$
$$[P_a, P_b] = \epsilon_{ab}{}^c J_c. \tag{851}$$

Here the P_a have been chosen as dimensionless AdS translation generators. The length scale ℓ will instead appear explicitly in the gauge connection.

The Lorentz generators J_a are related to the conventional antisymmetric generators J_{ab} through

$$J_a = \frac{1}{2}\epsilon_{abc}J^{bc}. \tag{852}$$

Unlike a simple Lie algebra, $\mathfrak{so}(2,2)$ admits two independent invariant symmetric bilinear forms. A general invariant bilinear form can be parametrized as

$$\langle J_a, J_b\rangle = \alpha_1\eta_{ab}, \tag{853}$$
$$\langle J_a, P_b\rangle = \alpha_2\eta_{ab}, \tag{854}$$
$$\langle P_a, P_b\rangle = \alpha_1\eta_{ab}. \tag{855}$$

Different choices of α_1 and α_2 lead to different Chern–Simons actions. To reproduce the ordinary Einstein–Cartan action, we choose

$$\alpha_1 = 0, \qquad \alpha_2 = 1. \tag{856}$$

Therefore,

$$\langle J_a, J_b\rangle = 0, \qquad \langle P_a, P_b\rangle = 0, \qquad \langle J_a, P_b\rangle = \eta_{ab}. \tag{857}$$

Notice that this bilinear form is nevertheless non-degenerate: it pairs the Lorentz generators with the translation generators.

The Gravitational Gauge Connection

We now introduce the $\mathfrak{so}(2,2)$ gauge connection

$$A = \omega^a J_a + \frac{1}{\ell}e^a P_a. \tag{858}$$

The factor $1/\ell$ is required so that the two pieces of the connection have the same physical dimension. We now substitute this connection into the Chern–Simons action.

We have

$$\langle A \wedge dA\rangle = \left\langle \left(\omega^a J_a + \frac{1}{\ell}e^a P_a\right) \wedge \left(d\omega^b J_b + \frac{1}{\ell}de^b P_b\right)\right\rangle. \tag{859}$$

Since

$$\langle J_a, J_b\rangle = \langle P_a, P_b\rangle = 0, \tag{860}$$

only the mixed terms survive:

$$\begin{aligned}\langle A \wedge dA\rangle &= \frac{1}{\ell}\left(e^a \wedge d\omega^b\langle P_a, J_b\rangle + \omega^a \wedge de^b\langle J_a, P_b\rangle\right)\\ &= \frac{1}{\ell}\left(e^a \wedge d\omega_a + \omega^a \wedge de_a\right).\end{aligned} \tag{861}$$

Now,

$$d(\omega^a \wedge e_a) = d\omega^a \wedge e_a - \omega^a \wedge de_a. \tag{862}$$

Since

$$d\omega^a \wedge e_a = e_a \wedge d\omega^a, \tag{863}$$

we find

$$\omega^a \wedge de_a = e_a \wedge d\omega^a - d(\omega^a \wedge e_a). \tag{864}$$

Therefore

$$\langle A \wedge dA \rangle = \frac{2}{\ell} e_a \wedge d\omega^a - \frac{1}{\ell} d(\omega^a \wedge e_a). \tag{865}$$

The last term is a total derivative. In the bulk calculation we may temporarily omit it, remembering that it can become important when the manifold has a boundary.

Thus,

$$\langle A \wedge dA \rangle \simeq \frac{2}{\ell} e_a \wedge d\omega^a, \tag{866}$$

where $\simeq$ denotes equality up to a boundary term. We next calculate

$$\frac{2}{3} \langle A \wedge A \wedge A \rangle. \tag{867}$$

Using invariance of the bilinear form, this may equivalently be written as

$$\frac{2}{3} \langle A \wedge A \wedge A \rangle = \frac{1}{3} \langle [A, A] \wedge A \rangle. \tag{868}$$

For

$$A = \omega^a J_a + \frac{1}{\ell} e^a P_a, \tag{869}$$

the commutator contains

$$[A, A] = \omega^a \wedge \omega^b [J_a, J_b] + \frac{2}{\ell} \omega^a \wedge e^b [J_a, P_b] + \frac{1}{\ell^2} e^a \wedge e^b [P_a, P_b]. \tag{870}$$

Using the $\mathfrak{so}(2,2)$ algebra,

$$\begin{aligned} [A, A] = \epsilon_{ab}{}^c \left(\omega^a \wedge \omega^b + \frac{1}{\ell^2} e^a \wedge e^b \right) J_c \\ + \frac{2}{\ell} \epsilon_{ab}{}^c \omega^a \wedge e^b P_c. \end{aligned} \tag{871}$$

Contracting with the remaining A and using

$$\langle J_a, P_b \rangle = \eta_{ab}, \tag{872}$$

one obtains

$$\frac{2}{3} \langle A \wedge A \wedge A \rangle = \frac{1}{\ell} \epsilon_{abc} e^a \wedge \omega^b \wedge \omega^c + \frac{1}{3\ell^3} \epsilon_{abc} e^a \wedge e^b \wedge e^c. \tag{873}$$

The first term will combine with the $e \wedge d\omega$ term obtained above to produce the Lorentz curvature.

Recovering the Einstein–Cartan Action

Putting the quadratic and cubic pieces together, and dropping the total derivative for the moment, gives

$$S_{\rm CS} = \frac{k}{4\pi}\int_M \left[\frac{2}{\ell}e_a \wedge d\omega^a + \frac{1}{\ell}\epsilon_{abc}e^a \wedge \omega^b \wedge \omega^c + \frac{1}{3\ell^3}\epsilon_{abc}e^a \wedge e^b \wedge e^c\right]. \tag{874}$$

The first two terms can be combined using

$$R^a = d\omega^a + \frac{1}{2}\epsilon^a{}_{bc}\omega^b \wedge \omega^c. \tag{875}$$

Indeed,

$$2e_a \wedge R^a = 2e_a \wedge d\omega^a + \epsilon_{abc}e^a \wedge \omega^b \wedge \omega^c. \tag{876}$$

Therefore,

$$S_{\rm CS} = \frac{k}{4\pi\ell}\int_M \left[2e_a \wedge R^a + \frac{1}{3\ell^2}\epsilon_{abc}e^a \wedge e^b \wedge e^c\right] \tag{877}$$

up to the boundary term coming from $d(\omega^a \wedge e_a)$. Since

$$\Lambda = -\frac{1}{\ell^2}, \tag{878}$$

we may equivalently write

$$S_{\rm CS} = \frac{k}{4\pi\ell}\int_M \left[2e_a \wedge R^a - \frac{\Lambda}{3}\epsilon_{abc}e^a \wedge e^b \wedge e^c\right]. \tag{879}$$

But the Einstein–Cartan action is

$$S_{\rm EC} = \frac{1}{16\pi G}\int_M \left[2e_a \wedge R^a - \frac{\Lambda}{3}\epsilon_{abc}e^a \wedge e^b \wedge e^c\right]. \tag{880}$$

Thus the two theories coincide provided

$$\frac{k}{4\pi\ell} = \frac{1}{16\pi G}. \tag{881}$$

Consequently,

$$k = \frac{\ell}{4G}. \tag{882}$$

This is the relation between the Chern–Simons level and thengravitational parameters of AdS_3 gravity. We have therefore shown explicitly that

$$\boxed{2+1 \text{ Einstein gravity with } \Lambda < 0 \quad \longleftrightarrow \quad SO(2,2) \text{ Chern–Simons theory.}} \tag{883}$$

The Chern–Simons equation is simply $F = 0$. For the gravitational connection

$$A = \omega^a J_a + \frac{1}{\ell} e^a P_a, \tag{884}$$

the curvature decomposes as

$$F = \left(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \right) J_a + \frac{1}{\ell} T^a P_a. \tag{885}$$

Hence

$$F = 0 \tag{886}$$

is equivalent to

$$T^a = 0, \tag{887}$$

together with

$$R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c = 0. \tag{888}$$

Equivalently,

$$R^{ab} + \frac{1}{\ell^2} e^a \wedge e^b = 0. \tag{889}$$

Thus the flatness condition of the $SO(2,2)$ gauge connection is precisely the statement that the spacetime is torsionless and locally of constant negative curvature.

This also makes transparent why pure Einstein gravity in three dimensions has no local propagating gravitational degrees of freedom: locally, its solutions correspond to flat Chern–Simons connections. Global topology and boundaries, however, may still carry nontrivial physical information.

Appendices

A Groups

Modern mathematics largely involves studying patterns and symmetries. Group theory is the study of symmetry. It allows us to say interesting, sometimes very detailed things about physical systems even when we don't understand exactly what the systems are. A group is a set G in which a rule of composition (some product in general) satisfy

- Associativity a,b,c $\in G$ then $(ab)c = a(bc)$.
- $\exists\ e \in G$ such that $\forall a \in G\ ae = ea = a$.
- $\exists$ inverse element $a^{-1} \in G$ for each $a \in G$ such that $a^{-1}a = aa^{-1} = e$.

If the operation of the group is commutative ($ab = ba$) the group is said to be *abelian*. In the opposite is *non-abelian*. G_1 and G_2 are groups. The are *isomorphic* (denoted by $G_1 = G_2$) if there exists a bijective mapping $f : G_1 \to G_2$ consistent with the multiplication operators

$$f(g_1, g_2) = f(g_1)f(g_2)\,,\ f(g^{-1}) = [f(g)]^{-1}. \tag{890}$$

We consider that we cannot distinguish between isomorphic groups. Subgroups H of G is a subset which is itself a group by the definitions above w.r.t the product of G. The center of G consists in the elements $w \in G$ that commutes with all of the elements of the group. The center is itself a subgroup.

Direct product of groups The direct product of G_1 and G_2 (denoted $G_1 \times G_2$) is the set of pairs $\{g, h\}$ in which the product and the inverse element take the form

$$\{g, h\}\{g', h'\} = \{gg', hh'\}$$
$$\{g, h\}^{-1} = \{g^{-1}, h^{-1}\}$$

and the unit element is $\{e_1, e_2\}$. Thus $G_1 \times G_2$ is a group.

A group *homomorphism* is a mapping $f : G \to G'$ ($g_1, g_2 \in G$)

$$f(g_1 g_2) = f(g_1)f(g_2), \tag{891}$$

consistent with the product operation.Here we consider that $f(e) = e'$, $f(g^{-1}) = [f(g)]^{-1}$. The set of all elements $g \in G$ such that $f(g) = e'$ is called the kernel of the homomorphism (Ker f).

Be the H a subgroup of G. A element is equivalent to another one $g_1 \sim g_2$ if $g_1 = g_2 h$, for $h \in H$. The equivalence relation must fulfill

- if $g_1 \sim g_2$ then $g_2 \sim g_1$.
- if $g_1 \sim g_2 h_{12}$, $g_2 \sim g_3 h_{23}$ then $g_1 \sim g_3(h_{12}h_{23})$ so $g_1 \sim g_3$.

This equivalence relation allows to divide G into disjoint sets called *cosets*. A coset consist in elements of G which are all equivalent to one another. The (right) *coset space* G/H is the set of cosets. The definition of equivalence $g_1 \sim g_2$ if $g_1 = hg_2$ lead us to the left coset. This concept is related to homogeneous spaces.

A.1 Lie groups and algebras

(For subgroups of $GL(n, \mathbf{C})$), i.e, matrix groups whose elements whose elements are matrices. The neighborhood (topology) for $n \times n$ matrices space is naturally introduced. Two matrices are nearby if all their elements are nearby. For matrices $M(t)$ are differentiable w.r.t a real parameter by the derivatives of the matrix elements $((\frac{dM(t)}{dt})_{ij}$ is $\frac{d}{dt}M_{ij}(t))$. In general the space of $n \times n$ complex matrices can be viewed as a $2n^2$-dimensional (Euclidean) space R^{2n^2} (with coordinates $2n^2$ matrix elements ReM_{ij} and ImM_{ij}). The smooth matrices are manifolds embedded in this space.
Example : A smooth family of matrices depending on a real parameter $M(t)$ is a curve in R^{2n^2}. Smooth groups are groups which are smooth manifolds in the space R^{2n^2}. They are *Lie groups*.

The simplest (non-trivial) Lie group is $U(1)$. It can be understood considering the complex numbers as 1×1 matrices. The group $U(1)$ is a circle in the complex plane. Other Lie groups are $U(n), SU(n), O(n), SO(n)$ for example. For each point of a manifold of dimension k in the $2n^2$-dim Euclidean space we have a tangent space at that point: a real vector space of dimension k consisting of vectors tangent to the manifold at that point.
Since the identity is an important element in the group, it is useful to parameterize the elements (at least those close to the identity element) in such a way that $\alpha = 0$ corresponds to the identity element. If we find a representation of the group, the linear operators will be parametrized in the same way.
The tangent space for a Lie group at the unit element is the *Lie algebra* of that Lie group. Any curve $g(\alpha)$ can be expressed near to the unity as

$$g(\alpha)|_{\alpha=0} = e, \tag{892}$$

and the linear operators of the representation

$$D(\alpha)|_{\alpha} = 1. \tag{893}$$

If we Taylor expand in some neighborhood of the identity element

$$D(d\alpha) = 1 + id\alpha_a X_A + \cdots, \tag{894}$$

where the parameter α is infinitesimal and

$$X_a = -i\frac{\partial}{\partial \alpha_a} D(\alpha)|_{\alpha=0}. \tag{895}$$

This X_a are the **generators** of the group. If the representation is unitary, the generators will be hermitian operators. One can show that the Lie algebra is a real vector space by performing a

product of a element of the algebra with a real number and the sum of two elements. Also the commutation belongs to the Lie algebra $[A_1, A_2] = A_1A_2 - A_2A_1$.
Example: The $U(n)$ algebra. Unitary matrices close to unity have the propertie

$$\left(1 + At + O(t^2)\right)\left(1 + A^\dagger t + O(t^2)\right) = 1, \tag{896}$$

therefore $A^\dagger = -A$, i.e, the Lie algebra of this group is the algebra of all anti-Hermitian matrices. Two lie algebras are isomorphic if there exists a one-to-one correspondence which preserves addition, multiplication by a real number and commutation. In the Lie algebras one can choose a basis, where the elements of the basis are the generators of the Lie algebra and the corresponding Lie group. Since the commutator belongs to the algebra

$$[T_i, T_j] = C_{ijk}T_k, \tag{897}$$

where C_{ijk} are anti-symmetric in the first two indices and real, and they are the **structure constants**. From (895) we can see that a possibility to go away from the identity in some fixed direction is taking an infinitesimal group element

$$D(d_\alpha) = 1 + id\alpha_a X_a, \tag{898}$$

and raising to a large power. Because the group properties this always gives another group element. This suggest that we can define the representation as

$$D(\alpha) = \lim_{k\to\infty}(1 + i\alpha_a X_a/k)^k = e^{i\alpha_a X_a}. \tag{899}$$

This is a particular parametrization of the representation, called exponential parametrization.

A.2 Representations

A *representation* of a group G is a mapping (D) of the elements of G onto a set invertible operators $T(g)$. This mapping need to be consistent with the group operations. In the case of Lie algebras we need to remember the commutation property. A representation of a Lie group can be used ti a representation of the Lie algebra (AG). $T(AG)$ is given by

$$T(1 + \epsilon A) = 1 + \epsilon T(A). \tag{900}$$

Not all representations of algebras are **generated** by a representation of the group. If the linear operators are unitary, for example, the representation of the group is an unitary representation.
A representation of G in a linear space V can fix a basis e_i in V. Then the linear operator action can be decomposed in this basis

$$T(g)e_i = T_{ji}(g)e_j, \tag{901}$$

taking e_i to some vector of V. Thus for every element is mapped to a matrix, in some basis and the matrix are n-dimensional with $Dim(V) = n$. For vectors $\psi \in V$

$$\psi = \psi_i e_i, \tag{902}$$

where the ψ_i are the components, then

$$T(g)\psi = \psi_i(T(g)e_i) = \psi_i T_{ji}e_j. \tag{903}$$

The components of $T(g)\psi$ are $T_{ij}(g)\psi_i$. The representations on the same space V are equivalent if $\exists\ S \in V$, with S an invertible operator such that

$$T'(g) = ST(g)S^{-1}. \tag{904}$$

For a subspace W of V. It's an invariant subspace of the representation if $\forall\, \psi \in W$ and $g \in G$

$$T(g)\psi \in W. \tag{905}$$

The representation $T(G)$ is said to be a *irreducible representation* if there is no non-trivial invariant subspaces.

The **fundamental representation** $T(g)$ of a Lie group G consisting of $n \times n$ matrices (for example SU(n) or SO(n)) and a (column) vector space V, acts on this space as

$$(T(g)\psi)_i = g_{ij}\psi_j. \tag{906}$$

Or, using a fixed basis in V. the action of $T(g)$ on the basis vectors is

$$T(g)e_i = g_{ji}e_j\,. \tag{907}$$

For $SU(n)$ the fundamental representation is complex, while for $SO(n)$ it is real.
A Lie group G and the Lie algebra AG, both which consists of $n \times n$ matrices. The algebra is a real vector space and also the space of the **adjoint representation**. Linear operators $Ad(g)$ in this representation (A a matrix on AG) acts as

$$Ad(g)A = gAg^{-1}. \tag{908}$$

To be gAg^{-1} a element of the algebra, we construct a curve on the group

$$h(t) = gg_A(t)g^{-1}, \tag{909}$$

expanding $g_a = 1 + tA + \cdots$, we see this is the curve defining $A \in AG$. Close to the identity also we have $h(t) = 1 + tA_h + \cdots$, so the requirement is given by

$$gAg^{-1} = A_h \in AG. \tag{910}$$

The matrices of the adjoint representation coincides with the structure constants. By the definition of a matrix of a representation

$$ad(t_i)t_j = T^{(i)}_{kj}t_k, \tag{911}$$

with t_j basis elements of AG (generators). $T^{(i)}_{kj}$ is the matrix of linear operators corresponding to the generators. We consider

$$ad(t_i)t_j = [t_i, t_j] = C_{ijk}t_k, \tag{912}$$

so directly we relate the structure constants as

$$T^{(i)}_{jk} = C_{ijk}. \tag{913}$$

A.3 Compact Lie groups and algebras

Lie groups are smooth manifolds. Compact Lie groups corresponds with those manifolds that are compact and the compact Lie algebras are the corresponding algebras associated to that groups.

Theorem 1. *A Lie algebra is compact if and only if it has a positive-definite scalar product, which is invariant under the action of the adjoint representation of the group.*

It means that in any compact Lie algebra (only in compact Lie algebras) there exists a bilinear form (A, B) such that for all $g \in G$ and $A, B \in AG$

$$(Ad(g)A, Ad(g)B) = (A, B), \tag{914}$$

and for all A

$$(A, A) \geq 0, \tag{915}$$

with being zero only for the zero element of the algebra. For matrix groups the scalar product in the algebra is the trace

$$(A, B) = -\mathrm{Tr}(AB). \tag{916}$$

It easy to check the invariance under the adjoint representation by permutation of matrices in the trace

$$\left(gAg{-}1, gBg^{-1}\right) = -\mathrm{Tr}\left(gAg^{-1}gBg^{-1}\right) = \mathrm{Tr}(AB). \tag{917}$$

It is non trivial to show the positive definiteness of the trace. An algebra generators can be chosen in a orthonormal basis

$$\mathrm{Tr}(t_i t_j) = -\frac{1}{2}\delta_{ij}. \tag{918}$$

In this basis the structure constants are full antisymmetric

$$[t_i, t_j] = C_{ijk}t_k, \tag{919}$$

so we can directly see this in the basis

$$C_{ijk} = -2\mathrm{Tr}[t_i, t_j]t_k = -2\left[\mathrm{Tr}(t_i t_j t_k) - \mathrm{Tr}(t_j t_i t_k)\right]. \tag{920}$$

If we compare this with the same but $(k \leftrightarrow j)$ we see that

$$C_{ikj} = -C_{ijk}. \tag{921}$$

All abelian compact lie algebras are direct sums of $U(1)$ algebras. A compact Lie algebra is *semi-simple* if it does not contain an Abelian invariant subalgebra. In the case of *simple* Lie algebras, they do not contain any invariant subalgebra.
Any compact Lie algebra A is uniquely representable in the form of certain $U(1)$ subalgebras and A_n simple subalgebras

$$A = U(1) + U(1) + \cdots + A_1 + \cdots + A_n. \tag{922}$$

So the study of compact Lie algebras reduces to the study of simple Lie algebras. Locally every compact Lie group is represented in a unique way in the form of a direct product

$$G = U(1) \times U(1) \times \cdots \times G_1 \times \cdots \times G_n, \tag{923}$$

with G_n simple groups. In the case of simple compact Lie algebra, there exists just one invariant positive-definite scalar product (up to a scalar global factor). If the algebra is semi-simple the set of invariants is more extense. Consider

$$A = A_1 + A_2, \tag{924}$$

then any vector $B \in A$

$$B = B_1 + B_2, B_1 \in A_1\,,\, B_2 \in A_2. \tag{925}$$

If $(,)_i$ is the scalar product in A_1, A_2 $(i = 1, 2)$, then all invariant scalar products have the form

$$(B, B') = \alpha_i (B_i, B_i')_i, \text{with}\, i = 1, 2, \tag{926}$$

and α_i arbitrary positive constants. The positive definite quadratic invariants (adjoint representation) are linear combinations of quadratic invariants of each simple algebra.
Any representation of a compact Lie group is equivalent to a unitary representation, and representations of Lie algebras are equivalent to anti-Hermitian representations. When we consider the group $SU(n)$ we use Hermitian generators($A = iB$ with B hermitian), so every element of the algebra

$$A = iA^a t_a, \tag{927}$$

where t_a are Hermitian matrices and A^a real coefficients. Elements close to the unit element group can be expanded

$$g = 1 + i\epsilon^a t_a, \tag{928}$$

with ϵ^a small parameters. The generators satisfy

$$[t_a, t_b] = iC_{abc} t_c, \tag{929}$$

where C_{abc} is real and fully antisymmetric. For complex representations we use Hermitian generators $T(t_a) = T_a$

$$[T_a . T_b] = iC_{abc} T_c. \tag{930}$$

B Pullbacks and Stokes Theorem

It is useful to recall briefly the geometrical statement of Stokes' theorem.

Consider a smooth map

$$X : M^{(d)} \longrightarrow N^{(D)}, \qquad D \geq d, \tag{931}$$

with local coordinates u^i on M and x^μ on N,

$$x^\mu = X^\mu(u). \tag{932}$$

For an embedding, X is smooth, injective, and has injective differential.

The differential of X defines the *pushforward*

$$X_* : T_p M \longrightarrow T_{X(p)} N. \tag{933}$$

If

$$\xi = \xi^i \frac{\partial}{\partial u^i}, \tag{934}$$

then

$$\boxed{X_* \xi = \xi^i \frac{\partial X^\mu}{\partial u^i} \frac{\partial}{\partial x^\mu}.} \tag{935}$$

For example, consider the unit circle embedded in the plane,

$$X : S^1 \longrightarrow \mathbb{R}^2, \qquad \theta \longmapsto (\cos\theta, \sin\theta). \tag{936}$$

Then

$$\begin{aligned} X_* \left(\frac{\partial}{\partial \theta} \right) &= \frac{\partial x}{\partial \theta} \partial_x + \frac{\partial y}{\partial \theta} \partial_y \\ &= -\sin\theta\, \partial_x + \cos\theta\, \partial_y, \end{aligned} \tag{937}$$

which is the tangent vector to the unit circle.

The dual construction is the *pullback*. For a one-form

$$\alpha = \alpha_\mu dx^\mu \in T^*_{X(p)} N, \tag{938}$$

the pullback is

$$\boxed{X^* \alpha = \alpha_\mu (X(u)) \frac{\partial X^\mu}{\partial u^i} du^i.} \tag{939}$$

More generally, for a p-form

$$B = \frac{1}{p!} B_{\mu_1 \cdots \mu_p} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_p}, \tag{940}$$

we have

$$\begin{aligned} X^* B = \frac{1}{p!} B_{\mu_1 \cdots \mu_p} (X(u)) &\frac{\partial X^{\mu_1}}{\partial u^{i_1}} \cdots \frac{\partial X^{\mu_p}}{\partial u^{i_p}} \\ &\times du^{i_1} \wedge \cdots \wedge du^{i_p}. \end{aligned} \tag{941}$$

For an oriented d-dimensional manifold M with boundary, Stokes' theorem states

$$\boxed{\int_M d\alpha = \int_{\partial M} \iota^* \alpha,} \tag{942}$$

where

$$\iota : \partial M \hookrightarrow M \tag{943}$$

is the inclusion map.

Thus the first term in (511) is a genuine boundary term.

If we impose

$$\delta \omega^{ab} \Big|_{\partial M} = 0, \tag{944}$$

or suitable asymptotic boundary conditions, this term does not contribute to the equations of motion.

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